



**GREEN
CLIMATE
FUND**

Meeting of the Board
26 – 28 February 2019
Songdo, Incheon, Republic of Korea
Provisional agenda item 34

GCF/B.22/10/Add.03

1 February 2019

Consideration of funding proposals - Addendum III

Funding proposal package for FP101

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled “Resilient Rural Belize (Be-Resilient)”;
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat’s assessment;
- e) Independent Technical Advisory Panel’s assessment;
- f) Response of the accredited entity to the independent Technical Advisory Panel’s assessment; and
- g) Gender documentation.

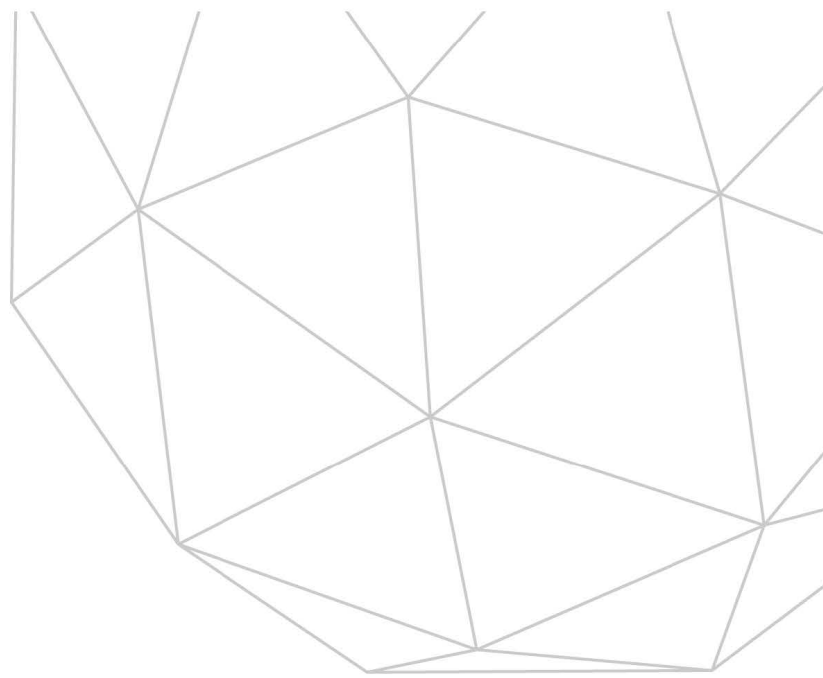
The funding proposal package for FP101 is being submitted for the first time for the Board's consideration at its twenty-second meeting.

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Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: Resilient Rural Belize (Be-Resilient)

Country/Region: Belize

Accredited Entity: International Fund for Agricultural Development - IFAD

Date of Submission: _____

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Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"[FP]-[Agency Short Name]-[Date]-[Serial Number]"

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		Resilient Rural Belize (Be-Resilient)
A.1.2. Project or programme		Project
A.1.3. Country (ies) / region		Belize
A.1.4. National designated authority (ies)		Ministry of Economic Development
A.1.5. Accredited entity		IFAD
A.1.5.a. Access modality		☐ Direct ☒ International
A.1.6. Executing entity / beneficiary		Executing Entities: Ministry of Economic Development, Ministry of Finance
A.1.7. Project size category (Total investment, million USD)		☐ Micro (≤ 10) ☐ Medium ($50 < x \leq 250$) ☒ Small ($10 < x \leq 50$) ☐ Large (> 250)
A.1.8. Mitigation / adaptation focus		☐ Mitigation ☒ Adaptation ☐ Cross-cutting
A.1.9. Date of submission		Initial Submission Feb 16, 2018. Resubmission Oct 31, 2018
A.1.10. Project contact details	Contact person, position	Margarita Astralaga, Director, Environment and Climate Division
	Organization	International Fund for Agricultural Development (IFAD)
	Email address	m.astralaga@ifad.org
	Telephone number	+39 06 5459 2151
	Mailing address	Via Paolo di Dono 44, Rome, Italy, 00142
A.1.11. Results areas <i>(mark all that apply)</i>		
Reduced emissions from: ☐ Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.) ☐ Low emission transport (E.g. high-speed rail, rapid bus system, etc.) ☐ Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.) ☐ Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.) Increased resilience of: ☒ Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.) ☒ Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.) ☒ Infrastructure and built environment (E.g. sea walls, resilient road networks, etc.) ☐ Ecosystem and ecosystem services (E.g. ecosystem conservation and management, ecotourism, etc.)		

A.2. Project / Programme Executive Summary (max 300 words)

1. As a Small Island Developing State in the Caribbean hurricane belt, Belize is highly vulnerable to the impacts of climate change. While the most extreme and visible effect is the increased intensity of tropical storms and hurricanes, climate change also manifests itself through increased temperature and sea level rise, as well as documented trends of increased droughts, flooding, and significant rainfall pattern variations. Smallholder farmers are amongst the most vulnerable as they suffer devastating losses, most dramatically in the occurrence of an extreme event but similarly through the impacts of persistent and unpredictable seasonal variations. Due to the high cost of such events, smallholders' limited investment capacity is fully focused on recovery and purchase of inputs, thus preventing them from investing in assets and measures that would reduce their vulnerability. As a result, the impacts of climatic events are more profound, and the recovery time is prolonged. This vulnerability is exacerbated by the poor market conditions for smallholder farming in Belize. With their current assets, capacities, and connections to markets, smallholders are not equipped to deliver a regular supply of adequate volumes of quality produce required for sustainable market access. As such, markets have turned towards imported goods to supply a large proportion of foods, including vegetables and fruits, that have traditionally been supplied by smallholders and can be competitively produced in Belize.

2. The project aims to introduce climate resilient agricultural practices that will allow smallholder farmers to have a sustainable production process and improved market access for their produce, even under the stress of climate change and extreme climatic events; therefore increasing their economic, social, and environmental resilience. In the context of smallholder farmers in Belize, improved resilience is defined as the capacity to minimize the impacts of climatic and economic shocks, enabling farmers to better confront periodic variations and strengthening their capacity to cope and recover in times of extreme stress. To achieve this in a manner that meaningfully improves smallholders adaptive capacity, significant improvements across the entire value chain are required, as weaknesses in production technologies and practices, fragile and variable physical access to markets, and limited capacity to respond to market demands are all factors that contribute to farmers' vulnerability. The complementarity of the IFAD loan, targeting productive development, with a GCF loan and grant that promotes increased resilience and directly addresses climate change threats, has encouraged the country to commit to an innovative approach to agricultural development that can significantly change the development path for rural smallholders in Belize. The Government of Belize, represented by The Ministry of Finance (MoF) for all legal and financial matters, is the responsible party for the approval of both loan and grant components from both IFAD and GCF.

3. The proposed Project has two complementary and mutually-reinforcing components: Climate Resilient Value Chains Development and Climate Resilient Rural Infrastructural and Assets Development, with cross cutting activities supporting both components. It will initially work in 23 communities clustered in five priority Districts and activities will focus on the development of value chains of six vegetables (tomatoes, sweet peppers, hot peppers, cabbages, carrots, and onions), one fruit (pineapple), and beekeeping products (principally honey).

4. Component 1 will introduce/strengthen smallholder participation in select value chains through the promotion of climate resilient and environmentally sustainable production methods, product diversification, and related innovations. It will rely on technical support and the physical presence in the project Districts of climate resilient agriculture specialists working with Ministry of Agriculture (MoA) extension agents. A competitive matching grant fund (MGF), partially funded by the GCF loan, will be structured to support climate resilient production and value chain development investments. Producer Organizations (POs) as well as individual members receiving support from technical areas of the project will be eligible for funding. Additionally, this component will support food security, self-consumption and healthy food choices through implementation of backyard gardens.

5. Component 2 will support public climate resilient infrastructure enhancing smallholder farming business/ rural enterprise opportunities, while serving the largest number possible of direct and indirect project

participants. It includes investments in rehabilitation and provision of climate proof roads, drainage and irrigation schemes as well as a nationwide climate information system. Water User Associations (WUA) will be formed and provided with technical assistance for institutional development and management of irrigation and drainage schemes.

6. The productive investments in Component 1 and the public infrastructure in Component 2 will be guided by regional Infrastructure and Production Plans, based on climate vulnerability and market studies. Both Components are also supported by cross-cutting activities which include government authorities capacity building, research and development (R&D), and policy engagement/consultation.

7. The project will introduce climate resilient agricultural practices that will allow smallholder farmers to have a sustainable production process and improved market access for their produce, even under the stress of climate change and extreme climatic events; therefore increasing their economic, social, and environmental resilience. Directly benefiting 30,000 rural people (of which 40% are women and 20% youth) and indirectly benefitting up to 29% of the population.

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)	February 2019
Expected financial close (if applicable)	
Estimated implementation start and end date	Start: August 2019 End: November 2024
Project/programme lifespan	15 years 0 months

A.4. Project Abbreviations

5Cs	Caribbean Community Climate Change Centre	MNR	Ministry of Natural Resources
AEO	Agricultural Extension Officer	MOA	Ministry of Agriculture, Fisheries, Forestry, Environment and Sustainable Development
AMO	Agricultural & Marketing Officer		
AWPB	Annual Work Plan and Budget	MOCR	Market Oriented Climate Resilient Plans
BMDC	Belize Marketing and Development Corporation	MOF	Ministry of Finance
BOS	Bureau of Standards	MOM	Management, operation and maintenance
BPs	Business Plans	MoU	Memoranda of Understanding
BRFP	Belize Rural Finance Programme	MOW	Ministry of Works
BWS	Belize Water Services	MRD	Ministry of Rural Development
BYG	Backyard gardens	MTR	Mid-Term Review
BZ/BZD	Belize Dollar	NAFP	National Agricultural and Food Policy 2015-2030
CABEI	Central American Bank for Economic Integration	NCCO	National Climate Change Office
CARD	Community Initiated Agricultural and Resource Management	NCCPSAP	National Climate Change Policy, Strategy and Action Plan
CARICOM	Caribbean Community	NDA	National designated authority
CATIE	Centro Agronómico Tropical de Investigación y Enseñanza	NDC	Nationally Determined Contribution
CCA	Climate Change Adaptation	NEMO	National Emergency Management Organization
CCRIF	Caribbean Catastrophe Risk Insurance Facility	NFAP	National Food and Agricultural Policy 2015 -2030
CCVI	Climate Change Vulnerability Index	NGO	Non-Governmental Organization
CDB	Caribbean Development Bank	NIWRA	Natural Integrated Water Resource Authority
CEA	Climate Extension Agents	NMS	National Meteorological Service
CI	Core Indicators	NPV	Net Present Value
CIAT	International Center for Tropical Agriculture	NRM	Natural resources management
CIS	Climate Information System	ODP	Organizational Development Plans
CN	Concept Note	ORMS	Operational Results Measurement System
CO	Climate Officer	PA	Programme Assistant
CPA	Country Poverty Assessment	PACCT	Programme Accountant
CPM	Country Programme Manager	PBAS	Performance Based Allocation System
CRA	Climate Resilient Agriculture	PCB	Belize's Pesticides Control Board
		PCR	Programme Completion Report

CRAPMA	Climate Resilient Agricultural Production and Market Access	PDR	Programme Design Report
CRI	Climate Risk Index	PES	Payment for Ecosystem Services
CRRIA	Climate Resilient Rural Infrastructure and Assets Development	PFM	Public Financial Management
CRVC	Climate Resilient Value Chains Development	PFS	Programme Financial Statements
CSN	Country Strategy Note	PIM	Programme Implementation Manual
CU	Credit Union	PM	Programme Manager
CVA	Climate Vulnerability Assessment	PMU	Project Management Unit
DC	Department of Cooperatives	PO	Producer Organization
DE	Departments of Extension	POC	Programme Oversight Committee
DFC	Development Finance Corporation.	PP	Procurement Plan
EA	Extension Agents	PPP	Public Private Partnership
EAP	Economically Active Population	PPPPs	Public-Private-Producer Partnerships
EFA	Economic and Financial Analysis	PRO	Procurement Officer
EIA	Environmental Impact Assessment	QB	QuickBooks Premier Accounting Software
EO	Extension Officers	R&D	Research and Development
EU	European Union	RB-COSOP	Results-Based Country Strategic Opportunities Programme
FAO	Food and Agriculture Organization	RFP	Request for Proposal
FM	Financial Management	RIE	Rural Infrastructure Engineer
FO	Finance Officer	RIMS	Results and Impact Management System
FPM	Financial Procedures Manual	ROD	Rural Organization Development Specialist
FY	Fiscal Year	RODO	Rural Organization Development Officer
GAI	Global Adaptation Index	RPP	Resilient Production Plan
GCF	Green Climate Fund	RRB	Resilient Rural Belize
GDP	Gross Domestic Product	RRI	Rural Roads Improvements
GH	Greenhouse	RSP	Rural Sector Performance
GIDC	Grenada Investment Development Corporation	SDGs	Sustainable Development Goals
GNI	Gross National Income	SECAP	Social, Economic and Climate Assessment Paper
GOB	Government of Belize	SEDP	Socio-Economic Development Planning
GSDS	Growth and Sustainable Development Strategy	SIB	Statistical Institute of Belize
GST	General Sales Tax	SIDS	Small Island Developing States
HH	Household	SMEs	Small, Medium and Micro Enterprises
ICP	IFAD Client Portal	SO	Strategic Objective
IDB	Inter-American Development Bank	SOE	Statements of Expenditure
IFAC	International Federation of Accountants	SSID	Small-scale Irrigation and Drainage
IFAD	International Fund for Agricultural Development	STG	Storage
IIFA	Infrastructure Investment Funding Award	TA	Technical Assistance
IM	Investment Management	TCGA	Toledo Cacao Growers Association
IPM	Integrated Pest Management	TEC	Technical Evaluation Committee
IPP	Infrastructure and Production Plan	TI	Transparency International
IPs	Implementing Partners	TO	PMU Technical Officer
IPSAS	International Public Sector Accounting Standards	TOR	Terms of Reference
IRR	Internal Rate of Return	TSFD	Toledo Small Farmer Development
IWRM	Integrated Water Resource Management	TWP	Technical Work Plan
KM	Knowledge Management	UB	University of Belize
KMP	Knowledge Management Plan	UBCF	University of Belize Central Farm
LAC	Latin America and the Caribbean	UMIC	Upper Middle Income Country
LF	Logical Framework	UNDESA	Nations Department of Economic and Social Affairs
LFS	Labour Force Survey	UNDP	United Nations Development Programme
LPA	Lead Programme Agency	UNFCCC	United Nations Convention on Climate Change
LTB	Letter to Borrower	UNICEF	United Nations International Children's Emergency Fund
LVCC	Lead Value Chain Consultant.	USAID	United States Agency for International Development
M&E	Monitoring & Evaluation	USD	United States Dollar
MC	Management Committee	VC	Value Chain
MED	Ministry of Economic Development	VCAMA	Value Chains Analysis and Market Assessment
MEO	Monitoring & Evaluation Officer	VCS	Value Chain Specialist
MFFSD	Ministry of Forestry, Fisheries and Sustainable Development	WB	World Bank
MGF	Matching Grant Fun	WUA	Water User Associations
MICS	Multiple Indicator Cluster Survey	WUGs	Water Users' Groups
MIS	Management Information System		

B.1. Description of Financial Elements of the Project / Programme

The overall goal of the Project is to develop value chains of smallholder farmers that are resilient and adapted to the effects of climate change. The total Project cost is estimated at USD 20.0 million, including price and physical contingencies, to be executed in six years. There are two components to the project: the first includes activities related to improving climate resiliency and marketability of the value chain as well as strengthening producers organizations; the second component includes upgrading of the public infrastructure necessary for a resilient production, such as roads, drainage and information systems. In addition, there are crosscutting activities of institutional strengthening and research necessary for the development of both components. The table below summarizes project costs by component.

Activities will be financed as follows:

- (a) IFAD will primarily finance costs related to investment and technical assistance to improve production, value-added activities and improving market access. It will also cover the costs of strengthening organizations and public institutions. Its contribution will be allocated as follows: 39% of the total costs of Component 1 (US\$3.2 million); 21% of Component 2 (US\$1.8 million); and 87% of Project Management (US\$3.0 million).
- (b) The GCF will finance the costs of activities related to incorporating measures and improving knowledge on adaptation to climate change and to increase the productive resilience of the beneficiary households. Its contribution of US\$8.0 million will be allocated as follows: 45% of the costs of Component 1 (US\$3.6 million); 49% of the costs of Component 2, especially for irrigation, drainage works and climate information systems (US\$4.1 million); 3.3% of PMU and 15% of cross-cutting activities costs (US\$0.25 million).
- (c) The GoB will provide counterpart funds to cover the costs of taxes and will complement the financing of road infrastructure and irrigation and drainage works by 30%. The GoB's total contribution will be US\$3.2 million allocated as follows: 6% of the costs of Component 1 (US\$0.5 million); 30% of the costs of Component 2 (US\$2.5 million); and 5% of Project Management costs (US\$0.2 million).
- (d) The project participants and POs will present proposals for funding under the Project to improve resilient production and adaptation to climate change. They will provide a counterpart contribution in cash and/or in kind of at least 10%, totaling US\$0.8 million of the total cost of Component 1 (for investment and technical assistance).

Component	Sub-component	Amount (for entire project in USD)	Amount (for entire project in BZD)	GCF funding amount in USD	Currency of disbursement to recipient
1. Climate Resilient Value Chains Development (CRVC)	1.1 Infrastructure and Production Plans (IPP)	1,196,557	2,393,115	250,286	USD
	1.2 Strengthening of Producers' Organizations	591,548	1,183,095	103,435	USD
	1.3 Value Chains Development	6,072,266	12,144,532	3,184,264	USD
2. Climate Resilient Rural Infrastructure and Assets Development (CRRIA)	2.1 Investment in Rural Roads Improvements (RRI)	4,526,207	9,052,413	1,357,862	USD
	2.2 Investment in Small-scale Irrigation and Drainage (SSID)	3,700,449	7,400,899	2,547,609	USD
	2.3 Climate Information System	256,300	512,600	224,263	USD
Component 3	Cross Cutting Activities	295,138	590,275	129,899	USD
	Project Management Unit	3,364,433	6,731,649	202,382	USD
Total project financing		20,002,898	40,008,578	8,000,000	8,000,000

The integration of the incremental costs and benefits in the total cost of the Project for all the models provides a net cash flow from which the following efficiency indicators were obtained: an Internal Rate of Return (IRR) of 15.74% and a Net Present Value (NPV) discounted at 10% of BZ\$9.4 million, (US\$4.7 million). An integrated financial model is summarized in section F.1 of this proposal and detailed in Appendix 10: Economic and Financial Analysis, including the projection during the entire project, assumptions and rationale; as well as a sensitivity analysis of critical elements of the project.

The expenditure categories presented in the table below are those commonly used by IFAD in rural development projects. Appendix 9: Programme Cost and Financing details costs per component and activity over the six-year period and their funding sources.

	Y1	Y2	Y3	Y4	Y5	Y6	TOTAL
I. Investment Costs							
A. Vehicles, equipment and materials	330,470	-	90,000	-	-	-	420,470
B. Grants	-	530,000	1,085,000	2,120,000	1,590,000	-	5,325,000
C. Consultancies, training and TA	857,550	783,000	1,125,000	633,200	380,300	253,950	4,033,000
D. Works	-	1,400,000	2,100,000	2,800,000	700,000	-	7,000,000
Total Investment Costs	1,188,020	2,713,000	4,400,000	5,553,200	2,670,300	253,950	16,778,470
II. Recurrent Costs							
A. Salaries and allowance	283,200	333,600	333,600	333,600	333,600	317,100	1,934,700
B. Operating Costs	121,220	119,160	119,160	119,160	119,160	119,160	717,020
Total Recurrent Costs	404,420	452,760	452,760	452,760	452,760	436,260	2,651,720
Total BASELINE COSTS	1,592,440	3,165,760	4,852,760	6,005,960	3,123,060	690,210	19,430,190
Local Inflation	7,952	47,645	122,292	212,926	143,058	38,835	572,708
Total PROJECT COSTS	1,600,392	3,213,405	4,975,052	6,218,886	3,266,118	729,045	20,002,898

Parallel financing from the Caribbean Development Bank is also being sought for the project and may come into place during the project implementation period, subject to the approval of the GCF and IFAD funds. Such financing would be under concessional terms for financing allocated to climate change at CDB. The financing would focus on scaling up infrastructure investment in Component 2 of the project based on the experience of the IFAD/GCF investment.

B.2. Project Financing Information							
	Financial Instrument	Amount	Currency	Tenor	Pricing		
(a) Total project financing	(a) = (b) + (c)	20	million USD (\$)				
(b) GCF financing to recipient	(i) Senior Loans	1.9148	million USD (\$)	(40) years	(0.00) %		
	(ii) Subordinated Loans		Options	() years	() %		
	(iii) Equity		Options		() % IRR		
	(iv) Guarantees		Options				
	(v) Reimbursable grants *		Options				
	(vi) Grants *	6.0852	million USD (\$)				
<i>* Please provide economic and financial justification in section F.1 for the concessionality that GCF is expected to provide, particularly in the case of grants. Please specify difference in tenor and price between GCF financing and that of accredited entities. Please note that the level of concessionality should correspond to the level of the project/programme's expected performance against the investment criteria indicated in section E.</i>							
	Total requested (i+ii+iii+iv+v+vi)	8	million USD (\$)				
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority
	Senior Loans	8	million USD (\$)	IFAD	(18) years	(3.86)*1 %	Options
	Grant	3.2	million USD (\$)	GOB	() years	() %	Options
	Equity	0.8	million USD (\$)	Project Participants...		() % IRR	Options
	Options		Options				Options
	Lead financing institution: IFAD						
<i>* Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution.</i>							
(d) Financial terms between GCF and AE (if applicable)	Both are sovereign backed loans, it is the decision of the Government of Belize to accept each terms and conditions. IFAD will provide a loan on ordinary terms based on the performance based allocation system (PBAS) defined by IFAD executive board, and the GCF will provide a concessional loan at highly vulnerable rate based on GCF board decision B.09/04 for highly vulnerable SIDS. The funding award mechanisms to be utilized for deployment of funds are spelled out in attached appendix 11.						
B.3. Financial Markets Overview (if applicable)							
Not applicable							

¹ Pricing applicable for 1st Semester, 2019

Please fill out applicable sub-sections and provide additional information if necessary, as these requirements may vary depending on the nature of the project / programme.

C.1. Strategic Context

There are three important factors that shape Belize's development and are fundamental to contextualize this Project's proposal. First is the overarching condition of economic and demographic 'smallness' and a high level of public debt that results in reduced public sector capital expenditure allocations and diverts investment from the critical area of poverty reduction. Second, the agricultural sector contributes meaningfully to employment and foreign exchange earnings; paradoxically, the country imports 50% of its food and many smallholder farmers are near or below the poverty line. Finally, the aggravation of climate change, especially in the shape of droughts and floods, is generating annual average losses 2.87% of GDP, pressing the need to implement adaptation measures for the most vulnerable population².

Economy and Demography

Belize is a small developing "upper-middle income" country, classified by the United Nations Department of Economic and Social Affairs as a "small island developing state." It has a total land area of 8,867 square miles and an estimated population of about 385,766 inhabitants, of whom 52% are rural dwellers. The country is relatively sparsely populated, the total population density being about 16 people per km².

Nominal GDP is estimated in US\$1.7 billion in 2016 (ranked 172 out of 194 in The World Bank Development Indicators). Belize's per capita income is around US\$ 4,360³. Since 2005, per capita GDP has remained at a standstill. Overall growth of the Belizean economy has been cyclical but has trended downward, with significant declines in five of the last nine years. Since the economic and financial crisis of 2008, the economy is characterized by low but wildly fluctuating growth rates, high inflation rates, high levels of unemployment and underemployment, low investment, persistently negative balance of payments, high public debts (around 100% GDP) and high levels of poverty, with more than 40% of the population below the poverty line.

Due to inward migration of unskilled labor and outward migration of the better educated, poverty levels are high in rural areas, including some households of agricultural workers, particularly migrant workers without access to land. Rural poverty rates are highest (more than 70%) in Toledo and Corozal districts. Unemployment remains high, ranging between 10% and 16% in the last several years, affecting particularly the young (under 25 years) and women. When combined with underemployment, the employment situation is considered to be grim. Poverty is most likely to have increased in the last few years due to population growth, the decline in real per capita income and the physical losses and damages done to the productive sector and the overall economy by extreme weather conditions.

As an open economy, Belize is highly vulnerable to external shocks, and is also significantly affected by climate-related and natural disasters. Over the last 15 years, the country has been severely affected by Hurricane Keith in 2002, which inflicted economic losses totaling approximately 30% of GDP; Hurricane Dan, which struck in 2007 and caused losses equivalent to 90 million dollars (7% of GDP), 64% of which corresponded to the agricultural sector; and Hurricane Earl in 2016 which resulted in a GDP decline of almost 1.0% and the primary sector contracting by almost 25%. In between, the country has also experienced various storms and droughts that have caused many losses to various sectors, particularly agriculture and the livelihoods of rural people. Nearly 25% of the population has been impacted and there were significant declines in agricultural production from the last three hurricanes. Of more importance, however, are the adverse effects of these events on distinct groups of the population; (i) persons living in rural areas, many of whom were agriculturalists and subsistence farmers, and have little or no resources to easily recover; (ii) poor marginalized and indigent persons who could be found in both rural and urban areas; and (iii) single women who are head of households with few alternative sources of

² Germanwatch 2017 available at: <https://germanwatch.org/de/download/16411.pdf>

³ <https://data.worldbank.org/indicator/NY.GNP.PCAP.CD?locations=BZ>

income. These groups are ill-equipped and have insufficient resources to hold them over when there is extensive flooding or droughts over a long period, as well resources to reinvest and recuperate from their physical losses.

Agriculture Sector

Agriculture is the main source of rural livelihoods in Belize and is an important overall contributor to the national economy. The sector is key to ensuring food security for Belize, in addition to its wider role for rural socio-economic stability. It is divided into three main sub-sectors: 1) a well-organized traditional one that comprises the export commodities of sugar, banana, citrus and marine products; 2) a well-integrated large- medium-scale commercial sector; and 3) small-scale agriculture that have mainly mixed farming systems that produce food for local consumption. Belize has an excellent climate for agriculture and abundant water resources. The natural endowment of internally renewable water resources is about ten times that of Central America and the Caribbean as a whole. Land is abundant, with approximately 800,000 ha or 38% of the total land area is considered suitable for farming and raising livestock, and 15% currently used for agriculture.

The sector's contribution to employment and its participation in foreign trade are important for the economy. According to the latest Labour Force Survey in April 2017, the total number of employed persons was estimated to be 150,112, with agriculture accounting for 17% of the employment. Males accounted for 90 percent of the country's agricultural workers. Although the official data indicates that women's employment in the sector is low, it is widely recognized that female employment is under-reported. Agriculture and related activities contribute more than 60% of the country's foreign exchange earnings, and it has also played a significant role in foreign exchange savings, with the value of domestic production directed to the local market was estimated at nearly BZ\$650 million in 2015. Traditional agricultural exports contribute more than 70% of total domestic exports. Although Belize is still dependent on traditional export commodities such as sugar, citrus, bananas and marine products, it has become self-sufficient in a number of food products over the last two decades, namely poultry meat, eggs, beef, pork, beans, rice and corn, and is now exporting some of these products to countries in the CARICOM and Central America. Despite this positive performance in food production, Belize's import food bill has almost doubled in the last 10 years.

Notwithstanding the important role of agriculture, the sector faces a number of challenges, including: (i) high level of poverty, estimated at more than 40%; (ii) moderate and in some areas a high level of food insecurity and malnutrition; (iii) underdeveloped agriculture and food value chain systems; (iv) pests and disease problems that sometimes accompany extreme weather conditions; (v) inadequate rural area and territorial development; and (v) high and increasing exposure to climate risks and limited resilience capacity.

The new agricultural policy framework (National Food and Agricultural Policy 2015 -2030) and the National Adaptation Strategy to Address Climate Change in the Agriculture Sector (launched in 2015) emphasize the importance of the sector to Belize's economy and highlight the need to implement adaptation measures and increase product value to ensure maintenance of food security. Both policy documents also stress the vulnerability of small scale/family farmers and their crucial role in rural development.

The National Poverty Eradication Strategy Action Plan (NPESAP) for 2006-2010 outlines the Government's overall policy framework for rural development, and it focuses on poverty reduction as the centerpiece of the development strategy. This strategy still guides the GOB's efforts at rural poverty reduction through: (i) the development of small, medium-sized and micro enterprises (SMEs); (ii) the strengthening of capacities in the rural area; (iii) the promotion of small-scale agricultural and rural development projects; and (iv) the implementation of rural community based projects to improve physical and social infrastructure in the villages.

Climate Change Framework

Belize became a Party to the UNFCCC in 1992, having the status of a non-Annex 1 country. Belize also joined the Kyoto Protocol in 2003. As a party state to the United Nations Framework Convention on Climate Change (UNFCCC), Belize submitted its Initial National Communication (INC) to the UNFCCC in 2002, its Second National

Communication in the second quarter of 2012 and its Third National Communication in April 2016. On the 22nd April 2016, Belize ratified the Paris Agreement on Climate Change and submitted its Nationally Determined Contribution (NDC). Belize's NDC is guided by its commitment to strategically transition to low carbon development while strengthening its resilience to the effects of climate change.

In 2015, the Government of Belize adopted its first comprehensive National Climate Change Policy, Strategy and Action Plan (NCCPSAP) and has sought, through several line ministries, to initiate policy-based activities, at the sector level, to address (adapt and mitigate) the impending impacts of Climate Change. Belize's climate change policy guides short, medium and long-term processes of adaptation and mitigation of Climate Change in accordance with national prospects for sustainable development in addition to regional and international commitments.

The country's history has been marked by numerous devastating tropical depressions, storms, and hurricanes. The former capital, Belize City, was twice destroyed by hurricanes in the 20th century, spurring the capital to relocate to Belmopan City. Climate change has significantly increased the threat level and observed climatic trends in Belize are considered alarming. Data from the Philip Goldson International Airport shows that the temperature along the coast of Belize has risen by 1.6°F while in the interior of the country the temperature has risen by 1.8°F, suggesting that the temperature in Belize is rising faster than the global average of 1.4°F. A detailed analysis on the climate change impact in Belize can be found in section C.2.

Building on previous experiences

The Be-Resilient program will build on a wealth of IFAD experience in the region and particularly two projects developed in Belize; Belize Rural Finance Programme (BRFP) developed by IFAD, and a Value Chain Project developed by FAO. The BRFP was a seven-year programme implemented to provide access to financial services (savings and credit) to poor and low-income rural families through credit unions. Through its shared incentive facility, the BRFP was able to recruit over 9,000 new members (men, women and youth) for credit unions that participated in the programme. The Resilient Rural Belize (RRB) Programme's target population is expected to include some of these same members recruited by the BRFP. Not only will this previous experience allow to build up on targeting mechanisms and identified producer organizations, but will support the MGF of the RRB Programme which requires Producers' Organizations to make counterpart contributions of up to fifteen percent of the value of the total investment by the programme. It is expected that the members of the POs, many of whom were recruited by the BRFP, will be able to access credit from their credit unions to finance the counterpart contributions.

The FAO Value Chain Project aimed to reduce poverty and improve the living standards of the rural population in the northern region of Belize (Corozal and Orange Walk Districts) by facilitating commercial agriculture-based enterprises to improve income, reduce unemployment and strengthening the value chains and productivity of onion, honey and sheep. The RRB Programme will build on the FAO Value Chain Project by further strengthening two of the three value chains (onion and honey) in addition to six other select value chains, through the adoption of climate resilient agricultural technologies to improve quality, productivity and to build climate resiliency. At the same time the project will develop the necessary infrastructure to support the movement of the produce and establish market linkages between producers and developed markets to absorb the higher quality and increased production.

In sum, without an effective support to rural development, smallholders will still be associated with low incomes because of limited access to inputs and credit, vulnerability to climate events, lack of a good distribution network, limited access to support facilities and opportunities to add value to primary production. Particularly important is support for appropriate climate resilient innovations, organized market intelligence systems and accommodation in the established supply chains. Most noteworthy is the apparent inability of small producers to establish among themselves reliable measures of standards and quality in production which can open market access to their outputs beyond their immediate environment. The key is to promote a comprehensive approach

which addresses multiple drivers and creates an enabling environment for climate resilient agriculture that is socially acceptable, economically viable and environmentally sustainable. This requires the identification and implementation of locally specific climate resilient solutions supported by technical assistance and capacity building at various levels. The project aims to demonstrate that resilient agricultural production reduces exposure to climate and economic shocks and can establish smallholder farmers as reliable and competitive suppliers of domestic produce in Belize. The project will demonstrate an approach which can be scaled nation-wide and across countries experiencing similar climate related stress, particularly in the Caribbean.

C.2. Project / Programme Objective against Baseline

Overview

As a Small Island Developing State in the Caribbean hurricane belt, Belize is highly vulnerable to the impacts of climate change. While the most extreme and visible effect is an increased intensity of tropical storms and hurricanes, climate change also manifests itself through documented trends of increased droughts, flooding, and significant rainfall pattern variations. Where, according to the International Center for Tropical Agriculture: "The most detrimental effects on agriculture are likely to come from increased variability in the seasonal distribution of rainfall, which is expected to lead to more frequent droughts and floods. Additionally, projected rises in temperature of 1.3 °C by the 2030s will increase stress on crops and livestock, impacting agricultural systems, forcing changes in management practices, and threatening food production." (CIAT; World Bank. 2018)

Smallholder farmers are amongst the most vulnerable as they suffer devastating losses, most dramatically in the occurrence of an extreme event but similarly through the impacts of persistent and unpredictable seasonal variations. Due to the high cost of such events, smallholders' limited investment capacity is fully focused on recovery and purchase of inputs, thus preventing them from investing in assets and measures that would limit their exposure. As a result, the impacts of climatic events are more profound, and the recovery time is prolonged. This vulnerability is exacerbated by the poor market conditions for smallholder farming in Belize. With their current assets, capacities, and connections to markets, smallholders are not equipped to deliver a regular supply of adequate volumes of quality produce required for sustainable market access. As such, markets have turned towards imported goods to supply a large proportion of foods, including vegetables and fruits, that have traditionally been supplied by smallholders and can be competitively produced in Belize.

The project aims to introduce climate resilient agricultural practices that will allow smallholder farmers to have a sustainable production process and improved market access for their produce, even under the stress of climate change and extreme climatic events; therefore increasing their economic, social, and environmental resilience. In the context of smallholder farmers in Belize, improved resilience is defined as the capacity to minimize the impacts of climatic and economic shocks, enabling farmers to better confront periodic variations and strengthening their capacity to cope and recover in times of extreme stress. To achieve this in a manner that meaningfully improves smallholders adaptive capacity, significant improvements across the entire value chain are required, as weaknesses in production technologies and practices, fragile and variable physical access to markets, and limited capacity to respond to market demands are all factors that contribute to farmers' vulnerability. The project proposes a comprehensive, focalized, and climate resilient approach to reduce the exposure to climate and economic shocks by promoting climate resilient agricultural production, investing in climate proof infrastructure, supporting producer associations, and strengthening value chains to establish smallholder farmers as reliable and competitive suppliers of domestic produce in Belize for an expanding market for fruit and vegetables.

Background and Climate Change Information

Belize is classified by the United Nations Department of Economic and Social Affairs as a "small island developing state". According to the Poverty Assessment Report of 2010, rural poverty reached 55% of the total population in

2009, compared to 28% in urban areas. Agriculture is critical to Belize's development; the sector represents 13% of GDP, provides more than two-thirds of export receipts, and officially employs 17% of the population.

Belize is considered to be among the most vulnerable countries in the world in terms of climate. Two detailed measurements of individual countries' vulnerability to climate threats have been made, where: (a) over the twenty-year period 1996-2015, Belize was ranked 26th out of 181 countries in the Global Climate Risk Index (CRI) prepared by the NGO Germanwatch in Bonn⁴. The CRI estimates the average annual losses due to climate-related events to be approximately US\$57 million, representing 2.87% of GDP; and (b) the University of Notre Dame's Global Adaptation Index (GAI) rates Belize's adaptive capacity at 121 out of 181 countries⁵, clearly stating: Belize "has both a great need for investment and innovations to improve readiness and a great urgency for action. Belize is the 56th most vulnerable country and the 62nd least ready country" with the two worst scores being in projected change of cereal yields, and agricultural capacity. This denotes important vulnerabilities in the food sector, affecting food security, nutrition and livelihoods.

In the last few decades, the average annual temperature in Belize has shown a tendency to increase. Furthermore, Belize's geographic location leaves it exposed to the risk of rising sea levels and more frequent and more intense tropical storms, both of which lead to flooding that frequently results in human and material losses, including substantial losses within the agricultural sector. Between 2000 and 2016, the agricultural sector alone accounted for losses in the magnitude of US\$232 million due to hurricanes and tropical storms. (CIAT; World Bank. 2018)

According to Belize's Third National Communication to the UNFCCC (2016), an increase in air temperature ranging from 2°C - 4°C is projected by 2100 for Belize. Similarly, a general decrease in annual rainfall of about 10% is projected by 2100. In addition to an increased frequency and intensity of tropical storms and hurricanes that is forecast for the entire Caribbean region, analyses of recent trends and models conducted by the Caribbean Community Climate Change Center ⁶ and the International Center for Tropical Agriculture⁷ reveal important climatic projections:

Temperature: A broad nation-wide increase of approximately 1°C is indicated in the decade of the 2020s. Increases of 1-1.8°C in the 2030s, 1.8 – 2.9°C in the 2050s, 2.5 - 4.3°C in the 2070s and 3.2 – 4.9°C in the 2090s are projected over the 1961-1990 values. The rate of warming is projected to be consistently lowest along the coastal section of the Belize District.

Rainfall: Seasonal rainfall variability will increase. An overall trend of declining rainfall is compounded with increases in intense precipitation events. A slight increase in rainfall in the 2020s is suggested in the early (May) and late (Oct-Nov) parts of the wet season with peak increases of about 2-4 mm/day. The dry season and the mid-wet season dip (June), on the other hand, are characterized by further decreases with largest reductions of about 4 mm/day. By the 2030s, reduced precipitation characterizes the entire country with exceptions only in the early (May) and late (November) sections of the wet season. In the 2050s, an enhancement of the 2030s pattern of reduced rainfall is projected to continue (-1 to -4 mm/day) in the dry season (December – April). The implications of these projections for smallholder land users will be greater likelihood of variability, of mini droughts or floods within the growing season, of greater erosion and of conditions of potential rotting in the field and/or in storage and of pests and diseases.

Projected Temperature and rainfall variations for Belize by the year 2050 are presented in the following figure, showing an increase in temperature of up to 1.8 °C, and a decrease in rainfall of up to 10%.

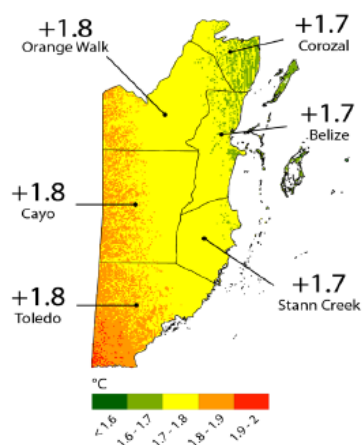
⁴ <https://germanwatch.org/de/download/16411.pdf>

⁵ <https://gain.nd.edu/our-work/country-index/>

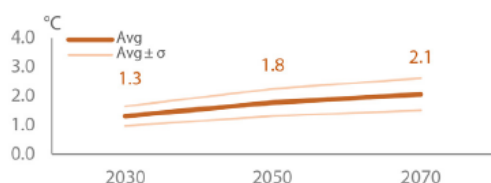
⁶ A National Climate Change Policy, Strategy And Action Plan To Address Climate Change In Belize, (Caribbean Community Climate Change Center, 2014)

⁷ CIAT; World Bank. 2018. Climate-Smart Agriculture in Belize. CSA Country Profiles for Latin America and the Caribbean Series. International Center for Tropical Agriculture (CIAT); World Bank, Washington, D.C. 24 p.

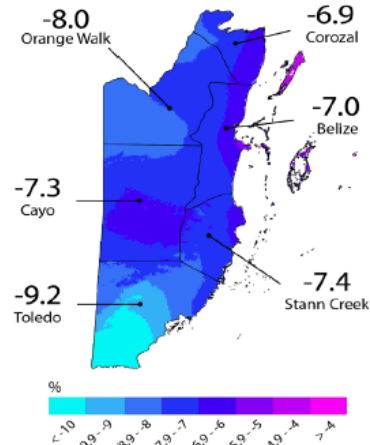
Changes in annual mean temperature (°C)



Average temperature (°C)



Changes in total precipitation (%)



Average precipitation (%)

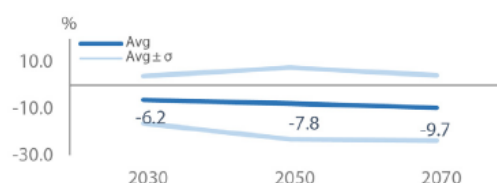


Figure 1 Projected Changes in temperature and precipitation in Belize by 2050 Source: (CIAT: World Bank. 2018)

Sea Level Rise: Sea level is projected to rise steadily along the coast of Belize. In the low, medium and high emission scenarios, sea level rise is projected to exceed 10 cm by the 2030s. Heights of 22, 23 and 38 cm respectively are projected for the low, medium and high emission scenarios by 2050 and 34, 56 and 120 cm respectively by the end of the Century.

Thus, Belize is expecting an increase in temperatures, higher rainfall variability and intensity and a sea level rise; all factors that will impact agricultural production in the country. These climate stressors combined with the vulnerability sources described in section C.1 will yield in significant climate change impacts to the beneficiary population, as can be seen below in Table 1.

Table 1 Climate Stressors, Hazards, Effects and Impacts

Climate Stressors	Climate Hazards	Effects on Smallholder Farmers	Impacts on Smallholder Farmers
Increase in hurricane frequency and intensity ^{8, 9}	- Storm surge - Heavy rainfall and inland flooding - High winds - Greater erosion	- Potential for large scale physical infrastructure damage - Produce loss/yield reduction - Reduction or loss of accessibility to markets, basic services	Economic losses affecting livelihoods
Seasonal rainfall variability will increase ¹⁰	- Increase in drought frequency and/or duration - Increase in flood frequency and/or duration - Greater erosion	- Produce loss/yield reduction - Physical infrastructure damage - Increase in incidence of pests and diseases - Reduction or loss of accessibility to markets, basic services	Reduced food security Health risks

Temperature increase ¹¹	- Loss of water/lower humidity - Increase severity of droughts	- Produce loss/yield reduction - Increase in incidence of pests and diseases	
Sea level rise ¹²	- Increase in flood frequency - Greater erosion	- Produce loss/yield reduction - Physical infrastructure damage	

Smallholder Agriculture and Climate Change in Belize

The agricultural sector in Belize is already severely affected by Climate Change; as compared to the 1960's there are nearly 70 more hot days or nights per year caused by an average increase of 0.10°C per decade. Annual rainfall on the other hand has decreased at an average rate of 3.1 mm per decade during the same period. Furthermore, the IPCC projects that "severe drought likelihood will increase 0-67% by 2050 and 0-95% by 2100 compared to the historical baseline; the number of hot days will increase 19-87 days by 2050 and 83-196 days by 2100 compared to the historical baseline; and the daily probability of heat wave will increase 13-41% by 2050 and 35-75% by 2100 compared to the historical baseline." (CIAT; World Bank. 2018) Climate change scenarios project impacts on area and yield for the vegetable sub sector in Belize of a decrease of up to 2% and 9.8% respectively, likely requiring an increase by 2% of vegetable imports.

According to FAO, the agricultural sector absorbs about 25% of total damage and losses caused by climate related disasters in developing countries, with additional social and humanitarian fall outs concentrated on the poorest populations whose main livelihood is agriculture, while at the same time putting food security and adequate nutrition at risk. Moreover, of all natural hazards, the agricultural sector is most severely impacted by floods, droughts and tropical storms; climate-induced disasters.¹³ (FAO, 2015) This relation is even more noticeable in the Caribbean where, for example, Hurricanes Irma and Maria accounted losses of 100% in crops, and significant damages to productive trees and death of livestock in Dominica. (ACAPS, OCHA and UNDP, 2017). The intensity and number of climate induced disasters has increased significantly over the last decade, and the Caribbean region has been harshly affected by them, hampering eradication of hunger and achievement of sustainable development. (FAO, 2015)

Historical data for Belize shows that the agricultural sector, particularly in Corozal, Orange Walk, Belize and Cayo, have been the most affected due to extreme weather events, in many cases significantly above global average estimates. During tropical depression 16 in 2008, economic impact was estimated at \$27.1 million USD, of which agriculture represented \$14.17M, assumed mainly by the crop sub sector which represented 95.7% of total agricultural loss. Just one year before, Belize suffered losses and damages amounting to \$89.5 million USD, during Hurricane Deane, of which agriculture represented \$57.9M being the most affected sector, accounting for 64.7% of total estimated economic impact. Even more so, the social and humanitarian fall out far outweighed the economic impact, since agriculture was the most crucial source of livelihoods for the poor populations that were most affected. These represented a macroeconomic impact of 25% and 63.8% of agricultural GDP respectively.¹⁴ The most affected crops in both events, as can be seen in the following table, were fruit trees and standing crops (Papaya, Sugarcane, Corn) while vegetables represented a very small percentage of losses; smallholder farmers in

⁸ National Hurricane Center, The major hazards associated with hurricanes. Source:

<https://www.nhc.noaa.gov/prepare/hazards.php>

⁹ Klotzbach, Philip J. "The influence of El Niño–Southern Oscillation and the Atlantic multidecadal oscillation on Caribbean tropical cyclone activity." *Journal of Climate* 24.3 (2011): 721-731.

¹⁰ A National Climate Change Policy, Strategy And Action Plan To Address Climate Change In Belize, (Caribbean Community Climate Change Center, 2014)

¹¹ A National Climate Change Policy, Strategy And Action Plan To Address Climate Change In Belize, (Caribbean Community Climate Change Center, 2014)

¹² A National Climate Change Policy, Strategy And Action Plan To Address Climate Change In Belize, (Caribbean Community Climate Change Center, 2014)

¹³ The impact of disasters on agriculture and food security. FAO, 2015

¹⁴ Macro socio-economic assessment of the damage and losses caused by tropical depression 16. ECLAC, 2009, and Macro socio-economic assessment of the damage and losses caused by hurricane Dean. ECLAC, 2007

the vegetable sub-sector however, were severely affected due to lack of capacity to recuperate and the use of inadequate production mechanisms to cope. The lack of crop diversification and the damages to road, and irrigation and drainage systems allowed for exponential losses to occur for the following periods.

Table 2 Summary of macro socio-economic damages and losses caused by Hurricane Deane and TD16 Belize

Damages and Losses caused by Hurricane Deane		Damages and Losses caused by TD 16	
Sector and subsector	Total impact (US\$ thousands)	Sector and subsector	Total impact (US\$ thousands)
Total	89 512 722.65	Total	27 052.07
Productive Sectors	62 603 422.65	Productive Sectors	16 110.75
Agriculture	57 917 287.00	Agriculture	14 091.85
Papaya	33 669 450.00	Papaya	1 232.10
Sugarcane	11 409 820.00	Sugarcane	1 653.73
Corn	2 189 380.00	Corn	4 593.60
Plantains	1 842 182.50	Plantain	20.25
Other crops	3 372 860.50	Rice	2 292.73
		Vegetables	1 951.00
		Banana	1 741.25
Livestock	607 916.00	Livestock	607.20
Fisheries	4 825 678.00		
Tourism	4 686 135.65	Tourism	1 501.20
		Petroleum	400.20
		Commerce	117.50
Social Sectors	17 166 650.00	Social Sectors	4 289.30
Housing	16 615 050.00	Housing	3 762.70
Education and Culture	399 950.00	Education and Culture	30.90
Health	151 650.00	Health	495.70
Infrastructure	5 842 650.00	Infrastructure	6 652.02
Road Transport	4 937 650.00	Road and Transport	6 281.85
Water supply and disposal	112 500.00	Water sanitation	219.82
Electricity Generation	480 000.00		
Communications	312 500.00	Bridges	150.35
Environment	3 900 000.00		

Furthermore, according to studies developed by CIAT, in Belize: "At any given time, up to US\$ 1.25 million of agricultural production is exposed to risk of floods, and up to US\$ 520,000 is exposed to risk of drought."¹⁵ The resulting effects of said loss on income, food security, family nutrition and health can be catastrophic, particularly for smallholder farmers whose adaptive capacity is particularly low; due to their difficult access to technical assistance and resources. Persistent losses due to climatic events has caused them to re-invest their limited resources in recovering their productive capacity, preventing them from adopting more resilient productive and marketing practices which would reduce their exposure.

¹⁵ CIAT; World Bank. 2018. Climate-Smart Agriculture in Belize: Identifying Investment Priorities. International Center for Tropical Agriculture (CIAT); World Bank, Washington, D.C. 16 p.

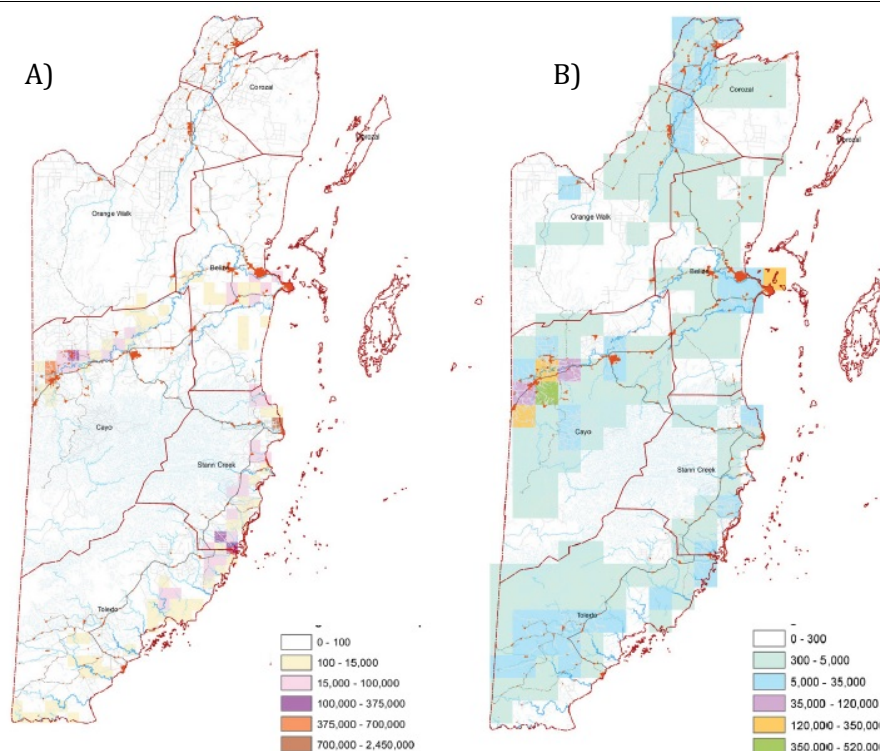


Figure 2 A) economic exposure to floods (US\$); (B) economic exposure to droughts (US\$).
Source: (CIAT; World Bank. 2018)

According to the National Agriculture Sector Adaptation Strategy to Address Climate Change in Belize (2015), climate change will “impact agriculture systems and practices such as soils and land preparation, pest and disease control, and water requirements (excess and deficits), while high temperatures will cause increased stress on current commodities such as livestock breeds, crop types and varieties.” A series of macro studies have been developed in Belize by national and international organizations alike to identify the “best-bet” adaptation and mitigation measures for the agricultural sector. Amongst which CIAT highlights practices such as i) the use of cover structures, ii) installation of drip irrigation systems, ii) application of nutrients through irrigation water (fertigation), iv) use of water-harvesting techniques, v) adjustment of planting dates to match rainfall patterns, vi) crop rotation, intercropping, planting of agroforestry systems, and vii) adoption of drought- and heat tolerant varieties. (CIAT; World Bank, 2018) For the vegetable sub-sector, the three climate resilient agricultural practices considered to be the most promising are the use of cover structures, drip irrigation and use of disease-tolerant varieties.

The following table summarizes the main findings of climate related exposures, specific vulnerabilities and adaptive capacity for the five priority areas. These will serve to guide development of CVAs and community specific action plans:

Table 3 Main identified exposures to climate change, vulnerabilities and adaptive capacity of 5 priority districts in Belize

TOLEDO		
Climate Change Exposure	Vulnerability	Adaptive Capacity (BaU)
- Region with highest risk of increased drought frequency and severity. - Reported Max. Temp. increase of 0.792° C between 1989-2017. - Projected increase in temperature between 1.8 to 2° C by 2050.	- Land productivity Dynamics: 68,304.28 (ha) with reported declining productivity, an additional 11,963.13 (ha) present early signs of decline; 29,706.78 (ha) are currently stable but stressed. - Accessibility to city in optimum conditions requires 6 – 16 hours. - 40% of population below poverty line.	- Selected VC (Pineapple) sensitive to rainfall variability but highly adaptable with irrigation. - High willingness to adopt different crop varieties that better adapt to droughts. - Medium organizational capacity.

- Reported Avg. annual precipitation reduction of 169.887 mm between 1980 - 2017. - Projected changes in rainfall between 7-10% by 2050. - Increase in average water deficit by 21.247 mm since 2001. - Significant rainfall pattern variations with 6-9 heavy rainfall days >50 mm/day in the rainy season (1981-2017). - Project areas are in low lying lands to moderate slope terrains. - 85.03% of area are broadleaved evergreen and/or semi-deciduous forest (>5m)	- Inadequately prepared for increased frequency and severity of droughts and floods. - Inadequate CIS for decision making and planning. - Underdeveloped agriculture and food value chain systems - Inadequate rural area and territorial development.	- Low investment capacity, access to resources for recuperation and no climate insurance. - Low accessibility to basic services in periods of extreme climatic and climate induced events. - Low technical training in climate resilience agricultural practices.
Stan Creek		
Climate Change Exposure	Vulnerability	Adaptive Capacity (BaU)
- Region at risk of increased drought frequency and severity, and hurricanes. - Reported Max. Temp. increase of 0.635° C between 1989-2017. - Projected increase in temperature between 1.6 to 1.8° C by 2050. - Reported Avg. annual precipitation reduction of 201.787 mm between (1980 - 2017). - Projected changes in rainfall between 6-8% by 2050. - Increase in average water deficit by 7.806 mm since 2001. - Significant rainfall pattern variations with 3-4 heavy rainfall days >50 mm/day in the rainy season (1981-2017). - Project areas are in low lying terrains to moderate slope terrains. - 79.67% of area are broadleaved evergreen and/or semi-deciduous forest (>5m)	- Land productivity Dynamics: 59,566.44 (ha) with reported declining productivity, an additional 7,202.42 (ha) present early signs of decline; 35,875.77 (ha) are currently stable but stressed. - Accessibility to city in optimum conditions requires 4 - 20 hours. - 20% of population below poverty line. - Inadequately prepared for increased frequency and severity of droughts and floods. - Inadequately prepared for increased frequency and severity hurricanes. - Inadequate CIS for decision making and planning. - Underdeveloped agriculture and food value chain systems - Inadequate rural area and territorial development.	- Selected VC (Hot pepper) Highly sensitive to salinity of soil and waters, and water stress. Better adapted to hot climates. - High willingness to adopt different crop varieties. - Medium organizational capacity. - Low investment capacity, access to resources for recuperation and no climate insurance. - Low accessibility to basic services in periods of extreme climatic and climate induced events. - Low technical training in climate resilience agricultural practices.
Cayo		
Climate Change Exposure	Vulnerability	Adaptive Capacity (BaU)
- Region with most area reporting declining productivity, at risk of increased flood/drought frequency and severity. - Reported Max. Temp. increase of 1.267° C between 1989-2017. - Projected increase in temperature between 1.8 to 2° C by 2050. - Reported Avg. annual precipitation reduction of 82.004 mm between (1980 - 2017). - Projected changes in rainfall between 6-7.9% by 2050. - Increase in average water deficit by 12.49 mm since 2001. - Significant rainfall pattern variations with 3-4 heavy rainfall days >50 mm/day in the rainy season (1981-2017). - Project areas are in low lying inland terrains vulnerable to floods.	- Land productivity Dynamics: 194.046.44 (ha) with reported declining productivity, an additional 5.085.50 (ha) present early signs of decline; 75,946.55 (ha) are currently stable but stressed. - Accessibility to city in optimum conditions requires 2-16 hours. - 34-65% of population below poverty line. - Inadequately prepared for increased frequency and severity of droughts and floods. - Inadequately prepared for increased frequency and severity hurricanes. - Inadequate CIS for decision making and planning. - Underdeveloped agriculture and food value chain systems - Inadequate rural area and territorial development.	- Selected VC (Sweet pepper) Highly sensitive to salinity of soil and waters, pests, diseases and water stress. - Selected VC (Cabbage) is irrigation dependant, highly sensitive to water limitation. Sensitive to flooding. But can benefit from warmer temp. - Selected VC (Tomatoes) is moderately sensitive to reduced irrigation and salinity; can benefit from warmer temp within tolerance range. Sensitive to flooding in high temperatures. - High willingness to adopt different more resistant crop varieties. - Medium organizational capacity. - Medium to Low accessibility to basic services in periods of extreme climatic and climate induced events. - Low investment capacity, access to resources for recuperation and no climate insurance.

- 84.76% of area are broadleaved evergreen and/or semi-deciduous forest (>5m) - Limited/no access to water bodies (max 25.63 ha).		- Low technical training in climate resilience agricultural practices.
Belize		
Climate Change Exposure	Vulnerability	Adaptive Capacity (BaU)
- Region at risk of increased flood frequency and severity, most at risk of increased frequency/intensity of hurricanes. - Reported Max. Temp. increase of 0.845° C between 1989-2017. - Projected increase in temperature between 1.6 to 1.8° C by 2050. - Reported Avg. annual precipitation reduction of 125.119 mm between (1980 – 2017). - Projected changes in rainfall between 5-7.9% by 2050. - Increase in average water deficit by 4.079 mm since 2001. With reported variations of up to 125.908 mm from one year to another. - Significant rainfall pattern variations with 3-4 heavy rainfall days >50 mm/day in the rainy season (1981-2017). - Project areas are in low lying coastal terrains. - 73.78% of area are broadleaved evergreen and/or semi-deciduous forest (>5m)	- Land productivity Dynamics: 101,968.86 (ha) with reported declining productivity, an additional 7,960.68(ha) present early signs of decline; 55,756.31 (ha) are currently stable but stressed. - Accessibility to city in optimum conditions requires 0 – 4 hours. - 20-25% of population below poverty line. - Inadequately prepared for increased frequency and severity of droughts and floods. - Inadequately prepared for increased frequency and severity hurricanes. - Inadequate CIS for decision making and planning. - Underdeveloped agriculture and food value chain systems - Inadequate rural area and territorial development.	- Selected VC (Onion) irrigation dependent, sensitive to saline soils, warmer temp. and excessive rainfall. Adaptable with shading/protective screens, controlled irrigation and adequate drainage systems. - Selected VC (Tomatoes) is moderately sensitive to reduced irrigation and salinity; can benefit from warmer temp within tolerance range. Sensitive to flooding in high temperatures. - Selected VC (Sweet pepper) Highly sensitive to salinity of soil and waters, pests, diseases and water stress. - High willingness to adopt different crop varieties. - Medium – High organizational capacity. - Medium investment capacity, access to resources for recuperation and no climate insurance. - Medium to High accessibility to basic services in periods of extreme climatic and climate induced events. - Low technical training in climate resilience agricultural practices.
Orange Walk		
Climate Change Exposure	Vulnerability	Adaptive Capacity (BaU)
- Region at risk of increased flood frequency and severity, most at risk of increased frequency/intensity of hurricanes. - Reported Max. Temp. increase of 1.247° C between 1989-2017. - Projected increase in temperature between 1.6 to 1.9° C by 2050. - Reported Avg. annual precipitation increase of 24.231 mm between (1980 – 2017). - Projected changes in rainfall between 7-8.9% by 2050. - Decrease in average water deficit by 7.796 mm since 2001. - Significant rainfall pattern variations with 2-3 heavy rainfall days >50 mm/day in the rainy season (1981-2017). - Project areas are in low lying inland terrains. - 80.70% of area are broadleaved evergreen and/or semi-deciduous forest (>5m)	- Land productivity Dynamics: 175,655.21 (ha) with reported declining productivity, an additional 2,871.84(ha) present early signs of decline; 50,710.83 (ha) are currently stable but stressed. - Accessibility to city in optimum conditions requires 2-4 hours. - 39-70% of population below poverty line. - Inadequately prepared for increased frequency and severity of droughts and floods. - Inadequately prepared for increased frequency and severity hurricanes. - Inadequate CIS for decision making and planning. - Underdeveloped agriculture and food value chain systems - Inadequate rural area and territorial development.	- Selected VC (Onion) irrigation dependent, sensitive to saline soils, warmer temp. and excessive rainfall. Adaptable with shading/protective screens, controlled irrigation and adequate drainage systems. - Selected VC (Carrots) irrigation dependent and salt sensitive, not particularly sensitive to storm/flooding. - High willingness to adopt different crop varieties. - Medium organizational capacity. - Medium-low investment capacity, access to resources for recuperation and no climate insurance. - Medium accessibility to basic services in periods of extreme climatic and climate induced events. - Low technical training in climate resilience agricultural practices.

Project Approach and Benefits

The project will assist Belize in strengthening and increasing climate resilient smallholder farming practices and assets across the agricultural value chain, allowing them to realize the productive potential of their available resources and to help them overcome the challenges imposed by climate change. This will be done through a step-wise approach starting from linked analyses of climate constraints and market opportunities at priority intervention areas and value chains selected on the basis of criteria clearly articulated and agreed with government. The co-investment of the GCF will focus on identifying and addressing climate vulnerability related challenges along the value chains and bring significant national and international technical assistance and investment to this process. The net effect will be a combination of increased income (= poverty reduction + greater capacity to address climate challenges) and greater stability of this income. This latter dimension, often overlooked, will also be monitored in the project's M&E system to test this hypothesis.

The project will build the capacity of existing producer organizations (POs) of smallholders who currently produce minor surpluses for the market to identify opportunities that make sense both from market and climate change resilience points of view. These will be analyzed with a focus on entry points along the (relatively short and simple in the case of Belize) value chains by crop. PO members will be assisted to map existing constraints, identify options to overcome them, and select the best option to access resources from a matching grant fund (MGF). This will be summarized in climate resilient Investment and Production Plans.

The MGF will cover – including a project participant contribution – investments in sustainable and climate resilient production, such as hurricane adapted low cost greenhouses, renewable energy pumps, agroforestry systems and technical assistance for climate resilient practices. These will be complemented with public infrastructure identified through an investment planning process in consultation with a range of stakeholders. This public infrastructure (i.e. physical access to fields during the rainy season, drainage of production areas) benefits the POs directly but also is accessible to indirect project participants. Other public goods supported by the project, such as a climate information system, have an even wider set of indirect project participants.

Project Strategy and Potential for Addressing Business-as-usual Pathway

The Project will follow the strategy laid out in the National Agriculture Sector Adaptation Strategy to Address Climate Change in Belize (2015) to foster resilience in a comprehensive manner, by directly addressing the climate induced threats faced by smallholders as well combating poverty and increasing competitiveness and productivity. To adapt to rainfall deficit/drought, excess/flooding and variability; temperature increase; changes in pests and diseases; and changes in soil fertility, the National Adaptation Strategy calls for:

- irrigation and drainage systems (addressed in this Project under sub-component 2.2);
- weather resistant feed roads, infrastructure to ensure access to farms (addressed in this Project under sub-component 2.1),
- use of renewable energy sources to reduce cost of pumping (addressed in this Project under sub-component 1.3),
- timely and localized weather information (addressed in this Project under sub-component 2.3), and
- implementation of climate resilient agricultural practices such as crop cover, organic fertilizers, raised beds, indigenous crops, polyculture systems, agroforestry, protective cropping structures and Biological Control Agents (addressed in this Project under sub-component 1.3).

Furthermore, the strategy emphasizes the importance of group resilience and inclusive as the most important element for successful implementation (addressed in this Project under sub-component 1.2) and indicates commodity insurance to maintain levels of production (addressed in this Project under cross-cutting activities).

By supporting smallholder farmers in providing markets with reliable, quality agricultural produce, the project will support a resurgence of domestic agricultural production for national consumption, reducing the vulnerability of smallholder farmers and, as a co-benefit, increasing the country's overall food security.

The current development pathway for smallholders in Belize is locked into a maladaptive climate response with low investment – low return logic as a result of a number of mutually reinforcing constraints which they are unable to address without supports. These include climate related (geography, meteorological characteristics), economic and policy related (dominance of large-scale producers and imports), institutional (weak producer organizations with limited involvement of women and youth) and infrastructural/technical/financial (inadequate and climate vulnerable road system, lack of irrigation and drainage). The project strategy hinges on identifying and addressing constraints in each target area and addressing them as a system. A Knowledge Management Plan will be developed to design the process of demonstrating, analyzing, documenting and disseminating the lessons learned in resolving each constraint, as discussed in Section E.2.2.

This project, which, in spite of a modest investment proposed from the GCF, operates country-wide and thus reflects the variety of agro-ecozones, smallholder production systems, value chains and market opportunities, represents an excellent opportunity to demonstrate an approach which could be institutionalized and scaled up in places with similar constraints, both in Belize and across the Caribbean which shares several characteristics.

Capacity Building and Institutionalization

The responsible Ministry, which also represents the NDA and is the GCF focal point, is the Ministry of Economic Development (MED), which is able to place the project objectives and activities in a multi-sectoral planning context. This increases the likelihood of impact and sustainability, including the linkages to development planning processes and the mainstreaming of climate risk management. Under guidance of the MED, specialized technical support for project delivery will be the responsibility of the Ministry of Agriculture, with some cooperating institutions from other government bodies and beyond.

Capacity building will be a major focus of the project and will be mutually reinforcing. For example, ongoing technical assistance (TA) will be embedded on site in each priority area rather than being concentrated in the PMU in the capital; this TA will comprise relatively young graduates who have a career path opportunity in the Ministry of Agriculture in the extension services, into which climate resilient agriculture approaches and materials will be mainstreamed via the project. This mainstreaming will be done in part by supporting a new program of Belize University / Central Farm to upgrade extension officers knowledge in the concepts and in particular practice of sustainable and climate resilient agriculture. This training will also benefit the project TA, who will also carry out applied research during the course of the project. However, as these practices are a relatively new concept in Belize, a suitable regional institution will provide long term and customized support, starting with a comprehensive needs assessment of the different stakeholders.

C.3. Project / Programme Description

The project aims to minimize the impacts of climatic and economic events on smallholder farmers while supporting continuous market access for their produce. The project will help farmers employ climate resilient and sustainable agricultural practices and technologies. It will integrate profitable smallholder participation into the development of select vegetable, fruit, and honey value chains. The project will support smallholder production capacity through investments in public and private climate resilient infrastructure, strengthening smallholder farmer POs and related/ relevant Government of Belize capacity development.

The complementarity of the IFAD loan, targeting productive development, with a GCF loan/grant that promotes increased resilience and directly addresses climate change threats, has encouraged the country to commit to an innovative approach to agricultural development that can significantly change the development path of rural smallholders in Belize. To accomplish the project's objective, climate change is fully mainstreamed into the project approach through, i) enhanced technical assistance and capacity building to farmers in sustainable and climate resilient agricultural practices, ii) co-financing producers organizations' business plans in all aspects related to increasing climate resilience and autonomy, iii) investment in climate-proofing of public infrastructure across the

value chain, and iv) strengthening national institutions for research, regulation, and education on climate adaptation and resilience in the agricultural sector. These interventions will have mutually reinforcing effects.

The project will initially work in 23 communities clustered in *five priority areas* in five Districts (five communities in Orange Walk District, six in Belize District; five in Cayo District, two in Stann Creek District and five in Toledo District). The main criteria for identifying communities include: i) presence of both male and female smallholder farmers; ii) potential integration of smallholders in targeted value chains; iii) market-oriented smallholder farmers (e.g., not just subsistence farming); iv) farmer interest in project; and, v) presence of formal or informal POs.

As stated above, Belize is considered to be among the most vulnerable countries in the world in terms of climate, in addition to being a very small country in terms of land area. Thus, the selection of priority areas is centered on IFAD's focus beneficiaries, which is the entry point for all IFAD supported projects, i.e., to improve the livelihoods of poor rural smallholders.

The target group includes: i) poor rural families (indigent and poor, with incomes below the poverty line); and ii) vulnerable rural families (whose income is $\leq 25\%$ above the poverty line, vulnerable to poverty). Additionally, households will have ≤ 25 acres of land, and be engaged part-time or full-time in farming. Direct project participants are 6,000 rural households, of which, 3,500 will be in the project's five priority area communities, and 2,500 will be in non-priority areas (to be identified during implementation). A further 24,000 households will benefit indirectly from the project.

Consistent with the importance of women in the Belize rural economy, and in smallholder farming generally (over 30% of farming is carried out by women), 40% of project participants will be women. For their similar importance, youth will comprise 20% of project participants.

Project Components and Outcomes

The proposed Project has two main complementary and mutually-reinforcing components: Climate Resilient Value Chains Development and Climate Resilient Rural Infrastructural and Assets Development, with a third component consisting of cross cutting activities and project management supporting both main components. The diagram below shows the links and complementarities between the project Components and key activities.

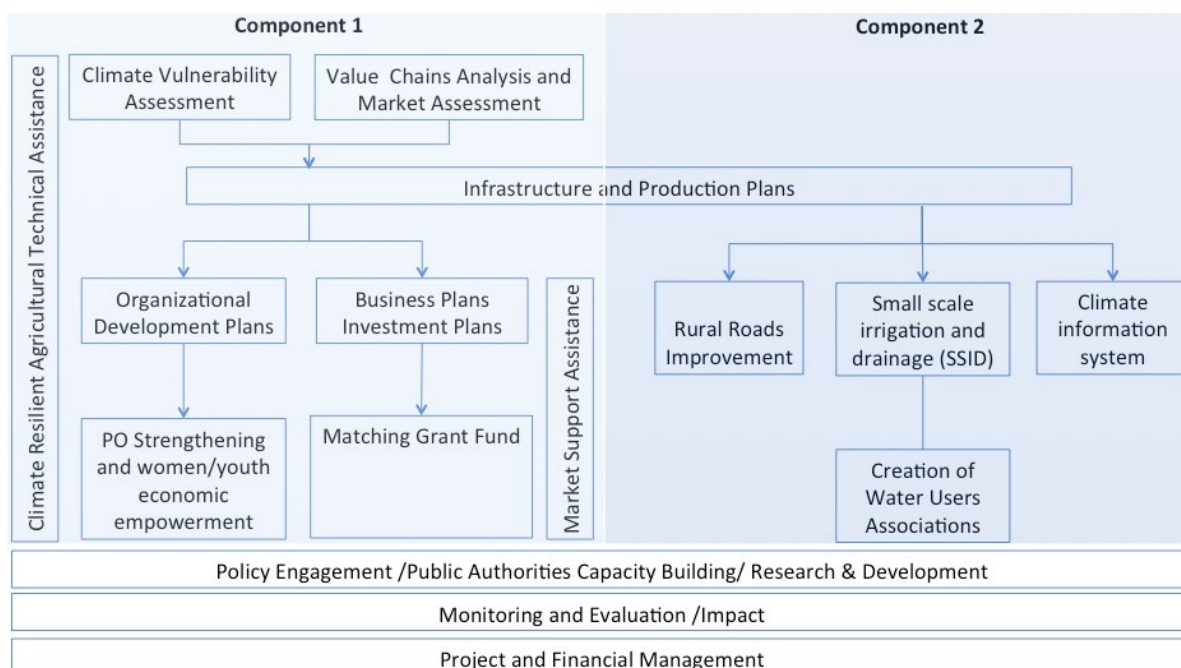


Figure 3. Project Structure

Component 1: Climate Resilient Value Chains Development (CRVC)

This component will introduce/strengthen smallholder participation in select value chains through the promotion of climate resilient and environmentally sustainable production methods, product diversification, and related innovations relying on technical support and the physical presence in the Districts of the project's climate specialist as well as the climate resilient agriculture specialists working with MoA extension agents. Value chain development will be participatory, with the objectives of: i) supporting high quality smallholders' production for commercialization; and, ii) enhancing sustainable smallholder farmer access to markets. Additionally, this component will support self-consumption and healthy food choices through support for backyard gardens.

Activities will initially focus on the development of six vegetables (tomatoes, sweet peppers, hot peppers, cabbages, carrots, and onions), one fruit (pineapples), and bee-keeping products (principally honey). These products were selected from over twenty commodities identified by the GoB and community stakeholders through a participatory process. These value chains are more resilient to extreme weather events, in particular when compared to tree based fruits which suffer severe damage (as can be seen in table 2). Nevertheless, the proposed commodities are vulnerable to climate variability, and climate proofing them through the means proposed in this project will ensure a more consistent output. These commodities are a relatively small proportion of domestic produce, hence the project is also supporting an production diversification strategy towards more climate resilient crops. Four criteria were used to prioritize the commodities: i) significant market opportunity; ii) scope for sustainable and profitable smallholder farmers participation; iii) value chain development potential (e.g., productivity, quality, and market access); and, iv) climate resiliency. Communities selected five priority products, from which POs will select two each for project support. Other value chains may be identified during implementation once the project has experience developing the initial eight products.

A summary of the proposed productive investments and their objectives in the context of Climate Change can be found in Figure 4 below. It must be noted, however, that the Climate Vulnerability Assessment (CVA) described in the next section will provide more precise information on crop and sites specific risks, which will be incorporated in the design of each Infrastructure and Production Plan.

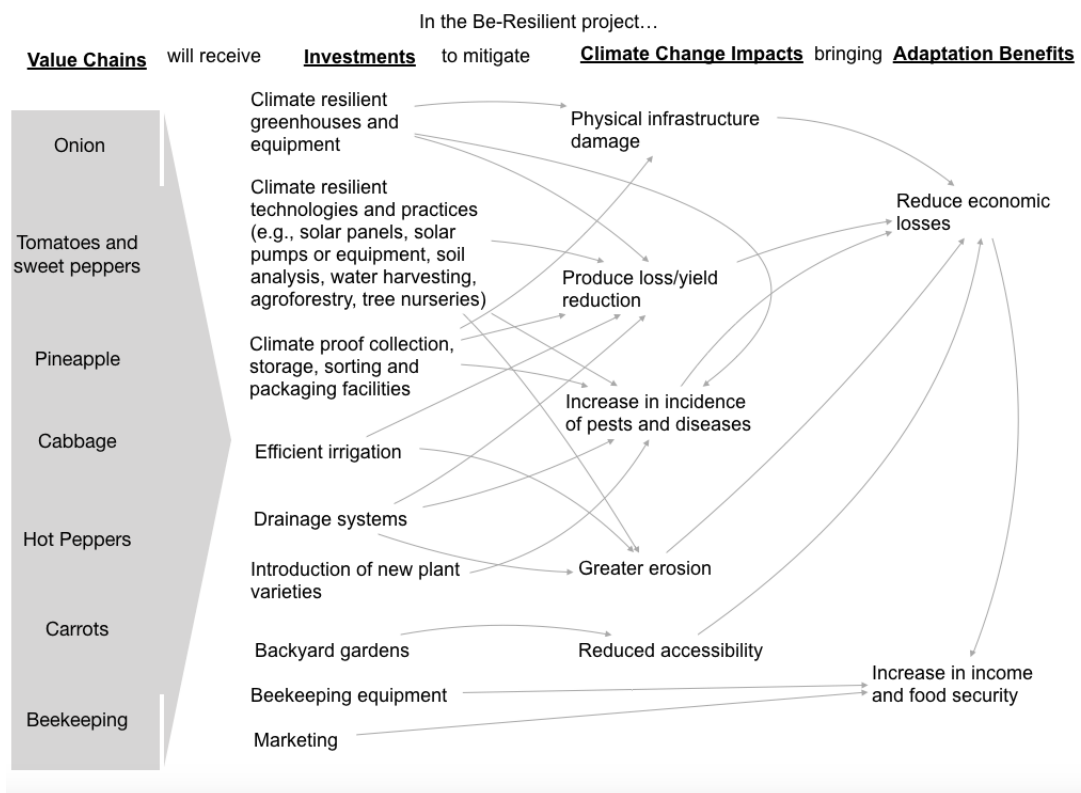


Figure 4. Value Chain Investments to Mitigate Climate Change Impacts

Availability and development of good simulation models for horticultural crops, especially vegetables is lacking in the Caribbean because these a minor portion of the agriculture sector, albeit a significant proportion of smallholder production. Table 3 below extracts from climate sensitivity models conducted in other regions as can be seen by the references provided. These studies and data provide guidance on management of these crops, but will be complemented by site specific climate vulnerability analysis (CVA), in the project.

Previous studies have forecast steep declines in numbers for a bee species in Latin America. However, recent studies are incorporating the idea that some species will show a measure of adaptation to changing conditions, and found a far less uniform outcome.¹⁶ In places with extreme weather events, beekeeping has been found to be an alternative source of livelihood that is less vulnerable to climate change. A single flood or deficient rain in one cropping season can ruin farmers' livelihoods. But bees are very mobile and as such, beekeeping can cope with most of the challenges it experiences. It can provide regular income less dependent on erratic weather pattern, and thus improve food security.

The crops selected for this project are climate change sensitive, but development of tolerant cultivars and adjustment in production system has made it possible to increase the productivity of these crops, even in warmer climate. Thus, supporting smallholder farmers in coping with the climate change impacts is the focus of this project.

Table 3 Proposed climate-resilient technological improvements

Value Chains	Climate Sensitivities and Adaptive Capacity	Climate Change Impacts	Investments for Climate Resilient Production	
Onion farm ^{17, 18}	Warmer temperatures shorten the duration of growth leading to lower crop yields. Onions are considered to have moderate sensitivity to increases in average or extreme temperature. Onions are very shallow-rooted and so they require frequent irrigation. Moderate to high sensitivity to water limitations. Unknown sensitivity to extreme heat events. Onions are sensitive to saline soils and excessive rainfall.	Increase in temperature between 1 to 1.8 °C	Shading screens	Facilities for collection and storage improve product quality and shelf life, and allow the supply period to be extended beyond the current 2 or 3 months. This prevents over-supply of the product for only a short period during the year at low prices, reduce post-harvest losses.
		Increase in frequency and severity of droughts (reduced rainfall of 10% during dry season)	Drip irrigation	Marketing assistance on climate resilient and sustainable practices
		Increase in frequency and severity of floods	Drainage systems Raised beds	Improvement in access roads
Tomato ^{19, 20}	High temperatures can cause losses in productivity due to reduced fruit set, with smaller size and lower quality fruits. However, may benefit from increase in average temperatures within tolerance range. Can be sensitive to extreme heat events depending on timing and crop growth stage. Moderate sensitivity to reduced irrigation and moderate sensitivity to salinity.	Increase in temperature between 1 to 1.8 °C	Greenhouses Temperature resilient varieties	Facilities for collection, storage and packaging improve product quality and shelf life.
		Increase in incidence of hurricanes and floods	Wind resilient greenhouses Drainage systems	Marketing

¹⁶ <http://www.pnas.org/content/114/39/10438>

¹⁷ Central Potato Research Institute (Indian Council of Agricultural Research, Impact Assessment of Climate Change for Research Priority Planning in Horticultural Crops, 2008. Available at: krishikosh.egranth.ac.in/bitstream/1/2054278/1/CPRI034.pdf

¹⁸ USDA, Onions and Garlic Crop Fact Sheet Series, 2015. Available at: <https://swclimatehub.info/system/files/Onions.pdf>

¹⁹ Central Potato Research Institute (Indian Council of Agricultural Research, Impact Assessment of Climate Change for Research Priority Planning in Horticultural Crops, 2008. Available at: krishikosh.egranth.ac.in/bitstream/1/2054278/1/CPRI034.pdf

²⁰ USDA, Tomatoes Crop Fact Sheet Series, 2015. Available at: <https://swclimatehub.info/system/files/Tomatoes.pdf>

	Rapid wilting and death of tomato plants is usually observed following a short period of flooding at high temperatures.	Increase in frequency and severity of droughts (reduced rainfall of 10% during dry season)	Drip irrigation Greenhouses	Technical assistance on climate resilient and sustainable practices Improvement in access roads
		Increase in Increase in incidence of pests and diseases	Greenhouses Selection of more resistant varieties	
Sweet and hot peppers ²¹	Highly sensitive to salinity of soil and waters. Peppers have adapted well to hot climates. Highly sensitive to water stress, either lacking or excessive. Sweet peppers are highly sensitive to pests and diseases.	Increase in frequency and severity of floods	Drainage systems	Facilities for collection, storage and packaging improve product quality and shelf life. Marketing Technical assistance on climate resilient and sustainable practices Improvement in access roads
		Increase in frequency and severity of droughts (reduced rainfall of 10% during dry season)	Drip irrigation	
		Increase in Increase in incidence of pests and diseases (sweet peppers)	Greenhouse Selection of more resistant varieties	
		Saline intrusion from sea-level rise	Raised bed Drainage systems	
Cabbage, ^{22,23}	Can benefit from slightly warmer temperatures, but can be harmed by extremely high temperatures. Moderate to high sensitivity to water limitation. Irrigation-dependent and somewhat salt-sensitive. Sensitivities may be reduced or amplified due to changes in pests and pathogens. Sensitive to flooding.	Increase in frequency and severity of droughts (reduced rainfall of 10% during dry season)	Drip irrigation system with small sprinklers	Facilities for collection, storage and packaging improve product quality and shelf life. Marketing Technical assistance on climate resilient and sustainable practices Improvement in access roads
		Increase in frequency and severity of floods	Drainage systems	
Carrots ^{24, 25,26}	Can benefit from slightly warmer temperatures, but can be harmed by extremely high temperatures. Moderate to high sensitivity to water limitations. Irrigation-dependent and salt-sensitive. Carrots are not particularly vulnerable to storm damage or flooding.	Increase in frequency and severity of droughts (reduced rainfall of 10% during dry season)	Drip irrigation	Facilities for collection and storage improve product quality and shelf life, and allow the supply period to be extended beyond the current 2 or 3 months. This prevents over-supply of the product for only a short period during the year at
		Saline intrusion from sea-level rise	Drainage systems Raised beds	

²¹ Penella, C. and Calatayud, A. Pepper Crop under Climate Change: Grafting as an Environmental Friendly Strategy. Submitted: June 7th 2017Reviewed: November 10th 2017Published: February 21st 2018. Available at: <https://www.intechopen.com/books/climate-resilient-agriculture-strategies-and-perspectives/pepper-crop-under-climate-change-grafting-as-an-environmental-friendly-strategy>

²² USDA, Broccoli, Cauliflower and Cabbage Crop Fact Sheet Series, 2015. Available at: <https://swclimatehub.info/system/files/Broccoli.pdf>

²³ Oxfam/CIAT, 2011. Impact of climate change on Jamaican hotel industry supply chains and on farmer's livelihoods Available at: http://dapa.ciat.cgiar.org/wp-content/uploads/2011/07/Case-Study_Jamaica.pdf

²⁴ Impacts of Climate Change on the Australian Agricultural Industry. Available at: <http://www.vegetableclimate.com/crop-impacts/carrots/>

²⁵ USDA, Carrots Crop Fact Sheet Series, 2015. Available at: <https://swclimatehub.info/system/files/Carrots.pdf>

²⁶ Oxfam/CIAT, 2011. Impact of climate change on Jamaican hotel industry supply chains and on farmer's livelihoods Available at: http://dapa.ciat.cgiar.org/wp-content/uploads/2011/07/Case-Study_Jamaica.pdf

				low prices, reduce post-harvest losses. Marketing Technical assistance on climate resilient and sustainable practices Improvement in access roads
Pineapple ^{27, 28}	Sensitivity to rainfall variability.	Increase in frequency and severity of droughts (reduced rainfall of 10% during dry season)	Irrigation systems	Facilities for collection, storage and packaging improve product quality and shelf life. New plant variety to improve productivity and marketability. Marketing Technical assistance on climate resilient and sustainable practices Improvement in access roads
		Increase in frequency and severity of floods	Drainage systems	
Beekeeping ²⁹	Recent studies are incorporating the idea that some species will show a measure of adaptation to changing conditions. Beekeeping has been found to be an alternative source of livelihood that is less vulnerable to climate change since bees are very mobile and it can cope with most of the challenges it experiences.	This activity was selected because of it's resiliency to climate change impacts and potential to contribute as an alternative source of income and food security.	Beekeeping equipment Technical assistance Access roads	

The CRVC has three sub-components to support climate resilient smallholder integration into value chains: i) Infrastructure and Production Plans; ii) Strengthening of POs; and iii) Value Chains Development.

Subcomponent 1.1: Infrastructure and Production Plans (IPP). This subcomponent will assess and facilitate a participatory diagnosis of value chain development needs in each of the five priority areas. It will result in an Infrastructure and Production Plan (IPP) for each area to guide resilient, smallholder focused value chain development. IPPs have two input studies, a *Climate Vulnerability Assessment* and a *Value Chains Analysis and Market Assessment*. These complementary assessments provide the foundation for preparing the IPPs, and will inform related GoB smallholder policy/ regulatory and public infrastructure investment plans found in Component 2.

The **Climate Vulnerability Assessment (CVA)** will analyze smallholder vulnerability to climate change in the five priority areas (and new areas to be selected) and identify measures to adapt to the climate change impacts. It will define climate change risks, identifying the most vulnerable communities, sectors, value chains, and social groups, as well mitigation activities. Together with the project's Climate Specialist, an international institution will conduct an evidence- based analysis of the relationships between climate variables and biophysical processes. This study will be complemented by a stakeholder self-assessment which will identify community level climatic threats.

²⁷ De Mondonca, Arnold ; Patterson-Andrews, Hazel. Investigation of the effects of rainfall (Climate Change) on pineapple production in Essequibo Tri-Lakes Area, 2015. Available at: <https://ageconsearch.umn.edu/record/242077?ln=en>

²⁸ Williams, et al. Impact of climate variability on pineapple production in Ghana. Agriculture & Food Security, 2017:26.

²⁹ Imbach et.al. Coupling of pollination services and coffee suitability under climate change. Available at: <http://www.pnas.org/content/114/39/10438>

The study and assessment will rank exposure to climate change between priority areas, value chains, producer groups, and social groups, identifying key constraints to resilience, levels of response capacity, and priority support areas. The analysis will assess factors influencing exposure and response across three core areas: i) exposure to climate-related natural disasters and rising sea-levels; ii) effects of population patterns, access to natural resources, agricultural dependency, and conflicts; and, iii) future vulnerability and adaptive capacity (e.g., infrastructure, social organization, and agro-ecological characteristics). Findings will be used to guide and prioritize area strategies for climate change adaptation/ resilience agro-ecosystems and public infrastructure.

Value Chains Analysis and Market Assessment (VCAMA) will define activities for the eight product value chains in each priority area. This includes definition of chain characteristics including: key actors (producers, transporters, packers, processors, traders, retailers, and consumers), market context/ potential, financial/ economic profile, product seasonality, and price trends etc. The objective of the analysis is to highlight strengths, weaknesses, opportunities, and threats to chain development, with a view to sustainably increasing profitable smallholder chain participation. The assessment will provide input to production intervention design, PO business plans, and public infrastructure planning. Data will also be used for end-market studies employing a USAID methodology designed for assessing demand and processing for high-end markets.³⁰

Together, the CVA and VCAMA provide key input to priority area **Infrastructure and Production Plans (IPP)**. The IPP is a “master plan” broadly defining smallholder value chain climate resilient production practices to be adopted (individually or in a collective, methods, and technologies), PO intervention imperatives, required infrastructure, and possible GOB regulatory/ policy needs.

Specifically, IPPs will identify: i) climate change adaptation capacity-strengthening needs; ii) infrastructure essential for value chain market development and disaster risk management; and, iii) value chains market opportunities. PO Business Plans (BPs) will be aligned with priority area IPPs, which have the goal of supporting climate resilience at the farm and community levels.

Subcomponent 1.2: Strengthening of Producers’ Organizations. Smallholders generally lack the resources and capacity to improve their production and to negotiate with market actors on an equitable basis. Despite this relative weakness, smallholder member-based POs offer a collective solution to overcoming these constraints. The project will strengthen PO capacity to improve resilient smallholder production and participation/ markets access in several value chains. POs require support to improve and coordinate membership services at scale, improve member market transactions representation/ negotiation skills, and enhance planning and project management capacities with a goal of achieving financial sustainability. In addition to organizational capacity building, the project will focus on social inclusion (youth and women), confidence building, leadership training, and rural empowerment. As a part of this commitment, the project will train local men and women in ten POs to become professional Local Managers. Local Managers will receive intensive six-month training, salary and mentorship support for two years, and their POs will receive some office support (e.g., computing and office equipment) after which POs will pay their salaries.

Activities under this sub-component will include *inter alia*: i) management and governance training; ii) development of internal administrative processes; iii) creation of Organizational Development Plans (ODPs); iv) national and international exchange visits; and, (v) youth (men and women) management/ leadership training. Gender member and leadership participation in POs will be increased (women are currently less than 30% of membership) through awareness-raising, and strengthening/ empowering the capacity of women to participate both as PO members and leaders.

³⁰ The Value Chain End-market Research is a toolkit addressing the process and value of end-market research for value chain development. Developed by the United States Agency for International Development in 2008, it provides a portfolio of tools proven through practical application.

Subcomponent 1.3: Value Chains Development. Smallholder farmers and POs have little or no experience participating in commercial value chains on an equitable and sustainable basis. This subcomponent will address smallholder value chain participation constraints by improving information to smallholders, enhancing linkages between farmers and buyers, improving marketing capacities, and providing technical assistance through the project's Climate Resilient technical staff, in coordination with the agricultural extension service of the MoA. Interventions will focus on local markets as well as on high-end domestic demand (e.g., hotels, supermarkets, restaurants etc.), strategically integrating smallholders/ POs into various parts of the value chain, including developing value-added functions (e.g., product aggregation collection, sorting, grading, storing, processing, transporting, marketing, etc.). Partnerships with key actors, such as industrial processors, hotels, and restaurants, will be explored to ensure smallholder can meet demand preferences.

Backyard gardens (BYGs) are also a part of this sub-component.³¹ When principle of climate resilient agriculture is applied, BYGs have the potential to improve food security and income/ expense savings. The Matching Grant Fund (MGF) (see below) will support groups of individuals from the same priority areas to invest in their individual BYGs. Collective application will foster cooperative activities and decrease MGF administrative burdens.

This subcomponent has three interlinked support activities.

Development of value chain Business Plans (BPs): This activity will provide TA to support for the development and implementation of PO BPs consistent with and supportive of priority areas IPPs. Support will include a basic PO profiling and diagnostic (from subcomponent 1.2), followed by an assessment of challenges (e.g., related to climate change, production, market development etc. based on the studies in subcomponent 1.1). POs will develop a plan to address priority problems, detailing required technical assistance, budget, counterpart funding, time lines, project participants' roles, expected results and benefits to the PO and smallholder members. Support will be provided to link POs with value chain market actors and technical assistance will be provided from the PMU and/ or specialized service provider for PO BP development.

Matching Grant Fund (MGF): Resilient production and value chain development investments defined in PO BPs will be financed by the MGF. This MGF will support investments with the goals of: i) increasing agricultural production climate resilience; ii) increasing production volume and quality; iii) climate-proofing value-chain infrastructure; and, iv) developing product value addition opportunities (e.g., via sorting, selection, packaging and labelling of products or the processing etc.).

Investment categories include: i) climate resilient technologies and practices (e.g., solar panels, solar pumps or equipment, soil analysis, water harvesting, agroforestry, tree nurseries); ii) climate resilient greenhouses and equipment; iii) climate proof collection storage, sorting and packaging facilities and equipment; iv) efficient irrigation (e.g., wells and water systems connections); v) drainage; vi) beekeeping equipment; vii) introduction of new plant varieties; and viii) backyard gardens. The MGF will not finance purchase or lease of land, refinancing or payment of debts, or activities harmful to the environment.

The MGF will be a competitive fund; open to formal and informal POs. Project target group members receiving support from technical areas of the project will be eligible for funding. The MGF will work with Credit Unions formerly supported by the BRFP to consider MGF funding as a risk reduction for loans to smallholder, groups of smallholders financing a PO project, and possibly to POs themselves. The MGF will promote/ support applications empowering women and youth. GCF loan component will be utilized to finance the MGF.

Support Assistance: This activity will support climate resilient production, product marketing, and market linkages. It has four activities:

- *Climate Resilient Production TA:* Significant crop and economic losses are *directly* related to unplanned, uninformed, and uncoordinated smallholder production decisions. A more coordinated and planned approach based on climate vulnerability assessment can avoid many of these losses. The project's Climate Resilience Extension Services together with Value Chain Specialist will support the implementation of PO coordinated

³¹ Backyard gardens are between 1/8 and 1/4 acres and located near the owners home.

demand-driven and climate informed smallholder crop planning process in priority areas. This will help smallholders plan which crops to grow, when to grow, and how to respond to market needs and anticipate climate events. The TA will facilitate the introduction of climate resilient production technologies that can: i) increase resilience to droughts and floods; ii) maintain soil fertility and water infiltration, iii) improve pest control, vi) increase carbon sequestration; and vii) reduce external inputs (and thus save energy, reduce emissions and lower costs). The technologies being considered include: crop rotation, integrated crop-livestock fertility, soil cover, raised beds, polyculture systems, using green manure and legumes for fertilization, agro-forestry, planting in contours, riparian planting and implementing community seedbanks.

- *Marketing TA:* Specialized marketing capacity development support will be provided to POs and smallholders, including: i) promoting farming as a business; ii) basic marketing skills and good marketing practices; iii) market analysis;³² iv) production and processing implications for meeting target market opportunities; and, v) access to/ use of market information technologies (e.g., the MoA's agricultural market price information system and the BMDC's wholesale and retail market prices data).
- *Development of Partnerships and Market Linkages:* The development and implementation of post-harvest and agro-processing facilities will be supported through the MGF and will assist POs to take on new value-added functions (e.g., product collection, sorting, grading, storing, processing, transporting, and marketing). The project will complement these investments by strengthening linkages between producers and buyers, and supporting PO partnerships with key market actors. Relationships will be initially build through PMU, BMDC, and MoA facilitated meetings between buyers and smallholders/POs with the objective of establishing direct contacts, information sharing, building PO/ smallholder confidence, and displaying PO smallholder products
- *Establishment of Intermediate Markets:* A large share of national market demand for many food products are met with legal and contraband imports, many of which can be substituted by local smallholder production. Responding to high-end market buyers interest in local supply, the project will finance two pilot farmers' markets in Belize City, and one each in San Pedro and Caye Caulker for POs members to sell directly to wholesale and retail consumers. These activities will be informed by and implemented concurrently with Partnerships and Market Linkage activities.

Component 2: Climate Resilient Rural Infrastructure and Assets Development (CRRIA)

This component supports climate resilient productivity and improved market access through rehabilitation and provision of road, drainage, and irrigation infrastructure in priority areas.

Infrastructure investment proposals will align with and support IPPs and PO BPs and demonstrate the potential to: i) reduce smallholder and infrastructure climate change vulnerability; and, ii) enhance smallholder economic opportunities/ incomes. The overarching goal is to support climate resilience infrastructure enhancing smallholder farming business/ rural enterprise opportunities, while serving the largest number possible of direct and indirect project participants.

The CRRIA component has three sub-components, which are explained in detail in the following subs-sections and further clarified in section 2 of appendix 11. Sub-components 2.1 and 2.2. will be financed through an Infrastructure Investment Funding Award (IIFA), with a total envelope of US\$8.2 million USD. The IIFA will award competitive funding for investments in a broad range of public (for common use) infrastructure that will reduce physical vulnerability and anticipated impacts of climate variability and enable and enhance private sector investments and activities in project priority areas.

A programmatic approach drives the component where project works are not pre-identified before the start of the operation but are selected on a periodic (annual) basis on specified criteria and demand. All applications for

³² The San Pedro and Placencia markets supply demand for 60 + vegetables and fruits (traditional, non-traditional, and organic fruits and vegetables).

IIFA will go through a three-step selection process of: (i) pre-qualification by PMU technical staff; (ii) screening and ranking including as minimum climate change vulnerability and number of beneficiaries per USD 1,000 spent; and (iii) final selection and inclusion into AWPB for approval. The main types of infrastructure eligible under the CRRIA Component include public infrastructure of common use such as:

- Last mileage of local or uncategorized feeder roads in rural areas. Eligible investments also include road ancillaries such as small bridges, drainage facilities and erosion protection systems to ensure climate resilience of the rehabilitated roads.
- Small-scale irrigation and drainage systems including intake structures (pumping station, rain harvesting ponds), main and secondary distribution networks (open channels or polyethylene pipelines), drainage networks and flood protection structures.

Subcomponent 2.1: Investment in Rural Roads Improvements (RRI) and in Small-scale Irrigation and Drainage (SSID). Investments will be directed at rural roads and ancillary structures most vulnerable to increase in rainfall intensity and flooding as well as those that complement the Component 1 objectives (e.g., by assuring project-supported value chains/commodity production year-round market access). Improved roads will largely be the “last mileage” of local or uncategorized feeder roads, and are often poorly made and maintained, rendered inaccessible to vehicular traffic in the rainy season. Eligible investments will also include climate proofing of roads with ancillary structures (e.g., small bridges, culverts, drainage facilities and erosion protection works etc.). The PMU will procure consultancy services for road design and supervision, as well as construction.

GCF funding will be utilized to finance the additional cost of climate proofing roads; beyond current standards in Belize for a well-built road to replace the existing dirt tracks (minimum required engineering standards such as proper subbase with +20 cm depth and a stabilized gravel layer of 6-10 cm, including minimum drainage facilities). The Climate Vulnerability Assessments (subcomponent 1.1) will provide more precise information on the site-specific risks and the road detailed engineering design will incorporate these criteria. This entails significantly more stringent technical parameters for designs that will ensure infrastructure specifications take into account the natural hazard risk and anticipated climate risk factors, and will therefore finance the engineering aspects of climate resilient rural roads identified: (i) slope stabilization structures such as dry stone or gabion wall dependent on the gradient of the road and road construction materials; (ii) paving of roads with durable materials; (iii) proper alignment of new roads to avoid vegetative loss; (iv) improved drainage systems (including cross drainage structures such as cascades, small check walls, culverts and causeway) to avoid erosion of road materials; (v) improved planning of roads with proper cross section and standard dimensions; and (vi) raising road level to adapt to climate change events, especially flooding.

The climate additionality provided by GCF in the road works, generates a significant improvement in road stability in the face of predicted adverse weather events. Including these adjustments will allow for road investments to last longer in good conditions, increasing accessibility and therefore community resilience. While at the same time reducing loss of products, and high expenditure of vehicles maintenance. Therefore, every investment will have to be analyzed on technical parameters that specifically address climate proofing and include activities such as: sections of the road surface level raised to an elevation of about 0.5 m higher than expected flood level to reduce risk of road damage and improve accessibility during flood event. Side slope adjusted from 1:2 to 1:3 or flatter to prevent flood damage and erosion from road surface runoff. Road drainage ensuring it will have the capacity to i) convey water from the surface of the road, as well from the different layers of the road structure, to a safe exit; ii) intercept surface water flowing towards the road and convey water across the road in a controlled fashion; and iii) not allow water to develop sufficient volume or velocity so as to cause excessive wear along ditches, at culverts or along exposed running surfaces, cuts or fills preventing excess water within the roadway that will adversely affect the properties of the materials with which it was constructed.

The SSID section of this subcomponent will ensure better climate, environmental, social, and economic resilience as the reliability of water supply and agricultural land management improves and directly addresses the effects of

continued, documented rainfall variability which results in droughts and floods.³³ Therefore, irrigation and drainage investments will directly address varying rainfall patterns caused by climate change and include measures to increase climate resilience, including: (a) high over-all efficiencies of irrigation-dependent production systems; (b) high output and high value per cub. m of water; (c) adequate hydraulic feasibility (d) adequate drainage systems to cope with excess water and erosion; (e) good control of water allocation over time and within the scheme; (f) limited losses, for the sake of flow capacity and scour control; (g) predictable and reliable water allocation.

Small-scale irrigation and drainage systems funded will include *inter alia* intake structures, pumping stations, rain harvesting ponds, main and secondary distribution networks, drainage networks and flood protection structures. On-farm irrigation/ drainage systems for connecting to main or secondary networks (e.g., drip, sprinkler, gravity ditches or field open drains) will be provided by project participant finance through their own resources or MGF funding. Works for the small-scale irrigation and drainage schemes, design, the PMU will procure supervision consultancies. Ten Water User Associations (WUA) will be formed and provided technical assistance for institutional development and management of irrigation and drainage schemes. All SSID investments will be climate proofed to withstand expected future precipitation scenarios.

Subcomponent 2.2: Climate Information System

The subcomponent will focus on the creation of the Climate Information System (CIS) which has the purpose of providing farmers with timely and accurate climate information, allowing them to plan production activities and minimize climate related production losses. Its functions will include climate analysis and monitoring, assessment and attribution, prediction (monthly, seasonal, decadal) and projection (centennial scale). The CIS will authenticate and communicate climate information at across the country primarily via TV and radio, although cellular services may also be considered. The Climate Resilient Production TA will strengthen farmers' capacity to use the information for decision-making and production planning.

This component will be informed and support the betterment of current systems provided by The National Meteorological Service (NMS) and the National Emergency Management Organization (NEMO). NMS collects, analyzes and provides meteorological and climate-related data and information to the general public. When major climatic events are forecasted to affect the country, NEMO works with the NMS to provide more frequent updates on the forecasted event to the general public. Nonetheless, there are important gaps in systematic dissemination of information and in specialization of information (i.e. to support the agricultural sector in defining irrigation patterns and crop selection).

Component 3: Cross-Cutting Activities and Project Management

The final component consists of the cross cutting activities and project management. Cross cutting activities are those which support interventions found in Component 1 and 2, of which there are three.

Public Authorities Capacity Building: The MoA will have a central role support project implementation and related sustainable outcomes. Institutional strengthening of the MoA's capacity to provide technical services to smallholder farmers, both for project implementation and in the future, is essential, particularly in the Departments of Extension (DE), the Department of Cooperatives (DC), and the Belize Marketing and Development Corporation BMDC.

The DE has a presence in each District and is responsible for executing MoA programs and operational activities at the local level. Intensive training/ capacity building of its personnel is planned for a range of strategic topics:

³³ Belize is identified one of the most vulnerable countries to the adverse impacts of climate change. Climate change forecasts for Belize for the period 2001-2080 for two basic meteorological parameters, temperature and precipitation, show increases in average seasonal temperature until 2080 between 2°C and 4 °C depending on the season and area. The forecast predicts a decrease of average rainfall by 100 mm and changes in rainfall distribution. Although annual rainfall volumes are more than adequate for most crops, the increasing intensity of rainfall within the marked rainy and dry seasons mean that many parts of the country will increasingly suffer from a distribution of rainfall that provokes floods and droughts. More details on climate change projections are provided in the SECAP paper in Appendix 13.

climate change and resilient production practices and technologies; value chain development; marketing information communication/ management; rural organization management/ business development; social inclusion; development communications audio-visual training materials. The project's Climate Resilient Agriculture Specialists will work with MoA extensions agents to develop and support service provision to target project participants.

The DC has regulatory oversight of all non-financial co-operative enterprises including *inter alia* registration, management training, supervision and audit functions, and member enterprise development. The DC suffers from weak/ antiquated legislation and low staff capacity which limits its effectiveness. The project will work to strengthen its capacities through training and capacity-building, resilient agriculture, social inclusion (gender and youth particularly) and value chain development.

The PMU's Rural Organization Development Specialist, along with two contract specialists, will work closely with the DE and DC to develop and execute a TA plan to strengthen MoA capacities, and to improve intra Ministerial/ departmental cooperation and collaboration. The plan will incorporate aspects of IFAD Learning Routes, and offer national and international study tours/exchange visits. As the MoA is one of a few GoB ministries with a Gender Focal Point, the project will support activities designed/ led by the focal point team (e.g., research, workshops, etc.).

Applied Research and Development (R&D). The project will complement MoA research activities by undertaking targeted research supporting sustainable smallholder climate change resilience production and market access (e.g., supporting research on extension-farmer linkages fostering access to locally-validated productivity and resilience enhancing agricultural technologies). Specific planned activities/ studies include:

- *Adaptation Strategies:* Support for the University of Belize's Faculty of Agriculture to conduct applied research on adaptation strategies and climate resilient practices including on diversified production systems, agroforestry, protein banks, and windbreaks. In a collaborative effort between University of Belize Central Farm (UBCF) and international experts coordinated by the Climate Specialist, develop training courses, and manuals on climate resilient practices, disaster risk-reduction and preparedness, post-harvest solutions and transformation, and storage solutions directed to farmers, MOA staff and extension agents.
- *Index-based insurance:* The CDB and CCRIF recently launched the "Integrated Sovereign Risk Management in the Caribbean" Project. The project includes a Disaster Recovery Fund for Caribbean countries, but has not yet defined a mechanism for smallholder farmers to access the fund. The proposed study will assess mechanisms which provide Belizean smallholders to climate disaster insurance, allowing them to make production investments with greater financial security. Potential mechanisms include an index-based or conventional indemnity insurance scheme.
- *Ecosystem Services Payments:* This study will assess the feasibility of using ecosystems payment services to stimulate maintenance and restoration of local ecologies (e.g., native forests) which provide essential environmental services but do not currently generate revenue for smallholder farmers and other landowners.
- *Smallholder Cane Farmer Production Diversification:* Many small "out-growers" in the Corozal District provide Tower Hill refinery cane employing low-input, low-output, marginally profitable production systems (due to low yields, poor quality cane, and high cutting, and transport costs). The end of preferred access to EU sugar markets for Belize will further depress smallholder incomes. The project will support participatory research to examine with smallholder farmers, their interest in and potential for diversification from cane. The study could also lead to the inclusion of communities in the Corozal District into the project.

Policy Engagement/Consultation: Several policy constraints to climate resilient value chain development were identified during project design, the most notable relating to import tariffs and permits, product quotas, and outdated agricultural and commercial laws and regulations. The project will fund public consultations involving a broad set of stakeholders (e.g., project participants, POs, the private and public sectors, etc.) on barriers to private sector value chain development. Discussions will likely lead commissioned research on targeted policy/regulatory

issues/ themes, or the strengthening relationships/ coordination among public and private institutions (e.g., updating and management/ enforcement of agricultural import laws and regulations).³⁴

Please see Appendix 4 -Detailed Programme Description for more information on the project activities.

Outcomes

The project will reach a total of 6,000 households, or 30,000 persons, all of which will have strengthened climate resilience.

The outcome targeted under Component 1 is that target project participants produce and market larger and more reliable supply of agricultural products via climate-resilient practices. Key results include:

- At least 30 formal and informal POs will be strengthened, benefitting at least 4,500 members, through improvement in their by-laws and in organizational and administrative skills. They will be able to increase their membership and generate revenues for their financial sustainability;
- Approximately 6,000 households will be engaged with vulnerability assessments and trained in climate resilient agricultural technologies and practices for application to their production and marketing systems;
- At least 2,000 smallholders will be able to produce and market incremental volumes of agricultural products and to adopt climate-resilient practices;
- Investments will be made in the rehabilitation/ construction of approximately 300 structures (covered structures, storage facilities, etc.) and other climate resilient technologies and infrastructure;
- Approximately 400 hectares of land will be brought under climate resilient management, benefiting from improved drainage and irrigation infrastructures, including about 240 ha under the newly developed public systems; and
- At least 700 smallholder farmers will build their entrepreneurial skills with training in business management and marketing.
- At least 2000 households will have increased resilience through backyard gardens

The Outcome targeted under Component 2 is that climate resilient public infrastructure and services, support production and access to markets for targeted project participants. Key results include:

- Physical vulnerability will be reduced and resilience to natural hazards and the anticipated impacts of climate variability strengthened in an estimated 23 communities in five priority areas. Quality of life and efficiency of agricultural production will be improved in these communities;
- Approximately ten community -level Water Users' Associations/Informal Water Users' Groups will be established with the participation of men and women, and trained for irrigation and drainage systems management, operation, and maintenance;
- Approximately 400 rural households (2,000 people) will directly benefit from improved road infrastructure; and
- Approximately 350 rural households (1,750 people) will be provided with reliable irrigation water supply and adequate drainage networks.
- Approximately 95,000 people will receive climate information for decision-making in agricultural production.

³⁴ POs agreements with MoA related to food imports (e.g., borders closed when there is sufficient local production) will need to be replicated and fully complied with in a timely manner. To this end, the MoA will be sensitized to the damage caused by non-timely enforcement of protocols, and support will be provided for efficient implementation.

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

The Ministry of Economic Development (MED) will be designated as Lead Programme Agency (LPA) with overall responsibility for project implementation. This arrangement is based on the following considerations: (i) the long standing successful partnership between IFAD and the MED; (ii) the multi-sectoral, cross-cutting nature of this project; (iii) the successful experience of the project execution model in the last IFAD project in Belize; and (iv) the GCF's clear position on their mandate to finance climate resilience projects (as opposed to co-financing rural development projects). The proposed structure replicates the structure of the previous IFAD-funded project, in which the MED served as the LPA, and to which much of the success of that project can be attributed.

The GOB, and especially the MED, has experience in the implementation of externally financed projects and is knowledgeable of the Categories of Expenditure for organizing disbursements and applying the percentages of financing according to the different funding sources. The MED is assigned overall fiduciary responsibility as the LPA. It will ensure proper financial management and implementation of the project through the creation of a dedicated Project Management Unit (PMU), which will operate with a high degree of independence from the MED. The principal mandate of the PMU will be to carry out the overall programming and budgeting and to take the lead for overall project implementation, which will include working with service providers, Government Ministries and Departments, individual project participants, producers' organizations and the municipalities of the Project's target areas. In addition, the PMU will be vested with financial and technical autonomy, guided by the PIM and reporting directly to the POC.

The PMU will comprise a total of 11 staff members: Project Manager, Finance Officer, Project Accountant, Procurement Officer, Administrative Assistant, Monitoring and Evaluation Specialist, Climate Agriculture Specialist, Rural Infrastructure Engineer, Rural Organization Development Specialist, Value Chain Specialist and Administrative Assistant. Please see Appendix 5. Institutional Aspects and Implementation Arrangements for the terms of reference detailing each positions profile and responsibilities.

As part of the design and in accordance with IFAD's Financial Management policies and guidelines, a Financial Management Assessment was undertaken for the proposed project's arrangements within the LPA. Although there are important weaknesses and challenges in Belize's public sector and specifically the MED having limitations in terms of capacity, systems, procedures and internal controls, the proposed independent PMU along with the systems and procedures requirements (please see Appendix 7: Financial Management and Disbursement arrangements for more details) would position the project at a low risk level from a financial management perspective and provide assurance of a satisfactory financial management function.

C.5. Market Overview (if applicable)

A preliminary review of information available indicates that no previous market studies were done on the various agricultural and food markets in Belize. A mapping of the market situation during the design mission suggested that there is much potential for small farmers to sell their products to local markets. IFAD developed nine production models of the selected value chains. Each incorporates the adoption of improved technology in the production process that requires an initial investment, and each model estimates a future projection of the expected results in the new production scenario. Details of the financial and economic analysis are presented in Appendix 10. In general, the models indicate that there is ample demand in the market for an increase in production. For instance, the expected production increase would replace 70% of the current imports of cabbages, 45% of onions, 40% of carrots and 100% of honey. The expected increase in production of pineapples would be absorbed by the processing plant for juice production.

The main constraints that small farmers face to access markets for their products include: (i) weak marketing system services such as inadequate transportation, storage, packaging and the lack of adequate standards that result in higher post-harvest losses and reduce farmers' ability to access several value-added markets (processed

and organic, supermarket and hotel outlets) because of unreliable quality, consistency and timeliness; (ii) an inadequate market information system to provide price, volume and other market data on a timely basis, resulting in limited effective decision-making and their efficient allocation of resources and the required dynamism to compete with imports and contraband products; (iii) limited incentives such as access to credit and lower cost input supplies which increases the level of risk involved in investing in existing and new agricultural ventures but which could reduce costs and related transaction costs; and (iv) inadequate attention to market development that results in continued dependence on a limited range of products and uncertain markets. These constraints are exacerbated by the impacts of climate change, as increased variability and intensity of rainfall strongly affects the growing season and therefore, producers' capacity to supply markets. Climate change imposes an additional cost to smallholder agricultural production in Belize, as producers have to invest in more resilient production capacities and technologies, and the country needs to provide the required infrastructure to ensure adequate production and access to markets.

During the design mission, two sets of buyers of products were identified with whom rural producers could be engaged: institutional buyers (hotels, restaurants, etc.) and individual buyers. These two segments in the various districts comprise the major component of the domestic market for vegetables and fruits, in which smallholder farmers have opportunities to establish sustainable linkages with buyers. The buyers, particularly the high end institutional ones indicated that lower quality, unreliability of supplies and low price competitiveness of the products are the three main factors that limit their purchases from local producers.

Belize has a large tourist sector which is an expanding market for food products. However, this market remains largely untapped for domestic producers. According to the Statistical Institute of Belize (2017), food and beverage (F&B) imports have been growing at an average of 9% per year from 2006-2016. In 2016, F&B exports totaled BZE \$243.71 Million, or 12% of the countries imports. Tourism has also been growing, the year 2016 culminated with another significant achievement for the tourism industry, boasting the highest overnight arrivals in twenty years. The potential for substitution of commodity imports combined with the demand for high-quality vegetables, fruits and other agricultural products by the tourism/hospitality sector provides much potential for increasing these products. This provides an expansion opportunity for small farmers if they can improve the quality, quantity and reliability of their supplies. Furthermore, land availability is not a constraining factor for agricultural production as the six districts of Corozal, Orange Walk, Belize, Cayo, Stann Creek and Toledo together has an agriculture potential of about 0.8 M ha (2 million ac), or 38% of the land area, that is suitable for agriculture. In these overall six districts of Belize that approx. 15% of this amount is under farming every year.

Except for the organized export sector (e.g. sugar), small farmers are poorly organized for both production and marketing of their products. As a result, their bargaining position is weak and they are often "forced" to sell their products to middlemen at lower prices. In addition, the lack of coordination among producers and the absence of storage facilities and agro processing activities often result in flooding the market at the same time after harvest and selling at low prices.

Although few smallholder farmers in Belize are organized as cooperatives or associations (formal or informal), existing producer organizations are very deficient in organizational, administrative and technical capabilities. Most do not provide services to their members and it's the main reason why they lose members. However, producers recognize the importance for them to be organized; on the one hand, because technical assistance can reach them more easily and on the other hand, it is the only way to improve their negotiating and competitive position in the local market that presently imports a large proportion of its food needs from abroad and is often flooded by contraband products mainly from Mexico.

C.6. Regulation, Taxation and Insurance (if applicable)

Regulation

According with the Department of the Environment (DOE) of the Ministry of Agriculture, Fisheries, Forestry, Environment and Sustainable Development (MOA) the EIA system in Belize establishes three categories or schedules (I, II and III), of proposed projects or activities, which may or may not require an Environmental Impact Assessment (EIA) of a Limited Level Environmental Study (LLES)

Information submitted is reviewed by the Project Evaluation/EIA Unit of the DOE. Review takes into consideration the Project Schedules of the Environmental Impact Assessment (Amendment) Regulations, 2007 as well as the criteria of the Environmental Impact Assessment (EIA) Procedural Manual (location, nature, magnitude, etc).

Our environmental and social due diligence process followed on the ground consultations, an internal IFAD screening process as well as a thorough analysis of the Environmental Impact Assessment Regulation (2007) concluding that none of the infrastructure proposed on the Project would be considered as Schedule I, therefore requiring a full EIA.

Nevertheless, some of the proposed activities by the project could be considered as Schedule II such as the improvement of roads less than 2000 meters in length, canalization or flood relief works or packing/canning of vegetable products – which may require a LLES. If a LLES is required, the nature and extent of such studies are determined by the DOE, in consultation with key agencies.

In fact DOE establishes that some Schedule II projects differ from Schedule I projects only in scale - which is the case of our proposed project. Large irrigation and drainage projects are usually Schedule I while small-scale project of the same type may fall into Schedule II. In a Schedule II project, the impact may not be as serious as a Schedule I project depending on size, location and other considerations.

All activities to be developed fall within IFADs B categorization, both internally and as accredited by the GCF; no activities qualifying above this categorization will be undertaken.

The Investment and Production Plans will inform the PMU which rural and ancillary structures are most subjected to climate variability as well as need for small scale irrigation and drainage systems. The required documentation (project description, land titles, and others) will be submitted separately to the Department of the Environment. It is expected that the first Environmental Clearances will be requested by the second semester of the Project and will be received in an average time of 6 months. For irrigation and drainage works, once the Environmental Clearance is received, the Project will have to submit documentation of a Water Abstraction License.

Taxation and other forms counterpart financing

As part of its contribution to the Project, the Government is assuming all tax costs associated with the Project expenditures. Specifically, the General Sales Tax (GST) of 12.5% for all goods and services to be acquired and on income taxes, the project considered a rate of 25% based on information provided to the design mission by the MED³⁵.

Regarding the expected contribution from the Project participants - i.e. Producers' Organisations (POs) - for Component 1, which amounts to USD 822,400 (see Appendix 4: Detailed Program Description - Matching Grant Fund - MGF), it is important to clarify that of this total amount, USD 440,000 will be contributed in cash to be spent directly by Project counterparts (and not through project accounts), while the remaining USD 382,400 will be contributed in kind. All purchases made through the MGF will be owned directly by POs; applicable taxes on purchases will be the responsibility of the project participant and will count towards their in cash contribution.

³⁵ The rate varies for independent consultants depending on their residence, whether from Belize, outside of Belize but from CARICOM-member countries (15%) or from other countries (25%).

Therefore, there will be no contribution transfer from project participants to IFAD. The anticipated local contribution will be used/exhausted locally. Without a financial transfer, IFAD cannot conduct AML checks. Yet, in the remote event a financial transfer to IFAD may be executed, the banks involved in the transaction will ensure, through their internal policies and standards, that anti-money laundering and counter terrorism financing regulations are adhered to both internationally and in the jurisdictions where the payments are executed.

Furthermore, a statement of precedence of funds for in cash contribution will be requested as part of the proposal for MGF evaluation at POC. Any suspicious or irregular documentation will be presented to the Financial Intelligence Unit of Belize which controls all potential illegal activities of all financial institutions regulated at Central Bank and any reports presented for ML/TF. All credit unions which could potentially provide financing to project participants are regulated by the Central Bank of Belize and overseen by the Financial Intelligence unit.

C.7. Institutional / Implementation Arrangements

The Ministry of Finance (MoF) will represent the Borrower for all legal and financial matters. The Ministry of Economic Development (MED) will be designated as Lead Programme Agency (LPA) with overall responsibility for Project implementation. The proposed structure takes into account the cross-cutting nature of this Project and replicates the similar successful model of the IFAD-funded BRFP, in which the MED served as the LPA. The Ministry of Agriculture (MoA) will play a key role in Project implementation and will work closely with the MED and the PMU to ensure that agricultural sector priorities and strategies are duly taken into account throughout the implementation of the Project. Ultimate decision-making power will remain with the MED.

Strategic direction and oversight will be provided by a Programme Oversight Committee (POC), chaired by the MED and including one representative from each of MoF, MoA, Ministry of Works (MoW), Ministry of Rural Development (MRD), Ministry of Natural Resources (MNR) and the National Climate Change Office. The POC will also approve the Annual Work Plan and Budget (AWPB) and Procurement Plan (PP), and approve the procurement of consultants, goods and works. It will also monitor the progress of Project implementation *via* monthly POC Meetings and reports provided by the PMU. Prior to start-up, the POC will additionally be responsible for the procurement of key PMU staff including the Programme Manager, the Finance Officer and the Monitoring & Evaluation (M&E) Specialist. The overall view of the Organizational Framework is illustrated in Figure 3 below.

IFADs Role

IFADs operational model ensures for continuous implementation support and supervision of project activities providing MED and more specifically the PMU with the required technical and expert guidance in definition of mitigation measures should any issues arise throughout the project cycle. No objection will be required for the approval of the AWPB, which will be screened to ensure programmed activities build towards successful project completion in an effective and efficient manner. Additionally, based on project performance IFAD will, as minimum, annually develop: i) external audits, and ii) supervision missions focused on monitoring and evaluation. Finally, IFAD with support from the MED and PMU will head a mid-term review to ensure project alignment with defined targets or propose corrective measures, and will develop project completion report and impact assessments. The manner and specific requirements for implementation support and supervision missions are fully developed in the Programme Implementation Manual.

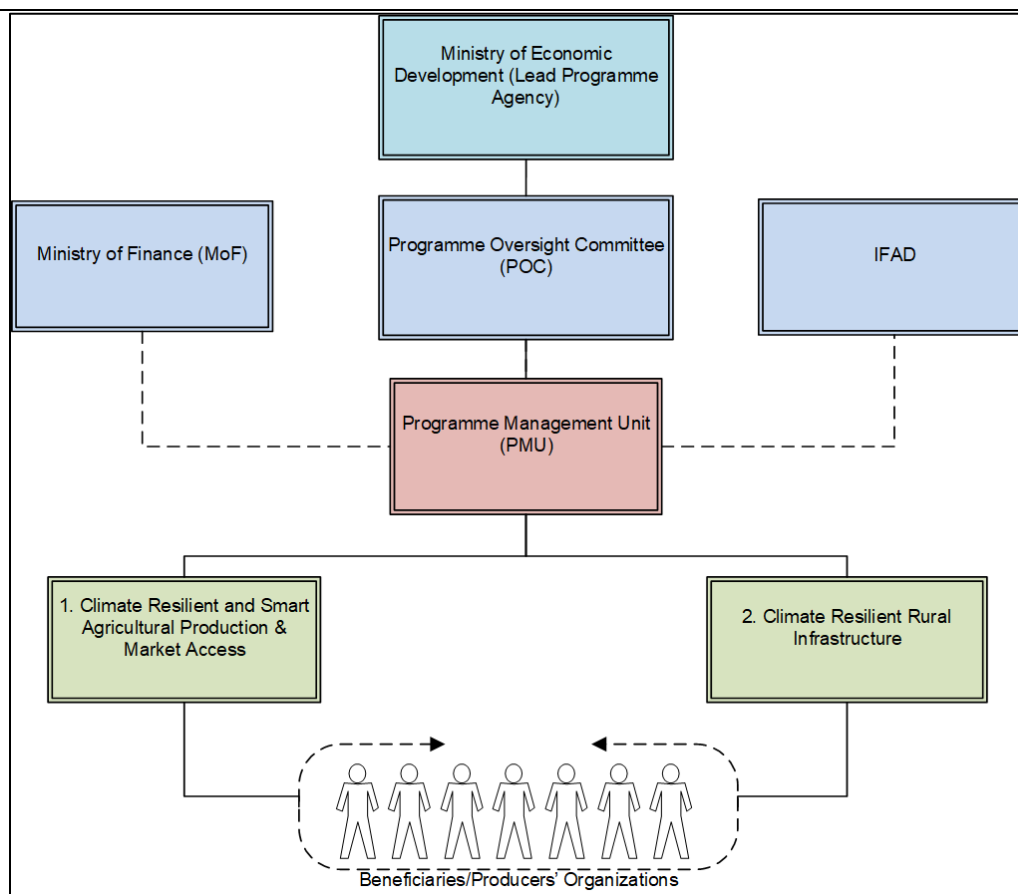


Figure 3. Organizational Framework

Day-to-day management and implementation of the Project will rest with the PMU, which will be anchored to the MED. The principal function of the PMU will be to carry out the overall programming and budgeting and to take the lead in overall Project implementation, which will include working with service providers, Government Ministries and Departments, individual project participants, producers' organizations and the municipalities of the Project's target areas. Specifically, the PMU will assume responsibility for the timely preparation of the AWPBs and PPs to be submitted to the POC for review and approval (and subsequently to IFAD for its "no-objection"). The PMU will lead in the implementation of the Project's two components and will work closely with the MoA and other IPs, to ensure that each component achieves its desired outcomes. In addition, the PMU will be vested with financial and technical autonomy, guided by the PIM and reporting directly to the POC.

The PMU will comprise a total of 11 staff members organized within the structure presented in Figure 4.

- (a) Project Coordination and Administration: Programme Manager, Finance Officer, Programme Accountant, Procurement Officer and Administrative Assistant;
- (b) Monitoring & Evaluation: Monitoring and Evaluation Specialist; and
- (c) Component Implementation: Specialist on Sustainable and Climate Resilient Agricultural Practices, Rural Infrastructure Engineer, Rural Organization Development Specialist, Value Chain Specialist and Administrative Assistant.

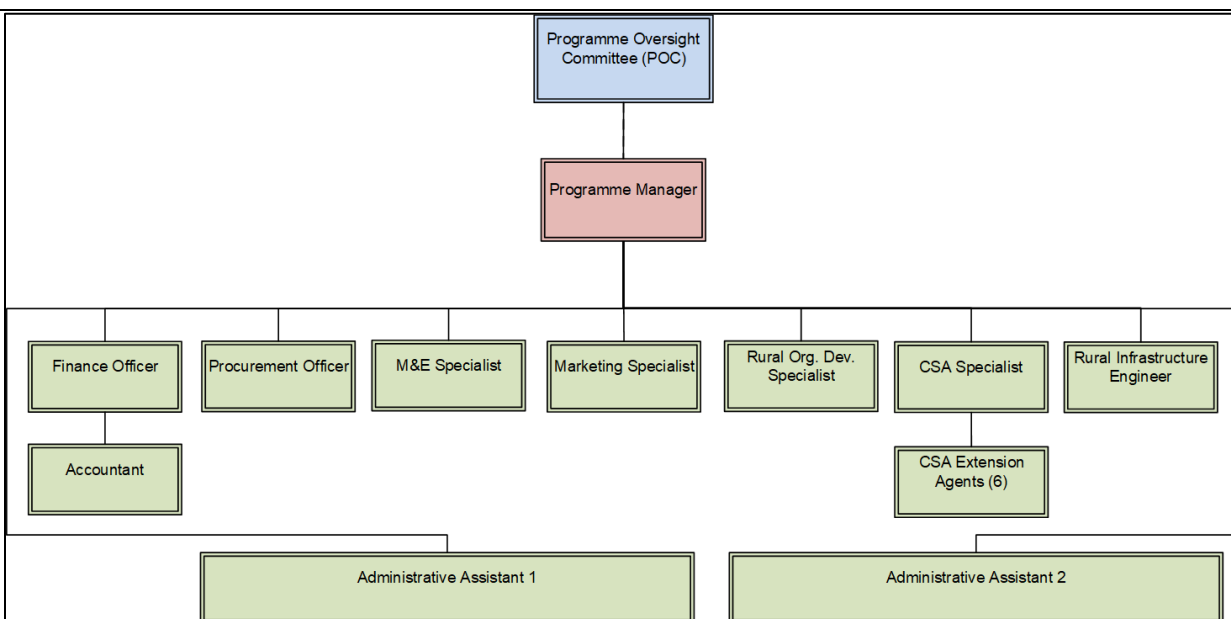


Figure 4. Project Management Unit

Implementing Partners. The success of the Project will depend largely on establishing viable relationships with IPs for the delivery of specialized services to the project participants, including the different governmental stakeholders. Relationships will be established *via* Memoranda of Understanding (MoU) to be signed between the Project and each IP. The PMU will be responsible for the development of the MoUs, the management of the relationships and the coordination of the activities and services to be provided by each IP for the respective components. With respect to project participants and POs, the PMU will facilitate the execution of formal agreements between these organizations and the MGF Trustee. These agreements will detail the responsibilities of each party in the implementation and performance of the MGF.

Funds Flow and Disbursement Arrangements. IFAD and GCF funds will be deposited in two designated accounts in US\$ opened and maintained by MOF, exclusively for the loan and grant proceeds, in the Central Bank of Belize. Only one designated account per financing institution will be opened (GCF loan and grant will be in GCF designated account), from which deposits to the specific operational accounts of the approved funded activities will be made, adequately allocating loan and grant totals accordingly. There will be one or more Operating Accounts in BZ\$ opened and maintained in a bank selected by the Borrower to process payments for the day-to-day activities of the Project. The PMU will submit requests to MOF for the transfer of IFAD and GCF funds from the designated accounts to the Operating Account(s). GOB counterpart contribution will also be deposited to the Operating Account(s) in quarterly tranches. The PMU will submit requests to MOF for the release of counterpart funds as soon as the AWPB is approved. All hiring and investments will be done directly by the PMU after approval of the AWPB. In that sense, the same financial arrangements and flow of funds apply for the MGF and IIFA, the selection and allocation mechanisms for Component 1 and Component 2 respectively. The POs local contribution will be used/exhausted locally; there will be no contribution transfer from POs to IFAD or to the PMUs operational accounts. GCF Loan will partially finance the MGF, and as such, an operational account will be opened and operated by the PMU, where funds will be deposited after approval of AWPB no other activities will be financed by GCF loan. Finally, as both loans are sovereign backed, all repayment responsibilities lie with the Government of Belize, in line with the agreed upon terms and conditions of each institution.

There will be an initial advance disbursement from the IFAD loan account to the corresponding designated account. Subsequent disbursements shall be based upon submission of withdrawal applications, mainly Statements of Expenditure, for at least 30% of the advanced amount. Further disbursement details and procedures are to be included in the Letter to the Borrower (LTB) which will be prepared by IFAD and sent to the Borrower

upon entry into force of the Financing Agreement. Disbursement procedures and requirements relating to the GCF will also be addressed in the LTB; IFAD will be responsible of channeling GCF Proceeds to the designated accounts held by MOF. It is expected that the Project will benefit from the use of the IFAD Client Portal (ICP) platform for submission by MOF of electronic Withdrawal Applications to IFAD for disbursements.

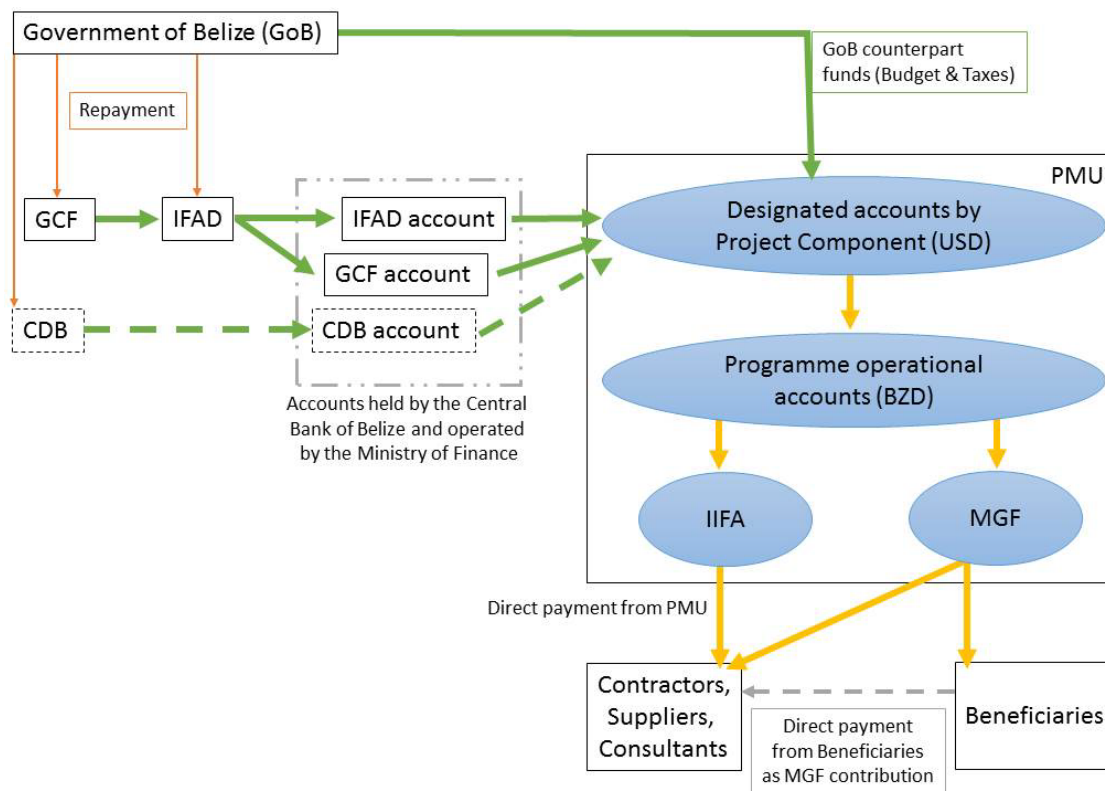


Figure 5. Flow of Funds

A detailed organizational structure, roles and responsibilities of the PMU, supervision and operation methodology with key contractual agreements can be found in Appendix 5.



Please provide a project/programme implementation timetable in [section I \(Annexes\)](#). The table below is for illustrative purposes. If the table format below is used, please refer to the activities as numbered in Section H. In the case of outputs, please mark when all the required activities will be completed.

Project timeline is designed considering activities will be developed on a rolling basis, that allow for simultaneous advances while avoiding bottlenecks throughout implementation, and allowing for investments to commence quickly. Therefore, the timeline considers the full period in which activities (i.e. individual IPPs for specific areas) can be developed and does not represent the start/end of the activity as a whole (IPPs of all areas). In this sense critical milestones have been identified as the completion of: i) and Climate Vulnerability and Value Chain & Market Assessments; ii) Infrastructure and Production Plans; iii) Agreement with CSA Consulting Firm; iv) GOB-PO Business Plan Agreement; v) Rural Roads Agreement; vi) Irrigation and Drainage Agreement; vii) Establishment of Water Users Association; and viii) Project Completion. Additional details including the AWPB for 2019 is presented in Appendix 24 “Detailed Programme Timeline, Critical Milestones and 2019 AWPB”.

[illegible]

2.3. Promote farmer to farmer exchange visits ³⁶				X	X		X	X			X	X			X	X			X	X				
Output 3. Climate resilient production and marketing																								
3.1. Design of Business Plans (BPs)				X	X	X	X	X	X	X	X	X												
3.2. Select and finance sustainable projects through Matching Grant Fund (MGF)						X	X	X	X	X	X	X	X	X	X	X	X	X	X	X				
3.3. Climate Resilient Technical assistance ³⁷				X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X				
3.4. Organize Market oriented meetings ³⁸							X	X			X	X			X	X								
3.5 Establish farmers' markets													X	X	X	X	X	X	X	X				
3.6. Training and TA for backyard gardens							X	X			X	X			X	X								
3.7. Acquire Backyard Gardens Assets						X	X	X	X	X	X	X	X	X	X	X	X	X	X	X				
Output 4. Rural Roads Improvements and Small-scale Irrigation and Drainage (SSID)																								
Activity 4.1 Select CRRIA investments and finance through Infrastructure Investment Funding Award (IIFA)				X				X				X				X				X				
Activity 4.2. Request Environmental Clearance			X	X	X		X	X	X		X	X	X		X	X	X		X	X	X			

³⁶ Includes farmer to farmer exchange visit at the national level, and one international visit to a more advanced producer organization implementing climate resilient agriculture in a neighbouring country.

³⁷ Includes the training of climate resilient agriculture extensionists and the continuous technical support provided by them to project participants.

³⁸ Structured meetings between buyers and producer's organizations to promote market access and establish business partnerships.

Activity 4.3. Design and construct ³⁹						X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Activity 4.4. Institutional development of Water User Associations						X	X	X	X	X	X	X			X	X			X	X				
Output 5. Climate Information System																								
5.1. Design of enhanced climate information system						X	X	X	X															
5.2 Pilot phase of implementation of CIS										X	X	X	X											
5.3 Full implementation of CIS														X	X	X	X	X	X	X	X	X	X	X
Output 6. Government Strengthened																								
6.1. Public consultation ⁴⁰				X	X	X																		
6.2 Invest in R&D ⁴¹			X	X	X	X	X	X	X															
6.3. Develop studies ⁴²		X	X	X	X	X	X	X	X	X														
6.4. Capacity building for MoA ⁴³			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X				

³⁹ Includes all design and construction of roads and irrigation works.

⁴⁰ Focused on identifying and addressing key regulatory barriers constraining climate resilient value chain development

⁴¹ Targeted research supporting sustainable smallholder climate change resilience production and market access

⁴² Four studies will be developed: Agricultural Adaptation Strategies, Index-based insurance, Ecosystem Services Payments, and Smallholder Cane Farmer Production Diversification:

⁴³ Institutional strengthening focused on the Departments of Extension (DE), the Department of Cooperatives (DC), and the Belize Marketing and Development Corporation BMDC.

D.1. Value Added for GCF Involvement

A major objective of the project is to reduce the vulnerability of project participants to the impacts of climate change. The United Nations Framework Convention on Climate Change recognizes the particularly high vulnerability to climate change of SIDSs and supports actions to address the special needs of these states, such as Belize. Despite strong efforts to put in place the necessary policies and strategies for climate adaptation, without GCF involvement to complement ongoing efforts and address gaps, GoB is not in a position to translate these plans into action. Only with involvement of the Green Climate Fund it is possible to:

- Shift to an adaptive pathway that fosters sustainable production systems. The baseline trend is to use more fertilizers and defensive chemicals, which in turn pollute the very environment the farmer is dependent upon, creating a vicious cycle of dependence. Current agricultural practices exacerbate climate change by land conversion, slash-and-burn, and poor manure management. With GCF support, the project will leave locked-in development paths and use an innovative approach in the design for sustainable and climate resilient agricultural practices.
- Tackle climate resilience in a comprehensive manner, by directly addressing the climate induced threats faced by smallholders as well combating poverty and increasing competitiveness and productivity. This is only possible bringing together stakeholders and knowledge from various sectors, as well as merging IFAD and GCF financing. With GCF Funding, the project has the calling power to mobilize the various areas within the GoB (such as MEF, MoF, MoA, MoW), academic sector, financial institutions, buyers as well as the producers organizations.
- Enable the country to implement its National Agriculture Sector Adaptation Strategy to Address Climate Change. This includes both short and long-term measures to address critical gaps in technological developments relevant to crop production, better soil management practices, diversification into drought resistant crops and livestock, and farm production adaptations which include, but is not limited to, land use, land topography and increasing use of low-water irrigation systems.
- Integrate women and youth in the agricultural supply chains. Currently women represent 30% of the formal workers in agriculture and are more likely to be unemployed and underemployed. The project will give special attention to fostering opportunities for women and youth, ensuring participation in POs and improve nutrition and food security.
- Use a proportion of funding for studies that facilitate achieving impacts beyond the Project itself, such as: i) study to explore potential mechanisms for the implementation of an Index-based insurance, and ii) a feasibility assessment and pilot program for Payments for Ecosystem Services (PES).
- Support the investment of resources for Research and Development (R&D) in order to advance further more toward acclimate change resilience, and toward a sustainable and profitable agriculture in Belize.
- Increase capacity in climate resilient agriculture and institutionalize climate resilient agricultural development in national educational institutions and extension services
- With the GCF funding, the Project unlocks potential for replication and scaling up to other Caribbean countries facing similar challenges, thus facilitating paradigm shift.

Belize is particularly vulnerable to hurricanes and tropical storms, which are expected to increase in both frequency and magnitude in the Caribbean. Due to these circumstances, the project is considered to be of high climate risk. As a result, it is impossible to dissociate strengthening livelihoods of the small farmers from increasing their resilience to cope with climate change.

D.2. Exit Strategy

The Project's long-term sustainability strategy in rural areas builds on experiences from the previous IFAD project in Belize, the BRFP. It was designed in close consultation with and involvement of relevant government agencies, including technical departments, as well as farmers' organizations to ensure ownership of the interventions and effectiveness of their impact. The government departments and local communities involved in the project preparation will be leading the implementation. Building on this foundation, the project ensures that the investments and results of the interventions are sustained in the longer-term beyond the project period. Furthermore, the three main elements are incorporated in the project design to ensure the durability and scalability:

- Institutional strengthening and capacity building for integrated, locally owned solutions. The strengthening of cooperatives and farmers' associations as well as institutional strengthening of the District-level structures of the MoA (i.e., cooperatives, agricultural extension) to implement the project at field-level. High level technical capacity building helps to improve extension services up to the level required to achieve adequate adaptive capacities. Local POs are involved to enhance sustainability of the efforts.
- Promoting private sector participation through enterprise development and enabling infrastructure. The project will promote the participation of the private sector through the co-financing required by the MGF and by developing partnerships and market linkages. Thus, entrepreneurship among communities to prepare integrated strategies to absorb and utilize the information provided concerning markets and market opportunities, and translate this into profitable decision-making concerning production. It will build capacity in these rural areas to engage in managing these ventures as enterprises in the communities.
- Knowledge building on the basis of lessons learnt from the implementation process of the Project. A monitoring and evaluation framework (M&E) system and data base-lines will be developed in such a way that M&E will be able to operate in real time from the beginning of project implementation. As communities are incorporated into the project, the GIS would be completed where necessary for all of the land holdings of the community, and that this electronic map would form the first level of information of the base line. Approaches to documenting and communicating experiences and lessons learnt will include a range of methods and tools which have proven to be very effective for training and training reinforcement. Dissemination will be done using a range of methods and platforms, such as capacity building sessions, learning and knowledge sharing events and workshops, as well as multiple media outlets.

The project will shift the way food is produced by smallholders in Belize, from the current business as usual scenario, where the smallholder is an unreliable source vulnerable to climatic and economic shocks. In the post-project scenario, project participants will have a business approach to food production that will be climate resilient, environmentally sustainable, socially equitable and economically feasible. The exit strategy relies on the success of the project itself to improve the livelihoods of project participants and to create an enabling environment at various levels that is supportive of climate resilient agricultural production.

In this section, the accredited entity is expected to provide a brief description of the expected performance of the proposed project/programme against each of the Fund's six investment criteria. Activity-specific sub-criteria and indicative assessment factors, which can be found in the Fund's [Investment Framework](#), should be addressed where relevant and applicable. This section should tie into any request for concessionality made in [section B.2](#).

E.1. Impact Potential

Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas

E.1.1. Mitigation / adaptation impact potential

In the context of smallholder agriculture in Belize, improved resilience is defined as the capacity to minimize the impacts of climatic and economic shocks, enabling farmers to better confront periodic variations and strengthening their capacity to cope and recover in times of extreme stress. This requires significant improvements across the entire value chain, as weaknesses in production systems and practices, fragile and variable physical access to markets, and limited capacity to respond to market demands are all factors that contribute to farmers' vulnerability. The project proposes a comprehensive approach to reducing the smallholders' exposure to climate and economic shocks by promoting climate-smart agricultural production, investing in climate proof infrastructure, supporting producer associations, and strengthening value chains to reinsert smallholders as reliable, competitive suppliers of farm products in the domestic market.

It should be noted that the Project's target group of poor and vulnerable farmers or agricultural workers, is scattered throughout the country and that IFAD together with the Ministry of Agriculture have made the effort of identifying their location, which has resulted in the first group of communities and villages that constitute the five priority areas. Any new community to be phased-in during project implementation will be identified by the PMU, in line with the targeting criteria, the GOB's priorities and the orientation from the Projects Oversight Committee (POC). Please see Section B of Appendix 2 for more detail on the targeting and description of the selected areas.

When possible, equipment requiring energy will be photovoltaic based such as solar panels, solar pumps and solar panels for drying fruits.

E.1.2. Key impact potential indicator

Provide specific numerical values for the indicators below.

GCF core indicators	Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (Mitigation only)	Annual	Not applicable
		Lifetime	Not applicable
	- Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience); - Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Total	Expected direct beneficiaries are estimated at 30,000 people – 12,000 females and 18,000 males. Expected Indirect beneficiaries will be up to 95,296 people, 50% male and 50% female
		Percentage (%)	Expected direct beneficiaries are estimated at 9.30% of the population –

			<i>7% of the female population and 11% of the male population.</i> <i>Expected Indirect beneficiaries will be up to 29.55% of the population</i>
<i>Other relevant indicators</i>	As described in detail in section H.1, other indicators that will be measured include: • Expected number of households reporting at least 30% increase in sales value. • Expected number of rural households engaged in climate vulnerability assessments and trained in climate resilience practices and strategies, improved agricultural productivity and market opportunities. • The number of rural people with strengthened resilience (males and females) • Number of households that have increased resilience and food security through backyard gardens • Expected increase in the number of people (males and females) with access to agricultural water supply ensure water supply during the growing season • Kilometers of roads constructed, rehabilitated or up-graded to all-weather status • Number of regulatory innovations and/or mechanisms for climate responsive planning and development • Number of households reporting adoption of environmentally sustainable and climate new/improved inputs, technologies or practices. • Number of facilities (covered structures, storage facilities) rehabilitated/constructed for climate change • Number of people receiving climate information for decision-making.		
The project participants will comprise male and female inhabitants and small-scale farmers from these households in selected rural communities that have the willingness and potential for improving productivity and access to markets. 3,500 rural households in the villages and communities that make up the priority areas jointly identified by GOB and IFAD for project implementation, and 2,500 households from non-priority areas, to be identified during implementation, thus a total of 6000 households. Of the target population, 40% are poor or indigent (25% and 50% below the poverty line), and 60% are vulnerable (25% above the poverty line), with a high potential of falling below the poverty line in case of climatic or economic shocks. According to the Country Poverty Assessment⁴⁴ poor households size in rural Belize is 5.0 people or more. Thus, we multiplied the 6000 households by the number of people per household to get to 30,000 project participants. Given that: (i) 30% of farmers in Belize are women; (ii) that when farming is a family business, women are engaged in several farming and marketing activities; and (iii) that women are the main responsible for backyard gardening, 40% of all project participants will be women and 60% will be men. The project will boost the rural economy and hence indirectly benefit 23,243⁴⁵ households engaged in farming in the country. The average household size is 4.1 people/household. Thus multiplying 24,243 households per 4.1 people per household 95,296 project participants. Total population size according to the 2010 Census is 322,453			

⁴⁴ Government of Belize and the Caribbean Development Bank. Country Poverty Assessment Final Report. Volume 1 and 2. August 2010

⁴⁵ Belize Population and Housing Census 2010. Country Report.

people, 161,277 males and 161,226 females. Since the population is split evenly between the genres, it is safe to assume that indirect project participants will be equally split among males and females.

Benchmarks

Belize is a small country in terms of Land Area (22,970 km²) and population density (16.1 people per sq. km of land)⁴⁶.

These simultaneous characteristics are difficult to emulate, usually only countries with large areas have a comparable population density as Belize. In the region, IFAD has agricultural development projects in El Salvador (21,041 km² / 342 people per km²), Granada (348 km²/ 318 people per km²) and Cuba (109,884 km²/102 people per km²), benefitting 23,800, 2,472 and 11,500 households, respectively. As discussed in section E.6.1, this project lies within the upper range of IFAD's projects in terms of costs per project participants.

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyze impact beyond a one-off project/programme investment

E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

The scaling up of this initiative can be achieved at a national level and, more broadly, across Caribbean SIDS facing similar constraints in their agricultural development. Since the project reflects a variety of agro-ecozones, smallholder production systems and value chains, the lessons learned can be taken to a wide range of circumstances.

At a national level, the project is designed to follow the strategy laid out in the National Agriculture Sector Adaptation Strategy to Address Climate Change in Belize (2015) by addressing climate risks constraining productive systems of smallholder farmers and strengthening smallholder producer organizations. These PO will in turn become more reliable providers of competitively priced domestic agricultural products to the local market. The market uptake of domestic production will stimulate smallholder farmers to adopt similar business models across Belize, with the potential to reach up to 24,000 farmers. For this to occur, however, the support provided by the project needs to be sustained beyond the project duration, the provisions for this to occur are the following:

- Institutional strengthening of the Ministry of Agriculture. The project is heavily investing in updating the Ministry's agricultural extension services to adopt climate resilient practices across the board. The project will integrate 6 extensionists specialized in climate resilient practices and will provide training to the existing MoA staff.
- Enhancing national technical capacity. The project will work with the University of Belize to consolidate an incipient technical course in agricultural practices that foster sustainability and climate resiliency and ensure that their Central Farms campus serves as a training and knowledge exchange center. This will be supported by a specialized international institution that will work with the University, the Ministry of Agriculture, and the project to introduce state of the art practices, technologies, and methodologies to implement climate resilient agriculture in the context of Belize.
- Policy support. The project is working with the government of Belize in four areas of policy development which will create an enabling environment for smallholder agriculture. A feasibility study for agricultural climate insurance will be conducted. This will be linked to Belize's participation in the Caribbean Catastrophe Risk Insurance Facility, focusing on how delivery mechanisms for smallholders to enhance

⁴⁶ World Bank, 2016 Data found in: <https://data.worldbank.org/indicator/EN.POP.DNST>

resilience can be enhanced. A second analysis will be conducted for establishment of payment for environmental services mechanisms to establish incentives for reforestation and sustainable land management, particularly in idle land. Thirdly, regulatory will be developed with the National Integrated Water Resource Authority to address gaps regarding responsibility for maintenance and support of water user associations for small scale irrigation. Finally, a study on smallholder livelihood diversification will be conducted, focusing primarily on the Corozal district, one of the potential districts for project expansion, which is currently primarily focused on sugarcane production.

- Provision of public goods. The project will establish a climate information system which be linked to customized seasonal agronomic advisories and will feed into the national system, with national dissemination via TV and radio. Furthermore, the project will collaborate with the Public Works Department in the design and construction of climate resilient roads and efficient small scale irrigation, leading the way towards revised design criteria and regulations for infrastructure in a changing climate.

The above actions are meant to create an enabling framework for resilient smallholder agriculture, with the primary driving force being the economic inclusion of smallholders. The investment support provided by the project will reduce the recurrent losses and recovery costs that currently keep smallholders as marginal actors in agricultural value chains. This model can be scaled up in the Caribbean region, as the comprehensive assessment, integration, and measurement of climate risk is a central aspect of agricultural rural development investments.

E.2.2. Potential for knowledge and learning

The main objective of knowledge management is to provide stakeholders with knowledge generated from Project implementation that can serve as inputs for scaling-up strategies and for policy discussion and development. It will be led by the Management and Evaluation (M&E) Specialist and will start with the development of a Knowledge Management Plan (KMP) during the first year of implementation. The KMP will encompass strategies and plans for the consolidation of knowledge information and its dissemination to Project participants and interested stakeholders. Approaches to documenting and communicating experiences and lessons learnt will include a range of methods and tools, such as the Project's website, print media (brochures and booklets with case studies, policy briefs, etc.), photographs, audio and video documentaries, which have proven to be very effective for training and training reinforcement. Dissemination will be done using a range of methods and platforms, such as capacity building sessions, learning and knowledge sharing events and workshops, as well as multiple media outlets (e.g. print publications such as the Agriculture Report, newspapers, media broadcasts and social media – Facebook, YouTube, Pinterest, Instagram). In addition, through the MOUs for establishing relationships with IPs such the UB's Faculty of Agriculture, the Project will be able to establish continuity in the dissemination and promotion of best practices and lessons learnt to project participants and to the wider community. KM products such as videos and literature will be supplied to the University Library so that information continues to be available for students and other interested parties to use as resources in their training and the development of their farming practices.

E.2.3. Contribution to the creation of an enabling environment

The project is designed to enhance the complementarity between public and private actors to support domestic agriculture in Belize. The underlying premise is that, with appropriate public support, the conditions exist to align private sector (producers, intermediaries, processors, and end purchasers) incentives in domestic agricultural value

chains. Addressing climate risk is a central aspect of this proposition as the exposure of the agricultural sector is high, with both public and private actors requiring a shift in their current practices.

The private sector, in particular agricultural producers, will be supported in reducing the impacts to climate exposure that undermines their productive activities. They will be supported in understanding the climate problem as reflected in their landscape, identifying their vulnerabilities and exposure, developing a territorial level response that acknowledges the need for community level action to enhance resilience, and identifying their investment needs for resilient production. While the end product will be direct project support to co-finance such investments, the methodology is equally important to allow producers to understand that climate risks must be internalized to become viable producers.

The public sector has an equally important role to play in incorporating climate risk into its approach to agriculture and rural development. First and foremost, climate risk will be fully integrated into the Ministry of Agriculture's support services, enabling producers to have access to the required technical know-how for sustainable farming. The provision of public services and infrastructure will be strengthened to ensure they are more resilient to climate exposure, be it through improved design parameters for roads and irrigation or through the establishment of essential climate information services. The development of a stronger policy environment will be supported through targeted studies in the key areas of climate insurance, environmental services, livelihood diversification, and water use regulation.

The demand driven approach supported by the project to drive climate resilient investment is an innovative approach to link private and public sector interests. Through public institutions, the project will provide the knowledge, technical support, and public infrastructure and services required for producer's associations to invest in climate resilient business models. While the project will also provide financial support to these investments, the producers will be the primary drivers of implementing such business plans and ensuring that they serve as a catalyst for strengthened access to markets. In this model, the public sector assumes the role of an enabler while the private actors take the lead and the responsibility in enhancing their livelihoods, which is an appropriate balance for enhanced prosperity and resilience.

Equally important to the methodological approach applied by the project is the introduction of new practices and technologies to Belizean farmers. The project will support cost effective investment, some of which will require additional upfront financing but will be more adaptive to climate fluctuations. Likewise, climate resilient practices to be adopted will focus on increasing the stability of production, adopting sustainable land management, and balancing commercial production with self-sustainability and diversification. Please see section F.2 for the description of the climate resilient practices to be applied in this project.

E.2.4. Contribution to regulatory framework and policies

This is the first project that will directly follow the strategy laid out in the National Agriculture Sector Adaptation Strategy to Address Climate Change in Belize (2015). By applying for the first time in Belize, a climate resilient approach to agricultural value chain development, it enables the country to prove the business case for sustainable and resilient agricultural practices. The government will benefit from the tools and knowledge to implement their strategy to make agricultural practices climate resilient and manage risks. The government will be able to perform a complete overhaul of the country's approach to agricultural extension services, shifting away from a focus on "green revolution" techniques and a narrow focus on productivity, which has exposed smallholders to increased climate risk. It also fosters climate proof rural investment, an innovative approach for Government that acknowledges and quantifies the additional cost of public investments that reduce exposure to climate related risks.

There are a number of policy bottlenecks to private sector development which have been identified by the project design team and partners, the most notable of which are import tariffs and permits, quotas and out-dated laws and regulations such as the Law on Cooperatives. Funding will be available to finance public consultations with a broad set of national and local stakeholders about the barriers to private sector development in applicable value chains.

These activities may also generate the need for additional studies / evidence, e.g. on specific regulatory or legal changes to laws such as the Law of Cooperatives or strengthening of the coordination between government institutions as regards illegal imports, increase of surprise controls, and enforcement of laws and regulations.⁴⁷ Complementary to the Policy Engagement activities, the program will also contribute with the following studies to help the government advance some key areas such as climate related insurances, environmental services, and production conversion:

- **Index-based insurance:** This study will explore and compare potential mechanisms to link vulnerable small farmers from Belize with the existing fund, with the aim of allowing small farmers to invest in increasing their resilience. Potential mechanisms include an index-based insurance or conventional indemnity insurance. While the implementation of such mechanism is beyond the scope of the proposed Project, additional funding could be pursued from partners like CBD and GCF for this purpose.
- **Payments for Environmental Services.** This study will conduct a feasibility assessment of establishing a pilot project on short-term Payments for Ecosystem Services (PES) as an economic incentive to stimulate the maintenance and restoration of native forests (mature, secondary, or riparian), which offer important environmental services but do not generate revenue for landowners. The study will assess the interest of landowners to participate in a pilot project, estimate payments per acre and costs of the entire pilot project, and potential partners and sources of funding. The study will identify core areas to establish connectivity corridors⁴⁸.

E.2.5. Theory of Change

The end goal of the Be-Resilient project is to build overall resilience to climate change by increasing and diversifying agricultural production by small farmers, and by facilitating their access to commercial market chains for the off-take of their surplus production. Figure 6 presents the project theory of change and the objectives to be developed through the two complementary and mutually reinforcing components, and supported by cross cutting activities that will allow to fill any gaps to successfully achieve the proposed paradigm shift.

Project outcomes in this Theory of Change must be understood as the result of an optimal development of capacity and resilience building activities assembled through the projects components explained in detail in section C.3 and represented in summary in the green boxes. These will provide the required tools to create the necessary conditions represented in red to achieve project outcomes and end goal. In themselves, each of these activities allow for a more informed and better prepared rural agricultural sector, with increased climate sensitivity and knowledge allowing for a sustained increase community resilience. Developed in close coordination through a participatory and holistic approach project outcomes and all the work done to ensure them, will allow for a successful paradigm shift in the countries rural poor agricultural sector.

⁴⁷ As regards legal imports, farmers' organizations' agreements with MoA (borders closed when there is sufficient local production) will need to be replicated and fully complied with in a timely manner. To this effect, MoA will be sensitized to the damage caused by non-timely enforcement of protocols, and support will be provided for their efficient implementation.

⁴⁸ Connectivity corridors are "stretches of tree or shrub vegetation connecting fragments of natural ecosystems through riparian strips, pastures with high tree density, living fences and other elements of the landscape". They may include buffer strips along streams and rivers, or agroforestry systems (scattered trees in crops and pastures and windbreaks).

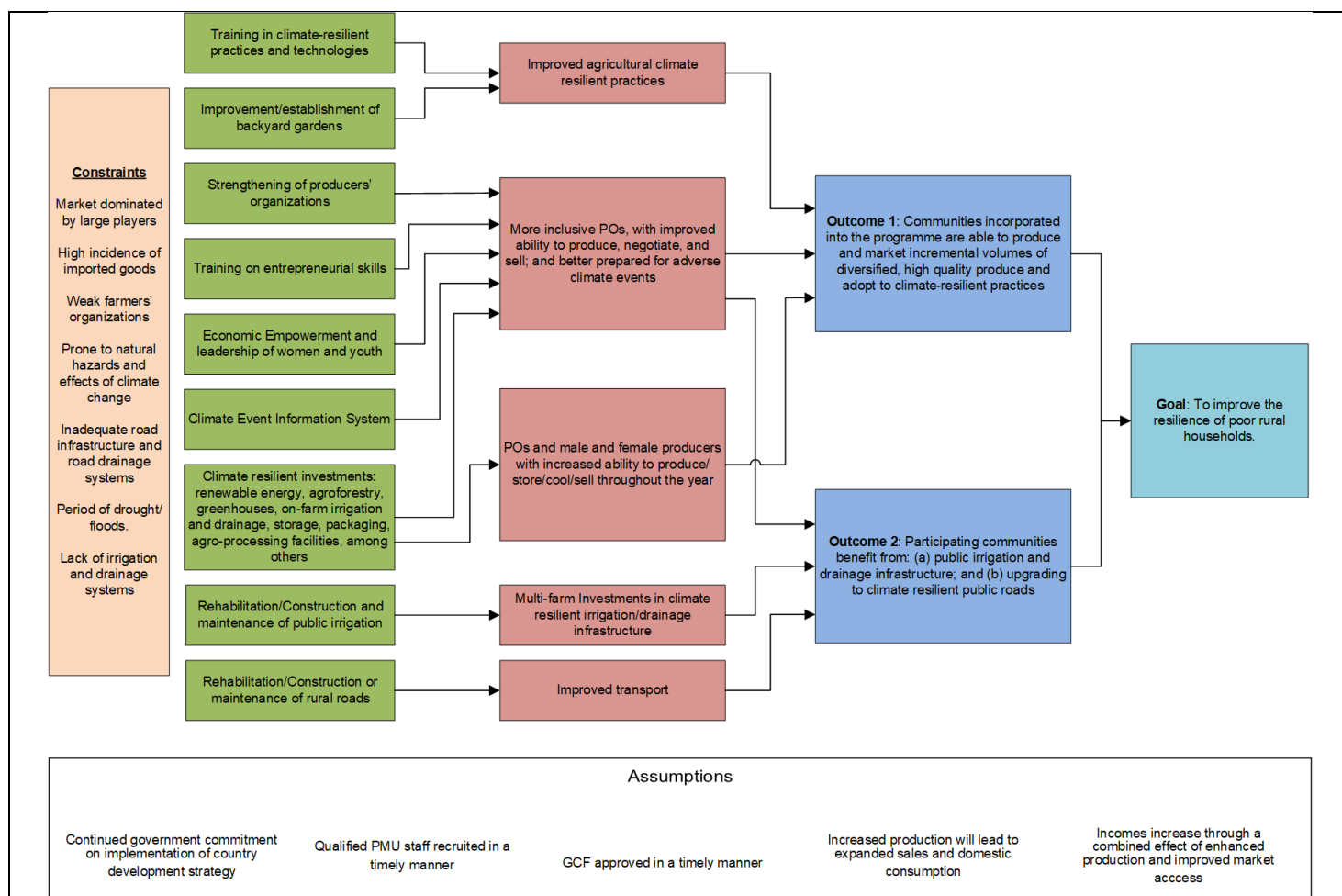


Figure 6. Theory of Change

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

The Project will yield many benefits, at the micro and macro levels, both direct and indirect. It will enhance the agricultural production of 6,000 smallholder farmer families as well as incomes through improved access to irrigation, adoption of climate resilient cropping patterns, and market linkages. The economic, social and environmental benefits are described below.

Economic

There will increased income of direct project participants as well as employment opportunities created as the local economy is stimulated by the activities of the Project, such as road infrastructure and agricultural facilities (milling, storage). There will be macro level indirect economic benefits derived by the contribution to food security and self-sufficiency, thus saving on potential imports. Reducing the vulnerability of the farming households will also reduce the likelihood of the households needing social protection/safety net pay-outs.

Social

The Project proposes to substantially improve decision-making among farmers, as they become active stakeholders in developing climate resilient agriculture and catering to local needs. Their capacity will be improved through training and engaging in implementing Project activities and interface with government officials for service provision and private sector for market negotiations. By working in partnership with other community groups and by acknowledging their existing knowledge and mobilization potential, the Project will create significant social capital. Collective decision making and the climate information system will enable advances in social and inter-community harmony. By improving food security and promoting sustainable agriculture; it may also lead to an improvement in nutrition of farm families via the diversification of output and increased disposable income. Institutional strengthening of producer's organizations will target inclusion of women and young people.

Project activities will specifically target women farmers, both in productive agriculture and, most directly in household level production and consumption. The project's training and technical assistance activities will engage young female and male farmers, as they tend to be more open and willing to learn and adopt new practices. Project indicators will be gender-disaggregated as appropriate, with an overall target of minimum 40% female participants in the project.

Distress economic migration is particularly acute among rural youth. Thus, the project will target the rural youth offering training, employment and empower them in the local producers organizations. While most of the Belize's food is produced by ageing smallholder farmers, older farmers are less likely to adopt the new technologies needed to sustainably increase agricultural productivity, and ultimately feed the growing world population while protecting the environment. Thus by providing rural youth with opportunities, the project not only curbs urban migration, but also maintains a vital and dynamic share of the workforce to give impulse the agricultural production.

Environmental

The promotion of sustainable and climate resilient agricultural practices will improve soil management and restoration. No expansion of agricultural land use is expected. The Project will foster environmentally-sound pesticide and fertilizer management techniques, including the application of organic alternatives as applicable. The Project will also have a positive impact on helping to promote sustainable communities and on "life on land" issues by promoting the sustainable use of natural resources and by reducing land degradation. It will have a modest impact in the area of affordable and clean energy; communities "off grid" will be encouraged to adopt photovoltaic panels and battery storage for domestic use.

The project will ensure that all infrastructure investment is fully compliant with IFAD and national environmental standards. In particular, any water harvesting and/or harvesting investment will be designed not to interfere with ecological water flows or natural drainage of water bodies. Should drilling wells be required, sound environmental practices will be respected to avoid cross contamination of aquifers and unsustainable extraction of groundwater. Life-cycle waste management practices will be applied in agro-processing facilities, with recycling and/or waste to energy facilities implemented where appropriate. Climate-resilient road rehabilitation will reduce interference with natural drainage and minimize erosion.

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

Several studies (BNMS, 2008; and GOB, 2002) have noted that Belize, on the whole, is highly vulnerable to Climate Change. They noted further, that the coastal zone of Belize, where a significant percentage of the population is located and where the bulk of economic activity takes place (tourism and agricultural production), is for the most part below the high tide level. As a result, this places the coastal zone in a very precarious position with regards to climate-driven sea level rise, especially when augmented by storm surges (CZMAI, 2013). Higher temperatures will adversely affect economically important crops in Belize such as sugarcane and citrus. Furthermore, warmer weather from high temperature will cause soil aridity, lead to proliferation of pests and diseases, and put pressure on water resources for water irrigation purposes. Sea level rise will cause flooding and saline intrusion and soil salinization. Rain-fed agricultural production will be affected, demanding improved management techniques and consequently increasing the cost of production. The combined impact is low agricultural yields, decrease in food production and higher food prices. Hence, there is a clear need for additional investment resources to confront the expected impacts of climate change and the additional costs this entails.

The target population of the project is amongst the most vulnerable in the country. They are highly exposed to climate risk and have low adaptive capacity. The limited government support this population receives is not geared towards increasing resilience; hence, they are constantly subject to social and economic losses due to climate induced events. The recovery process from such shocks is slow and painful, and does not allow them to prepare for future events. Without a significant overhaul to the country's approach to rural development, smallholder farmers will continue in a vicious cycle that reinforces their vulnerability.

E.4.2. Financial, economic, social and institutional needs

The macroeconomic context of the country is an additional critical factor. GDP growth has slowed substantially from 4.1% in 2014 to 1.0% in 2015 and -1.5% in 2016; projected growth in 2017 is a modest 1.9%. As described by the IMF in its most recent Article IV report, the economy faces multiple challenges: the macroeconomic outlook remains weak, public debt (at 100% of GDP) is elevated and medium-term economic growth is projected at less than 2% per year. In addition, the effective exchange rate appreciated since 2014, given the peg of the Belize dollar to the US dollar, reducing export competitiveness. Shocks in the export sector, the result of lower oil and agricultural prices, have widened the current account deficit of the balance of payments from 1.2% of GDP in 2012 to a projected peak of 14.0% in 2016 and 10.2% in 2017, with a consequent reduction in international reserves.

In the absence of a change in policies, rigid fiscal spending particularly concentrated in the public sector wage bill will continue to drive fiscal deficits and crowd out public investment. In that context, it is particularly difficult for the country to invest in rural development and to absorb the additional costs borne by climate induced shocks. As noted above, the aggravation of climate change is generating annual average losses 2.87% of GDP.⁴⁹

The MoA will have a key role during the implementation of the Project and responsibility for the sustainability of its achievements. The institutional strengthening of the MoA to provide technical services for small farmers' development is an important aspect of the Project not only for the purpose of sustainability but also to introduce new concepts and skills and to strengthen existing capacities in the DE, the DC and the BMDC.

⁴⁹ <https://germanwatch.org/de/download/16411.pdf>

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

The Government of Belize (GOB) has taken initiatives to mainstream Climate Change into its national development processes and mechanisms. These initiatives include the establishment of a National Climate Change Office under the Ministry of Forestry, Fisheries and Sustainable Development (MFFSD), the adoption of climate sensitive response mechanisms for some economic development sectors, and the preparation of a comprehensive National Climate Change Policy, Strategy and Action Plan which will provide policy guidance for the development of an appropriate administrative and legislative framework, in harmony with other sectorial policies, for the pursuance of a low-carbon development path for Belize.

As discussed in section C.2, the Project will follow the strategy laid out in the National Agriculture Sector Adaptation Strategy to Address Climate Change in Belize (2015) to foster resilience in a comprehensive manner, by directly addressing the climate induced threats faced by smallholders as well combating poverty and increasing competitiveness and productivity. The NDC considers agricultural resiliency as a main national strategy listing the following actions to be implemented to increase resilience in the agricultural sector: Improve both crop and livestock husbandry practices, increase access to drought tolerant crops and livestock breeds; adopt better soil and water management practices (addressed under Component 1 of the Project); reduce post-harvest losses and provide early warning/meteorological forecasts and related information to be competitive in the region (addressed under Component 2 of the Project).

The cross cutting nature of Climate Change means that several sectors not only are exposed to the impacts of climate variability and sea level rise, but also have a role to play in the management of policies being designed to address the problems identified. In recognition of this imperative, the GOB established, in 2009, the Belize National Climate Change Committee (BNCCC) and in 2012 a pro-tem National Climate Change Office (NCCO) in the Ministry of Forestry, Fisheries and Sustainable Development (MFFSD), with the assigned responsibility of coordinating the country's national, regional and international response to Climate Change and ensuring the development of a consistent framework across line ministries and agencies for mitigating the effects of Climate Change. The NCCO is also charged with the responsibility of coordinating Belize's external response to the various reporting and other requirements of UNFCCC and attracting or sourcing funds to support the national effort.

Notwithstanding the establishment of the NCCO, it is also recognized that there are several institutions whose functions and ministerial responsibilities are critical for the effective implementation of Climate Change.

The proposed Project forms part of a broader strategy for the agricultural sector in which many international institutions are involved, such as FAO, EU, IDB, UNDP, and the Caribbean Community Climate Change Centre (5Cs). The proposed Project would be working with other donors and private sector initiatives and building on the framework of a national strategy closely coordinated by the Government. The project management will be carried out by a project management unit (PMU) within the Ministry of Economic Development (MED).

E.5.2. Capacity of accredited entities and executing entities to deliver

IFAD has a long lasting history within Central America and the Caribbean dating back to 1978. Projects in the region have mainly focused on markets, business and value chains, yet have since evolved into further and much more regional specific development with a strong focus on resilience building, particularly in SIDS. The Caribbean region is one of the most vulnerable to economic downturns and climate change, demanding project planning to profoundly incorporate mitigation and adaptation measures in all project development; disaster risk management, climate resistant crop varieties, insurance schemes, and livelihood diversification are some of the most common examples, as well as constant capacity building and policy support. IFADs approach to Small Island Developing States, where the Caribbean region holds the largest percent of funding, mainly promotes: i) a reef-to-ridge approach to ecosystem conservation; ii) an integrated vision for disaster risk management and climate change adaptation; iii) poor people's access to land and the management of increasingly scarce natural resources, while reducing the potential for conflicts; iv) development of renewable energy sources capable of capitalizing on the islands' natural resources and reducing the significant costs associated with imports of fossil fuel; and v) appropriate insurance schemes.

Between 1978 and 2013 the Caribbean has received over \$275 million USD in funding for projects, making it the largest recipient within the SIDS; over \$75 million have been disbursed after 2005 benefiting over 530 000 people during this period. Within the LAC division, for the period ending in 2015, the Caribbean region reported 10 active country projects in Belize, Cuba, Dominican Republic, Grenada, Guyana, and Haiti. Additionally, a regional project developed by FIDA-Caribe in collaboration with FAO and ILO is underway; working on Youth Rural Employment Opportunities. The first stage is being developing Belize, Cuba, the Dominican Republic, Grenada, Guyana and Haiti while estimating impact to extended to youths from Barbados, Jamaica, Saint Lucia, Suriname, and Trinidad and Tobago. Even though IFADs role in the region, as compared to other International financial institutions, is a small one, its expertise in rural communities and smallholder agriculture plays a large advantage in the potential impact of projects due to its focus on the most vulnerable communities while promoting participatory approaches, gender equality, and market driven development that will not depend solely on international grants.

IFAD has financed three projects in Belize, the most recent of which, the Belize Rural Finance Programme (BRFP), had a strong focus on rural finance and did not directly involve agricultural value chains or rural infrastructure. Consequently, the extent to which lessons learned are available is limited. The principal lesson learned from the BRFP is that the Programme should make the most extensive use possible of partnerships, focusing on the GCF/GoB as co-financiers, and on other development partners, especially the FAO and the CDB, as sources of knowledge for the planned activities.

The three operations funded by IFAD since 1986 were: the Toledo Small Farmer Development (TSFD) Project (1986-1995); the Community Initiated Agricultural and Resource Management (CARD) Project (1999-2005); and the BRFP, a seven-year national programme that was completed in September 2016. The BRFP was approved in 2008 for US\$6.04 million including US\$3 million IFAD financing, co-financing by the Central American Bank for Economic Integration (CABEI) and the GoB. According to the Programme Completion Report (PCR) for the BRFP, the programme was a success in achieving its principal objective of putting in place an institutionally-driven and sustainable system of increased direct access to financial services for the rural poor in Belize.

The Project Management Unit will be an autonomous unit within the Ministry of Economic Development (MED). Section C.4 and Appendix 5- Institutional Aspects and Implementation Arrangements detail the institutional arrangement of the project.

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

The proposed project is informed by the several rounds of discussions with stakeholders at the national and local levels on climate change adaptation options and priorities in Belize. Three (3) Design Missions took place between April and November 2017. During these missions, IFAD staff and consultants met with potential project participants, Representatives of Producers' Organizations in all project areas, government representatives, non-governmental organizations, community leaders, educational institutions and other organizations that are engaged with developmental activities in Belize. During the field visits to targeted communities, discussions were held with stakeholders (community leaders, farmers and farmers' cooperatives and organizations, including female and young producers) to better understand their socio-economic situation, their production systems, competitiveness of the products in Belizean markets, the commodities with good production and market potential, farmers' constraints and their access to markets.

The GoB has been working closely with the project team from the beginning of project preparation, with staff of the MoA participating in many of the field visits. A participatory workshop was carried out with key staff of the MoA (including staff of the Department of Extension (DE) and Department of Cooperatives (DC) to discuss the strategic priorities of the Government for agricultural development and rural development, and to further define the commodities to be targeted by the Project. Representation of women was ensured through the participation of community-level organizations as well as by including female staff members of the relevant partner agencies.

The first Mission was carried out between April 24 and May 5, 2017. The purpose of the Mission was to understand the needs of the GoB, ensure IFAD could adequately respond and ensure the farmers organizations needs were included in the project design. The following is a list of stakeholders were consulted during the Inception Mission:

Government 1. Ministry of Economic Development 2. Ministry of Agriculture 3. The National Climate Change Office 4. Department of Cooperatives in the Ministry of Agriculture	International Financial Institutions 1. Inter-American Development Bank 2. United Nations Development Programme in Belize
Non-Governmental Organizations 1. Belize Enterprise for Sustainable Technology 2. Toledo Institute for Development and Environment 3. Caribbean Community Climate Change Center	Potential Project participants 1. Representatives of Producers' Organizations in the communities of: a. San Carlos, Orange Walk b. Nago Bank, Belize c. Valley of Peace

Private Sector Institutions	Quasi-Government
1. Belize Tourism Industry Association 2. Sugar Industry Research and Development Institute 3. Praxi5 Advisory Group	1. Development Finance Corporation 2. Public Utilities Commission 3. Belize Trade and Investment Development Service

By the end of the Inception Mission, four (4) priority areas were identified as potential targets for project intervention. In addition, a number of activities were identified to meet the needs of the priority areas, encompassing climate resilience and improvement in agricultural yield and access to markets. A preliminary Concept Note was developed, which provided a description of the intervention activities, including a preliminary identification of value chains to be targeted for development.

A second mission, the Formulation Mission, was undertaken from July 26 to August 11, 2017. The following is a list of stakeholders with whom consultations were carried out during the Formulation Mission:

Government	International Financial Institutions
1. Ministry of Economic Development 2. Ministry of Agriculture 3. Ministry of Finance 4. The National Climate Change Office 5. Department of Cooperatives in the Ministry of Agriculture 6. Ministry of Works 7. Statistical Institute of Belize 8. Ministry of Rural Development 9. Hydrology Department	1. United Nations Development Programme in Belize 2. Food and Agriculture Organization
Non-Governmental Organizations	Potential Project participants
1. Caribbean Community Climate Change Center 2. Maya Leaders Alliance 3. Toledo Alcalde Association	1. Representatives of Producers' Organizations in the communities of: a. San Carlos b. Nago Bank c. Valley of Peace d. Trio e. La Gracia f. Cow Pen g. Bella Vista 2. Independent discussions were held with women in the following communities: a. Valley of Peace b. Nago Bank c. Trio

Private Sector Institutions

1. Belize Roadway Construction
2. Anthony Thurton and Associates
3. Belize Tourism Industry Association
4. Marie Sharp's Foods

Quasi-Government

1. Development Finance Corporation
2. Belize Marketing and Development Corporation

Education Institutions

1. University of Belize – Faculty of Agriculture

By the end of the Formulation Mission, a preliminary design of the project was completed. This design included a description of the project targeting, an update to the project target areas, rationale for the project, a detailed description of project interventions, implementation arrangements, social, environmental and climate assessment procedures note (SECAP) and the project budget. A GCF concept was finalized and submitted.

The third and final mission, the Appraisal Mission, was undertaken from October 29 to November 4, 2017 to address the concerns raised by the quality enhancement review and GCF Secretariat comments, and to fill whatever gaps that was identified from the formulation mission. The following is a list of stakeholders with whom consultations were carried out during the Appraisal Mission:

Government

1. Ministry of Economic Development
2. Ministry of Agriculture
3. Department of the Environment

Education Institutions

1. University of Belize – Faculty of Agriculture

Non-Governmental Organizations

1. Representatives of Producers' Organizations in the communities of:
 - a. Valley of Peace

At the conclusion of the Appraisal Mission, the mission was able to address the concerns raised by the quality enhancement review and had undertaken to prepare the final draft of IFAD's Project Design Report (PDR) (Appendix 15) for submission to Quality Assurance. Appendix 19 presents the discussions and deliberations at the final public consultation workshop as well as the list of attendees.

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

The proposed financing structure is a mix of loans and grants that will tackle short and medium term objectives for a resilient agricultural production. The provision of the GCF loan/grant is of critical importance to provide the initial funding for a paradigm shift that enables smallholder producers to penetrate the market.

In Component 1, GCF Fund will be geared towards three main goals: 1. Strengthening the GoB capacity to respond to the climate challenges, 2. Ensure adequate climate vulnerability assessment to drive the project's investments; 3. Reducing the risk premium of smallholders to access funds to be more climate resilient. The vulnerability of the smallholder producers to climate change and their low recovery capacity to climate events require concessional loan financing to reduce the risk premium to make the investment viable. Furthermore, the GoB and smallholder producers will also necessitate extensive technical assistance to identify climate vulnerabilities and implement sustainable practices that promote climate resiliency. There is very little international funding available for the types of activities proposed and where it does exist (or if it becomes available during the Project lifetime) the Project will work towards ensuring complementarity. IFAD's past project in Belize helped develop rural financial services specifically targeted for community-initiated agriculture. The proposed project will promote local public and private investments since smallholder farmers will be required to present a co-finance to access the MGI resources. Most of the groups don't have capital to provide the co-financing requirements and thus will have to turn to local Credit Unions for the resources. Therefore, this project has been designed to crowd in potential private and public finance once the existing barriers are sufficiently removed.

In Component 2, GCF funds will cover additional costs imposed on infrastructure by climate variability and increased frequency of extreme climate events in addition to developing a nationwide climate information system. Thus, GCF funds will permit roads to be climate proof, increasing durability and allowing year-round access of smallholders to markets. In addition, the small irrigation schemes will provide smallholders increase resilience to longer drought periods and increased rainfall variability, extending production and productivity.

IFAD's project costs from USD\$1000-5000 per household project participant. Other similar projects to promote smallholder agriculture in the region vary from USD3500/household in Cuba to \$1300/household in Granada. The proposed project costs \$3,333/household, however, it is the only one which has significant investment in climate resilient infrastructure in addition to capacity building and institutional strengthening.

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

Please provide the co-financing ratio (total amount of co-financing divided by the Fund's investment in the project/programme) and/or the potential to catalyze indirect/long-term low emission investment.

Please make a reference to [E.6.5 \(core indicator for the expected volume of finance to be leveraged\)](#).

Not Applicable

E.6.3. Financial viability

To verify the financial viability of the project, the Annual Net Incremental Benefit Flow generated over a 15-year period was calculated for the project participants who are incorporating technological improvements in their production system promoted by the project (from the second year of execution). The Project was economically viable, since the economic analysis indicators that were estimated are positive: (i) the IRR is 11.64%; and (ii) the NPV discounted at 6% is BZ \$ 14.0 million. This level of return could be considered to be intermediate. Please see Appendix 10 for more details

E.6.4. Application of best practices

Section F.2 describes what is currently considered the best available technologies and practices to improve agricultural and infrastructure resiliency to climate change. In order to ensure the best available technologies will be applied, two institutions at the forefront in research and praxis on sustainable agriculture will be working in partnership with the PMU to conduct the training. The Project will execute an MOU with UB's Faculty of Agriculture, the leading local training institution on sustainable and climate resilient practices. UB will lead the training and capacity building efforts and it will be supported by a prominent international institution such as CATIE (to be procured by the PMU), CIAT and/or 5Cs that will enrich the training program with additional expertise and applied research from abroad. The Center for Training and Research in Tropical Agriculture (CATIE) located in Costa Rica is an international institution that has solid experience working on agricultural practices that promote sustainability and climate resiliency in Meso-America and other Latin countries, has developed methodologies on agricultural extension and has considerable experience in training extension staff, farm leaders and promoters in Latin American and Caribbean countries. The International Center for Tropical Agriculture (CIAT) is headquartered in Colombia and has 21 offices and field operation sites around the world, to develop technologies, methods, and knowledge that better enable farmers in developing countries to enhance eco-efficiency in agriculture. The Caribbean Community Climate Change Center (5Cs) is headquartered in Belize coordinates the Caribbean region's response to climate change and is the key node for information on climate change issues and on the region's response to managing and adapting to climate change. Thus, by contracting the best international experts, the project assures best practices and the latest available research methods will be applied.

E.6.5. Key efficiency and effectiveness indicators

<i>GCF core indicators</i>	Estimated cost per t CO₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)										
	<table> <tr> <td>(a) Total project financing</td> <td>US\$ _____</td> </tr> <tr> <td>(b) Requested GCF amount</td> <td>US\$ _____</td> </tr> <tr> <td>(c) Expected lifetime emission reductions overtime</td> <td>_____ tCO₂eq</td> </tr> <tr> <td>(d) Estimated cost per tCO₂eq (d = a / c)</td> <td>US\$ _____ / tCO₂eq</td> </tr> <tr> <td>(e) Estimated GCF cost per tCO₂eq removed (e = b / c)</td> <td>US\$ _____ / tCO₂eq</td> </tr> </table>	(a) Total project financing	US\$ _____	(b) Requested GCF amount	US\$ _____	(c) Expected lifetime emission reductions overtime	_____ tCO ₂ eq	(d) Estimated cost per tCO₂eq (d = a / c)	US\$ _____ / tCO ₂ eq	(e) Estimated GCF cost per tCO₂eq removed (e = b / c)	US\$ _____ / tCO ₂ eq
	(a) Total project financing	US\$ _____									
	(b) Requested GCF amount	US\$ _____									
	(c) Expected lifetime emission reductions overtime	_____ tCO ₂ eq									
(d) Estimated cost per tCO₂eq (d = a / c)	US\$ _____ / tCO ₂ eq										
(e) Estimated GCF cost per tCO₂eq removed (e = b / c)	US\$ _____ / tCO ₂ eq										
<i>Describe the detailed methodology used for calculating the indicators (d) and (e) above.</i> <i>Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.</i>											
Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only) Not Applicable											
Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)											

** The information can be drawn from the project/programme appraisal document.*

F.1. Economic and Financial Analysis

The project's financial and economic viability is highly sound. The Fund's intervention and support is vital, as the economic modelling carried out shows high economic and financial viability of the interventions.

Financial analysis The integration of the incremental costs and benefits in the total cost of the Project for all the models provides a net cash flow from which the following efficiency indicators were obtained: an Internal Rate of Return (IRR) of 15.74% and a Net Present Value (NPV) discounted at 10% of BZ\$9.4 million, US\$4.7 million).

The main objective of the financing provided by GCF is to incorporate in this Program of rural development the capacity of productive and economic resilience in the face of the consequences generated by climate change. Particularly in Belize, this problem is of great importance, since it is one of the countries in which climate change is expressed in events that generate significant and frequent damages (especially hurricanes and floods).

The main destination of these funds is that the investments foreseen both for the development of value chains and for investments in infrastructure, are adapted to climate change. The consequence of not contemplating this adaptation will cause a low impact and a poor sustainability of the program's intervention in the territory. An estimate of the expected results of the Project without the participation of the GCF shows a cash flow with a null return IRR = - 0.21% and a negative net present value (discounted at 6% per annum) NPV = BZ\$ -6.0 million (US\$ -3.0 million).

Economic analysis

For the economic analysis, the distortions that exist using market prices were adjusted by eliminating the effect of taxes and subsidies, market influences, unemployment and the exchange rate in order to reflect the opportunity cost in the use of the resources. Once economic prices were obtained for project costs, and for the sustainable production models, the following economic indicators were obtained: (a) IRR of 11.64%; and (b) NPV discounted at 6% of BZ \$ 14.0 million (USD 7.0 million). This level of return could be considered as intermediate.

Appendix 10 presents the detailed financial and economic analysis of the Project, and assesses the financial and economic feasibility and sustainability for the direct project participants and for wider society.

The proposed financing structure is a mix of loans and grants that will tackle short and medium term objectives for a resilient agricultural production. The provision of the GCF loan/grant is of critical importance to provide the initial funding for a paradigm shift that enables smallholder producers to penetrate the market. The vulnerability of the smallholder producers to climate change and their low recovery capacity to climate events require grants and concessional loans to reduce the risk premium to make the investment viable. Furthermore, the GoB and smallholder producers will also necessitate extensive technical assistance to identify climate vulnerabilities and implement practices that improve their resiliency to climate change.

IFAD's past project in Belize helped develop rural financial services specifically targeted for community-initiated agriculture. The proposed project will promote local private investments since smallholder farmers will be required to present a co-finance to access the MGI resources. Most of the groups don't have capital to provide the co-financing requirements and thus will have to turn to local Credit Unions for the resources.

The elevated Belize public debt (100% of the GDP) is a cause of concern to the International Monetary Fund⁵⁰ exacerbated by medium-term economic growth is projected at less than 2% per year. The grants resources will be dedicated to the part of the project that have a low and long-term profitability for which the costs cannot be supported by the current stakeholders: soft investments (technical assistant, training, etc.) and financial support (small producers) to carry out the innovative socio-economic development in this area, as well as the structuring of a Water Use Association (WUA) for the management and maintenance of the irrigation network. The concessional loan financing targeted to investments is to promote a transition in agricultural practices, to cover the incremental costs of climate proofing productive investments, and/or to directly address erratic weather patterns caused by climate change that affect target population. As such, the loan/grant modality is fully justifiable since the financing directly targets climate change threats that impose an additional cost to agricultural and rural development in the country. The Government of Belize is well aware that climate change imposes an additional cost to agricultural development in the country, and since project origination requested that IFAD seek to access a GCF grant to complement the IFAD loan the country is assuming. The complementarity of the IFAD loan, targeting productive development, with a GCF concessional loan and grant that promotes increased resilience and directly addresses climate change threats, has encouraged the country to commit to an innovative approach to agricultural development that can significantly change the development path or rural smallholders in Belize.

F.2. Technical Evaluation

The project will implement the strategy laid out in the National Agriculture Sector Adaptation Strategy to Address Climate Change in Belize (2015). As such, Climate-smart agriculture (CSA), as defined and presented by FAO at the Hague Conference on Agriculture, Food Security and Climate Change in 2010, jointly address food security and climate challenges. It is composed of three main pillars:

1. sustainably increasing agricultural productivity and incomes;
2. adapting and building resilience to climate change;
3. reducing and/or removing greenhouse gases emissions, where possible.

In this project, we consider techniques aligned which produce one or more of the following benefits:

- reduce external inputs, and thus save energy, reduce emissions and lower costs
- reduce erosion, maintaining soil fertility and infiltration
- improve pest control without the needs to pesticides and herbicides, thus decreasing possibility of water contamination
- increase soil water infiltration and thus reduce potential for flooding
- reduce water consumption, thus increase resilience to droughts
- increase carbon sequestration, contributing to mitigation

Below are techniques considered climate smart/resilient that will be applied by the project. Other activities may be considered at the time of the project, as long as they comply with the principles laid out by FAO and provide one or more of the benefits listed above.

1. Crop rotation – Rotating crops helps to improve soil stability and distribution of nutrients in the soil profile. The most effective rotations combine crops with different growth strategies: nutrient accumulating versus nutrient depleting; water accumulating versus water consuming; deep-rooted with shallow roots to bring up nutrients from below and increases biological activity.

⁵⁰ Staff Report: 2017 Article IV Consultation with Belize

2. Integrated crop-livestock fertility – In an integrated system, livestock and crops are produced within a coordinated framework. The waste products of one component serve as a resource for the other, reducing transportation costs and emissions. For example, manure is used to enhance crop production; crop residues and by-products feed the animals, supplementing often inadequate feed supplies, thus contributing to improved animal nutrition and productivity.
3. Soil cover – Protects the soil and the microbes within from the impact of sun heat, rain and wind. It stops the soil surface from sealing, and reduces the amount of rainwater that runs off, reduces evaporation and promotes fertilization from the decomposition of the organic matter.
4. Polyculture systems (such as MILPA Systems) – Polyculture is where several crops are grown in the same space in imitation of the diversity of natural ecosystems. It includes multi-cropping, intercropping, companion planting, beneficial weeds, and alley cropping. One of the main advantages of the polyculture systems is that different species coexist, sharing resources like water, light, soil and even ecological interactions, such as cross-fertilization and pest control.
5. Using green manure and legumes for fertilization – Green manure cover crops are mainly planted to provide cover, nitrogen and smother weeds. They can be planted as inter-crops, relay crops or, as sole crops during a land resting period. Their benefits include: effective control of weeds (smothering), the additional biomass produced when no other crop is able to grow and the nitrogen fixation if leguminous crops are used.
6. Agro-forestry - Agroforestry is the management and integration of trees, crops and/or livestock on the same plot of land. The biological and physical interaction between the crop and the livestock components are manipulated to enhance the agricultural production of the land base. Potential benefits from properly designed and managed agroforestry practices include: Increased crop production and economic gain, soil conservation and improved soil quality, carbon sequestration and increased biodiversity. Because of the deeper roots, trees aid in increasing soil infiltration and help prevent floods. Agroforestry also provides a diversification of income to the farmers, reducing vulnerability to climate change.
7. Planting in contours - Contours are level lines across a slope at a constant elevation. Vegetative barriers (such as grassy strips) are being located on the contour to control soil erosion. Water flowing down the slope picks up soil. When it reaches a contour barrier it slows down, the soil particles settle out, and more water enters into the soil.
8. Soil analysis - Proper nutrition is essential for satisfactory crop growth and production. The use of soil tests can help to determine the status of plant available nutrients to develop fertilizer recommendations to achieve optimum crop production and reduce excess usage that can be expensive and result in pollution of water resources.
9. Riparian planting - Riparian planting and management is the restoration and enhancement of wetlands, rivers or streams inside a property, improving the natural services provided by this type of ecosystems such as the fresh water needed for irrigation. Riparian plants filters water, prevents erosion. It minimizes erosion problems, especially if you plant native trees such as cabbages, moderates water flow (which helps prevent flooding), moderate water temperature and provide food to aquatic insects and fishes.
10. Community seedbanks- Community seedbanks are collective initiatives that empower farmers to use, exchange, and conserve local and improved plant varieties for food security and improved livelihoods. Informal

seed supply systems support the conservation of traditional farmer varieties; and thus, increase their resilience in face of climate change and pests; as well as maintain seed security at community levels.

11. Irrigation - The proposed irrigation investment is direct response to varying rainfall patterns caused by climate change. In addition, climate resilience will be supported by the following measures: (a) high over-all efficiencies of irrigation-dependent production systems; (b) high output and high value per cub. m of water; (c) adequate hydraulic feasibility; good design; and reasonable operation; (d) adequate drainage; (e) good control of water allocation over time and within the scheme; (f) limited losses, for the sake of flow capacity and scour control; (g) predictable and reliable water allocation; (h) good collaboration between the farmers; (i) good access to information about the normal and actual weather; and (j) good knowledge about management options, covering both cultivation and water management.

In the conditions of Belize, a climate resilient irrigation scheme would be characterized by the predictable availability of adequate water at the place and the time when the crops need it. Thus, the scheme must be: (i) hydraulically feasible (in terms of water availability); (ii) well designed (for example in terms of source capacity, conveyance capacities and control structures); and (iii) well operated (for example in terms of water allocation within the scheme). In addition to this, given the high level of precipitation in Belize, in some schemes there would be a need to consider efficient drainage systems to cope with excess water and erosion. The availability of sufficient water in the source will be assessed during the scheme selection phase. The design of required conveyance and regulatory structures, as well as the system operation set-up is designed to be supported by an International Technical Assistance.

12. Greenhouses – In order for greenhouses to resist the Caribbean weather conditions, with high frequency storms and sustained strong wind conditions, it needs to have a special engineering adaptation, such as: (i) galvanized steel frames, (ii) an easy system to remove the plastic, and (iii) a reliable anchoring system.

13. Roads - Some sections of the road surface level will be raised to an elevation of about 0.5 m higher than expected flood level to reduce risk of road damage and to prevent an inaccessible road during flood event. In terms of climate change, side slope should be adjusted from 1:2 to 1:3 or flatter to prevent flood damage and erosion from road surface runoff. A good road drainage system is vital for the type of roads under consideration. It shall have capacity to convey water from the surface of the road, as well from the different layers of the road structure, to a safe exit (stream or cross drainage structure). The drainage system also shall intercept surface water flowing towards the road and conveys water across the road in a controlled fashion. Furthermore, water must not be allowed to develop sufficient volume or velocity so as to cause excessive wear along ditches, at culverts or along exposed running surfaces, cuts or fills. This will prevent excess water within the roadway that will adversely affect the properties of the materials with which it was constructed. Cut or fill failures, road surface erosion and weakened subgrades followed by a mass failure are all products of inadequate or poorly - designed drainage. Different types of drainage structures that could be used in the case of Belize include cross drainage including pipe culverts, box culverts, spillways, small bridges, ditches and drains, scour checks, cut - off ditches; as well as retaining walls, gabions, rip-rap to protect embankments.

The existing standards in Belize allow to consider all these possibilities depending on the actual road conditions.

F.3. Environmental, Social Assessment, including Gender Considerations

The Social, Environmental and Climate Assessment Procedures (SECAP) is IFAD's main safeguards instrument to enhance the sustainability of results-based country strategic opportunities programmes, country strategy

notes, programmes and projects. It has risk assessment process that recognizes the heterogeneity of responses, given the widely different country and community circumstances.

Guiding principles and values in SECAP are aimed to evaluate risk and exposure of IFAD interventions to avoid, minimize and mitigate potential social and environmental impacts. SECAP has also a focus on climate change, providing assistance to project and programme managers in assessing the potential damages climate change will have over IFAD investments connecting adaptation with risk management and poverty reduction. It also has a focus on mitigation proposing measures to reduce projects carbon footprint and/or avoiding emissions with technology transfer and technical capacity.

The SECAP describes how to better mainstream environmental, social and climate change considerations into the project cycle, and demonstrates IFAD's commitment to go beyond "doing no harm" to maximizing development gains. It also takes into consideration the Sustainable Development Goals (SDGs) and other international relevant agreements. It seeks to ensure that IFAD's policies and strategies and its investments are designed to leave no one behind since sustainable development must be achieved for all – especially the poorest and most vulnerable to climate change.

The SECAP note is an integral part of project designs and should contain:

(i) An overview of the main environmental and natural resource management and social issues in the project area and identification of any significant impacts (positive and negative) and social concerns likely to be associated with the project. The overview should be informed by a rapid appraisal of environmental and, social issues to determine if an in-depth environmental and social impact assessment is warranted. The screening is considered a minimum requirement for all projects, including those which do not warrant further analysis (e.g. where the environmental/social risk of the project is category C or low, respectively).

A justification for the environmental category (A, B, C) is assigned to the project proposal on a preliminary basis, with references to country social and environmental policies, legislation and institutions, as well as the social, environmental screening requirements of any cofinancier i.e GCF.

(ii) An identification of the specific requirements for future project processing, including: (a) preliminary indications of the scope of the ESIA, climate risk analysis and Resettlement Action Plan and corresponding ESMP likely to be required during project formulation for category A and high climate risk project proposals, or the type of additional environmental analysis and corresponding ESMP required for category B and moderate climate risk proposals; (b) any consultation requirements on the environmental and social dimensions of the project proposal, including the views of interested parties on these dimensions of the proposal; and (c) budgetary requirements.

Key Environmental and Social risks and impacts anticipated by the project with salient measures

Project sub component	Anticipated potential risk/impact	Mitigation measure
1.1: Infrastructure and Production Plans (IPP)	Not applicable	--
1.2 Strengthening of Producers' Organizations	Not applicable	--
1.3 Value Chains Development		Careful selection of the type of agrochemicals and management of their use (timing, dosage, mode of application, etc.) can reduce to acceptable levels the environmental risks they pose while providing the needed benefits for increased production with lower financial and health risk costs. Sufficient technical and
<i>Backyard gardens (BYGs)</i>	Inappropriate agro-chemical handling and excessive use, resulting in health problems	

<i>Matching Grant Fund (MGF)</i> <i>Development of value chain Business Plans (BPs)</i>	• Agro-chemical runoff and contamination of water sources due to insufficient understanding of agro-chemical use • Risk of deforestation. Farmers prospering as a consequence of the Project might want to encroach on the forest to expand agricultural areas. This could be the case for the cooperative in Nago Bank • Waste management. Farmers may not implement a segregation plan and a safe storage and transportation plan for toxic waste from pesticides and fertilizers. • Potential ecosystems damage; pest resistance, soil acidification, damage to non-target species and water pollution. • Human health intoxication, allergies and other long term health problems. Neurological health effects, such as memory loss, loss of coordination, reduced stimulus response, impacts on vision, altered or uncontrollable mood and behaviour, and reduced motor skills. Pesticides have also been linked to cancer, hormone disruption, and problems with reproductive and foetal development • Greenhouse gases emission from fertilizers	managerial capacity-building is provided for the selection, application, storage, disposal and monitoring of pesticides, which are often hazardous if they are misused or handled improperly. Farmers will be capacitated in deforestation free agriculture and they will also be required to avoid encroaching in new areas to take part in the project. Live cycle management practices will be applied in processing facilities with recycling and/or waste to energy facilities implemented where necessary Reduction of synthetic pesticides in favour of bio pesticides Climate resilient agriculture will be implemented in favour of a sustainable production Law enforcement and compliance with international safety and security procedure will be required for all workers. Application of guidelines for pesticides and handle of toxic products. Training will also be provided to all stakeholders The project will reduce the use of synthetic fertilizer and also improve the production with sustainable agriculture.
2.1: Investment in Rural Roads Improvements (RRI)	• Labour conditions and workers safety might not comply with international standards • Soil compaction, erosion and loss of microbiome in the areas surrounding the infrastructure works • Noise and dust during the construction phase • Pollution of soil and water table by spilling and percolation of chemical products during the construction phase • Human health affected by increased traffic	Training on safety and security will be provided for workers with international standards Avoidance of wheel traffic and tillage of wet soils, minimize tractor or heavy oversized instruments. Review machinery maintenance and mounted system, reduce vibration and harshness. Reduce the dust by watering the area, use EPA filters for workers when necessary Traffic safety capacitation in the project influence area to all residents. Signals and road safety design such as speed bumps. Animal corridors feasibility will be analysed and signals will be installed in the roads. Farmers registration will be required for the participation in the project to avoid land

	• Animal crossings over roads affecting local fauna • Risk of land speculation forcing poor farmers out of the project areas; this is especially the case for the communities in Trio • Impacts of natural resource mining necessary to build the road (such as sands and rocks) • The improvement of the feeder road in Nago Bank could contribute to further deforestation and encroachment of the surrounding forest	speculation. Farmers will be supported throughout the whole process Materials for infrastructure construction will be brought from the area when possible, the project will also support recycling of construction waste. Infrastructure specifications take into account the natural hazard risk and anticipated climate variability risk factors, and will therefore finance the engineering aspects of climate resilient rural roads identified: (i) slope stabilization structures such as dry stone or gabion wall dependent on the gradient of the road and road construction materials; (ii) paving of roads with durable materials; (iii) proper alignment of new roads to avoid vegetative loss; (iv) improved drainage systems (including cross drainage structures such as cascades, small check walls, culverts and causeway) to avoid erosion of road materials; (v) improved planning of roads with proper cross section and standard dimensions; and (vi) raising road level to adapt to climate change events, especially flooding.
2.2: Investment in Small-scale Irrigation and Drainage (SSID)	• Risk of overuse of water resources and table depletion. Irrigation may ignore sufficient availability of groundwater due to lack of hydrological data that allow a correct assessment of available water for agricultural purposes • Labour conditions and workers safety might not comply with international standards • Soil compaction, erosion and loss of microbiome in the areas surrounding the infrastructure works • Conflicts over the use of water • Impacts over the environmental flows of water courses	Should drilling wells be required, sound environmental practices will be respected to avoid cross contamination of aquifers and unsustainable extraction of groundwater. Hydrologic studies will be done to evaluated aquifer characteristics The project will support studies for the evaluation of environmental services, the environmental flow of the water courses and water table will be assessed. Water rights and stakeholders will be identified and organized as part of the project
2.3: Climate Information System	Not applicable	

Environmental and social risk category of the project

The project is categorized by IFAD as a Category B as the it may have some adverse environmental and/or social impacts on human populations or environmentally significant areas, but the impacts:

(i) are less adverse than those for category A; (ii) are site specific and few are irreversible in nature; and (iii) can be readily remedied by appropriate preventive actions and/or mitigation measures. Category B projects require an ESMP, which is incorporated in the SECAP review note showing the output from the environmental and social analysis

IFADs categorization of the project is in line with GCF environmental and social risk category B which includes activities with potential limited adverse environmental and/or social risks or impacts that individually or cumulatively, are few, generally site specific, largely reversible, and readily addressed through mitigation

measures. This include projects such as adaptation of crop farming systems to climate change and small scale agriculture initiatives aligned with the Belize resilient project.

The due diligence process categorized the project based on an evaluation procedure and a standardized questionnaire that provides information on the likelihood, frequency, intensity, manageability, duration and reversibility determining that the type of risk and the significance of potential impacts were low or in few cases medium as indicated in the identification of risks and impacts table

The potential impacts are site specific, reversible in nature; and can be readily remedied by appropriate preventive actions and/or mitigation measures. The project aims to increase the economic, social, and environmental resilience of smallholders by promoting climate-smart agricultural production, investing in climate proof infrastructure, supporting producer associations, and strengthening value chains. Due to the nature of the Project's activities, it is expected that the main negative environmental impact can be prevented. The investments in public infrastructure will be site specific and of low probability of impact as it will consist of rehabilitation rather than new infrastructure. In fact, the project will not finance any activities that may generate significant adverse environmental and social risks and impacts.

The screening of the Resilient Rural Belize program has been conducted on a preliminary basis at the project concept stage for consideration at the Operational Strategy and Policy Guidance Committee (OSC) review stage, and then finalized in advance of the quality enhancement review. The assessment process involves a review and approval of project documentation by IFAD's Executive Board as part of the project cycle approval process. The Executive Board reviews the President's report and recommendations on the project proposal. In all of the steps there were a number of internal and external evaluators that provide inputs and peer review the process.

The Project does not envisage any adverse social impact and is not expected to engage with Indigenous Peoples. However, a complete set of activities including free and prior informed consultation will be part of the implementation process in order to minimize any possibility of indigenous people presence in the project area, avoiding and mitigating possible impacts on their lands. The ESMP has prepared an Indigenous Peoples Plan as a word of caution preventing any potential conflict or damage caused by the project during implementation

Stakeholder Engagement strategy

The strategy will be part of a plan prepared by MoA-PMU at the beginning of the project and must define the dissemination mechanisms that will be used so that the stakeholders have the necessary and sufficient information for their active participation in decision-making and as beneficiaries of the services and products offered by the project. Likewise, it should define in more detail, if necessary, the spaces and mechanisms for stakeholder participation in each of the project components. Furthermore, consultations must be developed ensuring a participatory approach. The strategy and the Plan, once formulated, will be shared with key actors, institutional and communities in order to receive feedback and agree on its application. Participation of stakeholders is essential and mandatory during the design and during the execution of the project.

The stakeholder engagement strategy has made an initial identification of main parties, although the number of parties may vary during the project implementation, the plan as well as its strategy need to actively manage expectations addressing concerns instead of problems by tackling issues in a fashionable, timely manner that respect cultural and social differences among stakeholders. The strategy will be based on the study of the types of participation expected in relation to each outcome and specific activity of the project, identify the mechanisms and instruments necessary to involve stakeholders, including consultation on language, timetables, places and times in which activities must be carried out to guarantee the participation of all, especially in the case of women

and young people. The engagement strategy should be mainstreaming during project implementation and needs to be live document with ongoing feedback from stakeholders and taking into account the following;

- Engagement during the project life will be carried out by addressing stakeholders' areas of influence and interest during the project design and implementation phases.
- The type of approach and the frequency of engagement will depend on the stakeholder role in the project.
- Engagement approaches will go from partnerships to consultations and communications fostering ownership at the different levels of the project.
- Identify institutional, governmental, non-governmental, commercial and community actors at the national, district and local levels that complement the identification already made during the design.
- Build capacities in the project teams (PMU and consultants) on the importance of stakeholder participation and how to promote it.
- Prepare a dissemination and information strategy that responds to the characteristics of the different types of institutional and community actors, including women and young people, including an analysis of their link with the available social media.
- Identify the actors that should be strengthened in their leadership and capacity to make decisions within the framework of the project and in their communities.
- Define procedures for the establishment of formal or informal commitments between the project and the stakeholders, mainly beneficiaries, for their continued participation until the end of the activities, once they are involved.
- If the project extends to indigenous communities, it will be necessary for the Plan in its entirety to respond specifically to the socio-cultural particularities of this population.

Gender

The Projects' Strategy of Social Inclusion and Gender is based on the mainstreaming of both concepts throughout most of the activities, in order to guarantee an increase in the participation of women and young people, the strengthening of their capacities in different fields and their access to all the benefits offered by the program. Some specific activities are carried out as affirmative actions and have a differentiated budget.

It is recognized there are important information gaps which would provide information that account for the differences between men, women and young people, allowing to clarify the situation and participation of women and young people in the areas, organizations and topics that the program will address. In this regard and following IFADs participatory approach and operational model, to respond to this limitation, the project will use three instruments throughout project implementation to gather the required information: rolling baseline studies at project beginning, the Climate Vulnerability Assessment and the Value Chains Analysis and Market Assessment. Complementing the Development of Infrastructure Production Plans (IPP), an analysis and consolidation of the information collected on gender and social inclusion will be carried out. Thereby supporting the definition of an opportunity for the increased and strengthened participation of women and young people in climate resilient actions and value chains. These in turn, will be used to develop project specific action plans under strict guidelines to be included in the Program Implementation Manual.

The elaboration of Plans of Action on Gender (and social inclusion) in each area, based on the information obtained from the studies mentioned above, during the first year of implementation of the program, will serve to achieve greater specificity and concreteness in the proposals of the Strategy.

Affirmative actions are specific activities designed to address inequalities in a society and help members of a disadvantaged group or a more vulnerable group to overcome obstacles to equal access to development opportunities. Implementation of the “Be Resilient” Project will take into consideration, amongst others:

- (a) Terms of Reference of all PMU staff as well as contracts and MOUs with implementing partners and service providers will include understanding and demonstrated experience in working from a gender equality perspective, as a requirement;
- (b) Participation of a male and a female representative of project participants in the Programme Oversight Committee (POC);
- (c) The creation and strengthening of Water users organizations (Component 2) will promote the participation of women;
- (d) Monitoring and evaluation will use disaggregated data (registration, collection, analysis and reporting) and the Project’s implementation strategy will be adjusted, if gender and poor population targets are not met;
- (e) Awareness raising and training materials will be adequate and understandable for its specific audiences;
- (f) Provisions would be made for training to be implemented as much as possible in rural communities to make them more accessible to women; and
- (g) Women’s and youth participation in technical assistance and exchange visits or study tours organized or financed by the Project will be facilitated.

The Matching Grants Fund (MGF) investments (Component 1) is a competitive fund to which male and female small scale farmers can submit business proposals in line with their reality, needs and opportunities. The MGF will also have certain gender considerations. These would include:

- (a) The Calls for proposals for the Matching Grants Fund, will explicitly mention the Project’s target group and specifically invite young and female farmers to participate;
- (b) Selection criteria for the proposals presented to the Matching Grants Fund will include a bonus for business proposals presented by young farmers, female farmers or organizations that have at least 40% women in their membership; and
- (c) Any business proposal that includes the reduction of women’s workload, either through the proposed technology to be introduced, or other measures (e.g. more equal distribution of domestic work amongst household members), would receive a bonus. This will be defined in the selection criteria for Business proposals for the MGF.

Addressing the food and nutrition security issues of the target group and the need to build resilience to the effects of climate change in food production systems, the Project would work with men and particularly with women to improve healthy food choices:

- (a) Improve or establish backyard or home-gardens and promote the selection of more nutrient-rich food to grow in the backyard
- (b) Provide nutrition training that will go beyond a food-based approach to nutrition. It will address amongst others: (i) the links between poverty and nutrition; (ii) nutrition as a family issue and not as a woman’s affair only; (iii) household economies for a healthy diet; and (iv) healthy diets and food choices. Nutrition education would also include counseling for men and women in the appropriate use of income that should be oriented towards nutrition, health and education.
- (c) Establish weekly/monthly farmers’ markets in urban environs to encourage consumption of fruit and vegetables and creating markets for small male and female farmers.

As female participation in agriculture is often underestimated and invisible, also by the agriculture extension workers, the MOA staff and extension agents (PMU) will receive training in gender equality concepts and

practical tips as how to ensure equal male and female participation in extension, training, farmer to farmer training, and technical assistance. The key farm activities in which more than 25% of women are directly involved include irrigation and water management, harvesting of own permanent crops and fruits, milling, grinding and sun drying, and livestock rearing: cattle, but also sheep, pigs and poultry. These data provide an important orientation to “Be Resilient” Project implementation, as it clearly shows that extension and technical assistance needs to reach and benefit both men and women. While women are less involved in marketing, the Project’s marketing support should take into consideration that between 14-18% of marketing of grains, permanent crops, and vegetables are carried out by women, and that their participation in training, fairs, buyer/seller encounters, is essential. Finally, new opportunities for young men and in particular women, to become engaged in an agro-processing business as part of the value chain approach, will continuously be identified and supported.

Under Component 1, there will be specific attention to strengthening gender equality in the Producers’ organizations as at present the female participation in these organizations is very low. The capacity building project to be developed and implemented with the Producers’ Organizations will not only focus on administrative, business and organizational skills, but also confidence building and rural empowerment. The Rural Organizational Strengthening Specialist will be responsible for achieving this and will work closely together with staff from MOAs Department of Cooperatives.

IFAD has developed several Learning Routes and participation in these as well as national and international study tours/exchange visits have been programmed for project participants and MOA staff in Gender Equality.

Women’s participation in irrigation and water management (27%) is also an indication that the rural infrastructure component and setting up of Water Users Groups, women need to be involved as well.

As one of the few ministries in Belize, the MOA has a recently appointed Gender Focal Point. The Project would support some technical activities as identified by the Gender Focal Point, such as studies, workshops, etc.

Grievance and Redress Mechanism

IFADs Grievance Redress Mechanism can be accessed when necessary to manage project-related grievances that cannot be resolved by the project’s Executing Entity. IFAD Complaints Procedure for alleged non-compliance with its social and environmental policies and mandatory aspects of its Social, Environmental and Climate Assessment Procedures (SECAP)

The objective of the IFAD Complaints Procedure is to ensure that appropriate mechanisms are in place to allow individuals and communities to contact IFAD directly and file a complaint if they believe they are or might be adversely affected by an IFAD-funded project/programme not complying with IFAD’s Social and Environmental Policies and mandatory aspects of SECAP.

Complaints must concern environmental, social and climate issues only and should not be accusations of fraudulent or corrupt activities in relation to project implementation – these are dealt with by IFAD’s Office of Audit and Oversight.

To file a complaint for alleged non-compliance with IFAD’s social and environmental policies and mandatory aspects of its SECAP, IFAD will consider only complaints meeting the following criteria:

- The complainants claim that IFAD has failed to apply its social and environmental policies and/or the mandatory provisions set out in SECAP.
- The complainants claim that they have been or will be adversely affected by IFAD’s failure to apply these policies.

- Complaints must be put forward by at least two people who are both nationals of the country concerned and/or living in the project area. Complaints from foreign locations or anonymous complaints will not be taken into account.
- Complaints must concern projects/programmes currently under design or implementation. Complaints concerning closed projects, or those that are more than 95 per cent disbursed, will not be considered.

The complainants should first bring the matter to the attention of the government or non-governmental organisation responsible for planning or executing the project or programme (the Lead Agency), or to any governmental body with the responsibility for overseeing the Lead Agency. If the Lead Agency does not adequately respond, then the matter may be brought to the attention of IFAD. The issue may be brought straight to IFAD if the complainants feel they might be subject to retaliation if they went to the Lead Agency directly. The Regional Division will examine the complaint and, if necessary, will contact the Lead Agency, or the governmental body with the responsibility for overseeing the Lead Agency, to decide if the complaints are justified.

At the PMU-Project level, it will establish one or more grievance mechanisms at field level to file complaints. Contact information and information on the process to file a complaint will be disclosed in all meetings, workshops and other related events throughout the life of the project. The project will include in the capacity building program information on the GRM and will organize consultations to determine the most suitable way for beneficiaries and stakeholders to communicate their concerns and ideas.

1. The Grievance Redress Mechanism and guidelines will be developed for the project taking into account IFADs corporate Complaints Procedure to receive and facilitate resolution of concerns and complaints with respect to alleged non-compliance of its environmental and social policies and the mandatory aspects of its Social, Environmental and Climate Assessment Procedures.
2. The project will also be responsible for documenting and reporting as part of the safeguards performance monitoring on any grievances received and how they were addressed.

Complaints can be raised either orally or in writing, directly to the Project Management Unit (PMU) and can be sent by email to beresilient.grievance@gmail.com.

Complaints must include the following information:

- Name, address, telephone number and other contact information
- Whether the complainants wish to keep their identity confidential, and if so, why
- Name, location, and nature of the IFAD project/programme (if known)
- How the Complainants believe they have been, or are likely to be, adversely affected by the IFAD-supported project or programme

Submitted complaints will be sent to the Programme Manager and M&E officer to assess whether the complaint is eligible. Eligible complaints will be addressed by the PMU. The PM with support from the M&E Officer will be responsible for recording the grievance and how it has been addressed if a resolution was agreed.

If the situation is too complex, or the complainer does not accept the resolution, the complaint must be sent to a higher level, until a solution or acceptance is reached:

- 1st level: At this level, received complaints will be registered, investigated and solved by the PMU.
- 2nd level: If the complaint has not been solved and could not be solved in level 1, the PMU must report it to the POC. Received complaints will be registered, investigated and solved by the POC.
- 3rd level: If the complaint has not been solved and could not be solved in level 2, the complaint must be submitted to IFAD following the procedure stipulated in section 3.10.3 above.

For every complaint received, a written proof will be sent within ten (10) working days; afterwards, a resolution proposal will be made within thirty (30) working days.

In compliance with the resolution, the person in charge of dealing with the complaint, may interact with the complainant, or may call for interviews and meetings, to better understand the reasons.

All complaint received, its response and resolutions, must be duly registered.

Upon acceptance of a solution by the complainer, a document with the agreement should be signed with the agreement.

F.4. Financial Management and Procurement

As part of the design and in accordance with IFAD's Financial Management policies and guidelines, a Financial Management Assessment was undertaken for the proposed project's arrangements within the Lead Programme Agency, the MED. The results of the assessment indicate that the proposed set up of the Programme Management Unit (PMU) along with the systems and procedures requirements identified in Appendix 11 would position the Project at a low risk level from a financial management perspective.

Please see Appendix 7: Financial management and disbursement arrangements and Appendix 8: Procurement for further details on management and procurement.

G.1. Risk Assessment Summary

The main risks of the Project relate to the need to create and significantly strengthen institutions on all levels, government and producers organizations alike. The introduction and adoption of a paradigm shift in sustainable and climate resilient agricultural practices is also a challenge. Overall, taking into account the mitigating measures as described below in G.2, the level of impacts of risks to the Project is medium/low with a medium/low probability.

G.2. Risk Factors and Mitigation Measures

Institutional Weakness

Description	Risk category	Level of impact	Probability of risk occurring
Weakness of the MoA and its ancillary services (e.g., extension, cooperatives) may hinder the project and offer divergent (and possibly not climate resilient) technical advice to farmers.	Technical and operational	Medium (5.1-20% of project value)	Medium

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The project was designed to mitigate this risk. An institutional strengthening of the MoA was included among the Cross Cutting Activities to provide technical services for small farmers' development and to introduce new concepts and skills and to strengthen capacity in the DE, the DC and the BMDC. In addition to the Capacity Building Program, the Project will employ six climate resilient agriculture specialists who will work in partnership with the MoA extensions agents (DE) in providing technical support to all project participants. This alongside work will ensure agility and uniformity in the technical advice. It is expected that these risks will be reduced to low with the proposed measures.

Beneficiaries Engagement

Description	Risk category	Level of impact	Probability of risk occurring
Lack of engagement of producers to coordinate production planning and market access may result in maintenance of the status quo (low prices, quality and reliability). Producers may be reluctant to offer an in kind/cash counterpart to finance production improvement through the MGF because of high risks, weather-related, environmental or market.	Technical and operational	High (>20% of project value)	Low

Mitigation Measure(s)

Support will only be given to farmers who are already in a PO and to demand-driven activities. These groups will be screened; their strengths and weaknesses identified, and the institutional strengthening activities of the Project (Sub-component 1.2) will improve their functioning. A Climate Vulnerability Assessment will be

performed in order to selected methodologies and production and post-production technologies to mitigate these risks. Improvement of value chains (strengthened linkages between input suppliers, producers and buyers; increased access to technical services, etc.) and upgrading product quality (use of better quality inputs, adoption of standards, better production and marketing practices, etc.) will mitigate market risks. In the consultation process it was clear that the project participants are already engaged with the project, thus the probability of occurrence is low.

Environmental Sustainability

Description	Risk category	Level of impact	Probability of risk occurring
Negative environmental impact of project investments such as inadequate handling of agrochemicals, excessive water withdrawal, deforestation among others listed in Table 4 of the SECAP Review Note in Appendix 13.	Social and environmental	High (>20% of project value)	Medium

Mitigation Measure(s)

The PMU will be responsible for ensuring that the Environmental Management Plan will be implemented and necessary Environmental Clearances have will be requested during the Project. The Environmental Management Plan lists the mitigation measures to minimize the negative environmental impacts. The Project's specialized extension agents will be trained to explore the potential for increasing production without the use of chemical fertilizers, especially using indigenous technologies (e.g., milpa systems); supporting integrated soil fertility systems like crop-livestock systems; avoiding chemical fertilizers in favor of local options that are available on-farm such as animal manure and crop residues; promote community education on improving indigenous practices to maximize production; use of nitrogen-fixing trees, where feasible (agroforestry); promote rotations (e.g., inclusion of legumes, multi-cropping); ensure the efficiency of fertilizer use through appropriate fertilizer selection, timing and split applications; the use and application of agro-chemicals should be carried out under the guidance of IFAD statements and Belize's Pesticides Control Board (PCB); ensure observance of the mandatory sixty-six feet riparian natural vegetation, and staggered planting on contours when utilizing sloping land for agricultural production. Furthermore, agroforestry practices will be encouraged particularly in the buffer zones of protected areas and forest reserves; identify areas that should not be cleared (e.g., streams and bodies of water) and limit the clearing of land to the areas that are most suitable for agricultural production; avoid the use of fire as a land clearing method; and increase awareness and education on sustainable forest management. It is expected that these risks will be reduced to low with the proposed measures included in the project Environmental and Social Management Plan.

Political Priorities

Description	Risk category	Level of impact	Probability of risk occurring
Political priorities may not always be consistent with the procedures established for the selection	Financial	High (>20% of project value)	Medium

of investment proposals. The available funds for investments in infrastructure could be used to supplement to investment from the other international donors, central or local budgets instead of serving the project's participants.			
Mitigation Measure(s)			
Component 2 (CRRIA) is built around the empowerment of the targeted communities in decision-making. The approval process of AWPBs, in conjunction with the approach in which Project works would be selected on an annual basis, provides opportunities to fine-tune decision-making during implementation and to minimize the risk that such adverse effects could occur. In addition, the Project includes funding to finance public consultations with a broad set of stakeholders about the barriers to development of the value chains. Wherever possible, these consultations should prioritize utilizing existing spaces for policy discussion to increase the chances that actions identified by the consultations will feed into the policy process. All proposals will have to demonstrate the need for climate change adaptation and commercial viability of the proposed venture (selected for support under Component 1 and included in the IPPs). Funds will be allocated through a participatory and demand-driven decision-making mechanism supported by a selection/ranking procedure. Current government priorities and plans are aligned with the project. It is expected that these risks will be reduced to low with the proposed measures.			
Other Potential Risks in the Horizon			
Past IFAD projects in Belize have suffered from start-up difficulties, personnel turnover and poor M&E. To prevent this from repeating in the Project, at start-up the Project will intensify the support to ensure that the design is effectively translated into implementation; and put more emphasis on installing and making operational an appropriate M&E system.			

H.1. Logic Framework.

Please specify the logic framework in accordance with the GCF's [Performance Measurement Framework](#) under the [Results Management Framework](#).

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level ⁵¹						
Paradigm shift objectives						
Increased climate-resilient sustainable development	To build overall resilience to climate change by increasing and diversifying agricultural production by small farmers, and by facilitating their access to commercial market chains for the off-take of their surplus production.					
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Fund-level impacts						
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions	Number of males and females benefiting from the adoption of diversified, climateresilient livelihood options (including fisheries, agriculture, tourism, etc.)	Household survey (mid-term and completion)	0	10,000 (40% women and 20% youth)	30,000 40% women and 20% youth)	Please see methodology description in section E.1.2.
	Number of rural households reporting high level of produce diversification through project activities	Household survey (mid-term and completion)	0	150	400	At least 20% of farming related household income will come from reported produce diversification due to project intervention.
	Number of households reporting at least 30% increase in sales value	Household survey (baseline, mid- term and completion)	0	800	1600	Increase in production will lead to expanded sales and domestic consumption. ⁵²

⁵¹ Information on the Fund's expected results and indicators can be found in its Performance Measurement Frameworks available at the following link (Please note that some indicators are under refinement): http://www.gcfund.org/fileadmin/00_customer/documents/Operations/5.3_Initial_PMF.pdf

⁵² Producers that implement investment projects are expected to increase at least 30% the value of their sales, estimated on the improvement of the family's net income. Each technician will assist the different families in the improvement of their main crop (30 - 50% of net family income). The weighted average of the improvement of the net income calculated in the 8 analysed models is 300%, which allows to infer increases between 90 and 150% in the beneficiary families. The effectiveness of these results is usually variable but should not be less than half of what was expected; goal could be surpassed.

	The number of formal rural PO's supported to become better integrated in selected Value Chains	M&E system, supervisions and completion study	0	10	30	
<i>A2.0 Increased resilience of health and well-being, and food and water security</i>	Number of males and females with year round access to reliable and safe water supply despite climate shocks and stresses	Household survey (baseline, mid- term and completion)	0	700	1750	Water supply for agricultural production.
	Number of food-secure households (in areas/periods at risk of climate change impacts)	Household survey (baseline, mid- term and completion)	0	500	2000	
<i>A3.0 Increased resilience of infrastructure and the built environment to climate change</i>	Number and value of physical assets made more resilient to climate variability and change, considering human benefits (reported where applicable)	Tracking of funded IIFA proposals, and completion study	0	50	300	Value to be included based on definition of IPPs and selected CRRIA to be funded by IIFA.
	Number of households directly benefitting from improved public infrastructure (in areas/periods at risk of climate change impacts) allowing year round access.	Tracking of CRRIA investments, household survey (baseline, mid- term and completion)	0	1,000	2,820	

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level

Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Project/programme Outcomes	Outcomes that contribute to Fund-level impacts					

A4.0 Strengthened institutional and regulatory systems for climate-responsive planning and development	Number of regulatory innovation and/or mechanism for climate responsive planning and development	M&E system and completion study	0	1	2	
A5.0 Increased generation and use of climate information in decision-making	Number of people receiving climate information for decision making	Household survey (baseline, mid- term and completion)	0	95,296	95,296	All rural households will receive climate information. Please see methodology description in section E.1.2.
A6.0 Strengthened adaptive capacity and reduced exposure to climate risks	Number of households reporting adoption of environmentally sustainable and climate new/improved inputs, technologies or practices.	Household survey (baseline, mid- term and completion), and supervisions	0	2,500	5,500	
	Number of households engaged in climate vulnerability assessments and trained in climate-resilient practices and strategies, improved agriculture productivity and market opportunities.	Tracking participation in CVA development and assistance to trainings in M&E system and completion study	0	2,500 (40% women and 20% youth)	6,000 (40% women and 20% youth)	
Project/programme outputs	Outputs that contribute to outcomes					
1. Infrastructure and Production Plan (IPP) (Subcomponent 1.1)	Number of IPPs developed at community level	Tracking submitted IPPs in M&E system and completion study	0	10	23	
2. Producers Organizations	Number of rural formal members supported.	M&E system and	0	150	450	

Strengthened (Subcomponent 1.2)	(by sex and age 40% women and 20% youth)	completion study				
3. Climate resilient production and marketing (Subcomponent 1.3)	Number of persons trained in business management and marketing. (by sex and age 40% women and 20% youth)	Tracking assistance to trainings in M&E system and completion study	0	200	700	
	Number of households developing backyard gardens	MGF investment tracking M&E system and completion study	0	500	2000	
4. Rural Roads Improvements and Small-scale Irrigation and Drainage (SSID) (Subcomponent 2.1)	Kilometers of road (in areas/periods at risk of climate change impacts) allowing year round access.	Tracking of funded IIFA proposals, and completion study	0	30	80	
	Hectares of rural agricultural land served by climate resilient irrigation and drainage systems (Disaggregated ha covered by i) irrigation and ii) drainage systems)	Tracking of funded IIFA proposals, and completion study	0	250	730	
5. Climate Information System (CIS). (Subcomponent 2.2)	Number of CIS systems for CRA developed and tested	Supervision, Mid- term and completion study	0	0	1	
6. Government Strengthened (Cross cutting Activities)	Specialized trainings developed (Climate resilient agriculture, Gender equality, Strengthening of rural organizations)	Tracking training and assistance in M&E system and completion study	0	15	30	

Activities	Description	Inputs	Description
Activity 1.1. Develop a Climate Vulnerability Assessment	A detailed assessment of the vulnerability to climate change will be conducted in each priority area (and new areas to be selected) to define the risks posed by climate change and to provide information for identifying measures to adapt to climate change impacts.	Academic Institution	Prominent international institution such as CATIE CIAT and/or 5Cs to be procured by the PMU.
Activity 1.2. Develop a value-chain analysis and market assessment	An assessment to understand the market situation of the value chain of the eight of commodities, the strengths and constraints for their production, and networks with other key players in the market.	Consultant	To be selected by PMU and supervised by program staff (see Appendix 5, Annex 1: Terms of Reference for Key Programme Staff)
Activity 1.3. Develop the Infrastructure and Production Plan (IPP)	A Plan to define the climate resilient practices and technology to be adopted, infrastructure designs and management practices.	Consultant	To be selected by PMU and supervised by program staff (see Appendix 5, Annex 1: Terms of Reference for Key Programme Staff)
Activity 2.1. Training	Training for Management Committee (MC) members and the regular members	PMU	
Activity 2.2. Develop internal administrative instruments	Development of instruments for the necessary of the functioning of the POs	Consultant	To be selected by PMU and supervised by program staff (see Appendix 5, Annex 1: Terms of Reference for Key Programme Staff)
Activity 2.3. Promote exchange visits	Exchange visits for PO's in international and national organizations	PMU	PMU structure and responsibilities are described in detail in Appendix 5.
Activity 3.1. Design of Business Plans (BPs)	Technical assistance will be provided to POs to develop BPs and identify specific investments to be made, and to support them in linking the regional IPP.	Consultants	To be selected by PMU and supervised by program staff (see Appendix 5, Annex 1: Terms of Reference for Key Programme Staff)
Activity 3.2. Select and finance through Matching Grant Fund (MGF)	Investments in resilient production and support to the value chain, as designed in the BPs, will be financed from the MGF.	Proposals from POs	

Activity 3.3. Technical assistance	Technical assistance for smallholder farmers on Climate Resilient Production and Marketing	Extension agents and consultant	Six extension agents will be working side by side with the technical staff of the MoA
Activity 3.4. Organize meetings	Meetings will be organized between buyers and sellers, initially to establish direct contacts and share information, to facilitate confidence building and to display products for sale, facilitated by the MoA and BMDC.	Consultants	To be selected by PMU and supervised by program staff (see Appendix 5, Annex 1: Terms of Reference for Key Programme Staff)
Activity 3.5 Establish farmers' markets	Establish two farmers' markets in Belize City, San Pedro and Caye Caulker, in which organized groups of small farmers can sell directly to both wholesale and retail consumers.	PMU	PMU structure and responsibilities are described in detail in Appendix 5.
Activity 3.6. Training and TA for backyard gardens	Participants developing backyard gardens receive specialized training to ensure adequate development, maintenance and usage of BYGs in the context of climate change and extreme climatic events.	TA and consultancy	To be selected by PMU and supervised by program staff (see Appendix 5, Annex 1: Terms of Reference for Key Programme Staff)
Activity 3.7. Backyard Gardens Assets	Assets for backyard gardens, selected through MGF, are bought by the PMU and delivered to project participants.	PMU	
Activity 4.1 Select CRRIA investments and finance through Infrastructure Investment Funding Award (IIFA)	Investments in CRRIA, as designed in the IPPs, will be financed from the IIFA.	Proposals from POs	
Activity 4.2. Request Environmental Clearance	Request Environmental Clearance for public infrastructure works.	Project design and consultancy	To be selected by PMU and supervised by program staff (see Appendix 5, Annex 1: Terms of Reference for Key Programme Staff)
Activity 4.3. Design and construct	Procure design, supervision and works for rural roads and ancillary structures that are most vulnerable to climate variability as well as small-scale irrigation and drainage systems including intake structures, pumping stations, rain harvesting ponds, main and secondary distribution networks, drainage networks and flood protection structures.	Works and services, consultancy	Works will be carried out by the PMU through private sector consultants and contractors selected on a competitive basis.

Activity 4.4. Institutional development	Establishment and development of Water Use Associations (WUA) for maintenance, operation and management of small-scale public irrigation and drainage systems.	PMU	PMU structure and responsibilities are described in detail in Appendix 5.
Activity 5.1. Design and develop a climate information system	Design and develop a Climate Information System (CIS) to provide farmers with timely and accurate climate information, allowing them to plan their activities accordingly and minimize climate related losses.	Consultancy	To be selected by PMU and supervised by program staff (see Appendix 5, Annex 1: Terms of Reference for Key Programme Staff)
Activity 6.1. Public consultation	Public consultation with a broad set of stakeholders about the barriers to private sector development in value chains. Participants include Project beneficiaries, their organizations, private sector and public sector actors.	PMU	PMU structure and responsibilities are described in detail in Appendix 5.
Activity 6.2 Invest in R&D	Applied R&D to advance towards climate change resilience and sustainable, profitable agriculture.	Universities	University of Belize's Faculty of Agriculture to conduct applied research on adaptation strategies and climate resilient practices including on diversified production systems, agroforestry, protein banks, and windbreaks.
Activity 6.3. Develop studies	Develop studies on i) adaptation strategies, ii) to explore and compare potential mechanisms (such as index-based insurance) to link vulnerable small farmers; and iii) to assess the feasibility of establishing a pilot project on short-term Payments for Ecosystem Services (PES) as an economic incentive to stimulate the maintenance and restoration of native forests that provide environmental services but don't generate revenue for landowners, and iv) Production Diversification for Smallholder Cane Farmers.	Consultant	To be selected by PMU and supervised by program staff (see Appendix 5, Annex 1: Terms of Reference for Key Programme Staff)
Activity 6.4. Capacity building for MoA	Institutional strengthening of the MoA to provide technical services for small farmers' development, introduce new concepts and skills	Extension agents and consultants.	Six extension agents will be working side by side with

	and to strengthen capacity in the Department of Extension (DE), the Department of Cooperatives (DC) and the Belize Market and Development Corporation (BMDC).		the technical staff of the MoA. The Rural Organization Development Specialist with the support of two consultants will work closely with officers from the DC to strengthen their capacities and to create synergy with other activities carried out by the Department.
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H.2. Arrangements for Monitoring, Reporting and Evaluation

Monitoring and Evaluation in IFAD's projects applies a participatory approach, in which Project participants are fully engaged in the collection of field data, discussion and analysis of this data, and decision-making regarding changes that may be required for more effective implementation. The PMU will be responsible for analyzing the data against the targets to monitor, assess implementation progress and effectiveness and reporting the results to the POC and IFAD.

The project will assess the changes in the well-being of the project participants, while trying to compare the situation ex-ante (from baseline information) with the ex-post (after project completion) situation. In the evaluation process, external factors that can interfere with the activities to achieve the desired impact on the target group are also taken into account. Furthermore, it will contribute to improving the stakeholders' capacity to know and understand the results of the Project's interventions, and identify best institutional practices and lessons learnt.

Monthly Status Reports will be prepared by the Project's M&E Specialist and submitted to the Project Manager for endorsement before submission to the POC. The M&E Specialist will prepare status reports using the data collected via the Management Information System (MIS). More detailed, mandatory semi-annual and annual reports will be prepared for submission to the Borrower and IFAD with input from all PMU staff and consolidated by the M&E Specialist. The latter would be presented to and approved by the POC before submission to IFAD. These reports would inform on the progress of execution of the Log Frame Indicators. A detailed information flow, reporting, monitoring and evaluation is presented in Appendix 6- Planning, M&E and learning and knowledge management.

I. Supporting Documents for Funding Proposal

- ☒ NDA No-objection Letter (Appendix 16)
- ☒ Feasibility Study (Appendix 15 with support from Appendix 1-14)
- ☒ Integrated Financial Model that provides sensitivity analysis of critical elements (xls format, if applicable) (Appendix 10)
- ☒ Confirmation letter or letter of commitment for co-financing commitment (If applicable) (Appendix 17)
- ☐ Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) – *see the Accreditation Master Agreement, Annex I* (AMA not yet agreed between IFAD and GCF)
- ☒ Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan (Appendix 13: SECAP Review Note)
- ☐ Appraisal Report or Due Diligence Report with recommendations (If applicable)
- ☐ Evaluation Report of the baseline project (If applicable)
- ☒ Map indicating the location of the project/programme (Appendix 18)
- ☒ Timetable of project/programme implementation (Section C.8)

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

Appendix 1: Country and Rural Context Background

Appendix 2: Poverty, Targeting and Gender

Appendix 3: Country Performance and Lessons Learned

Appendix 4: Detailed Programme Description

Appendix 5: Institutional Aspects and Implementation Arrangements

Appendix 6: Planning, M&E, and Learning and Knowledge Management

Appendix 7: Financial Management and Disbursement Arrangements

Appendix 8: Procurement

Appendix 9: Programme Cost and Financing

Appendix 10: Economic and Financial Analysis

Appendix 11: Draft Programme Implementation Manual

Appendix 12: Compliance with IFAD Policies

Appendix 13: SECAP Review Note

Appendix 14: Contents of the Project Life File

Appendix 15: IFAD's Programme Design Report

Appendix 16: NDA No-objection Letter

Appendix 17: Co-Financing Letter

Appendix 18: Project Map

Appendix 19: Stakeholder Consultation Minutes

Appendix 20: Budget Plan in GCF Template

Appendix 21: Certificate of Financing Agreement between the GoB and IFAD

Appendix 22: Executive Board Approval Be-Resilient

Appendix 23: Gender Assessment and Action Plan

Appendix 24: Detailed Programme Timeline, Critical Milestones and 2019 AWPB

Appendix 25: Be-Resilient Training Activities, Outline and Budget



GOVERNMENT OF BELIZE

Ministry of Economic Development, Petroleum,
Investment, Trade and Commerce

**ECONOMIC DEVELOPMENT AND
PETROLEUM
NATIONAL DESIGNATED AUTHORITY
GREEN CLIMATE FUND**

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**P.O. Box 42
Ground Floor, Sir Edney Cain Building
Belmopan City
Belize, Central America**

Our Ref: IA/GCF/1/18 (41)

January 30, 2018

Mr. Howard Bamsey
Executive Director
Green Climate Fund
Songdo Business District
175 Art center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Dear Mr. Bambsey:

**Re: Belize – Funding proposal for the GCF by IFAD regarding Resilient
Rural Belize (Be-Resilient)**

We refer to the project Resilient Rural Belize (Be-Resilient) as included in the funding proposal submitted by IFAD to the Government of Belize on the 24th January, 2018.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project Resilient Rural Belize (Be-Resilient) as included in the funding proposal.

By communicating our no - objection, it is implied that:

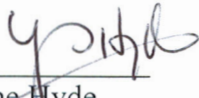
- (a) The Government of Belize has no-objection to the project - Resilient Rural Belize (Be-Resilient) - as included in the funding proposal;

- (b) The project - Resilient Rural Belize (Be-Resilient) - as included in the funding proposal is in conformity with Belize's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project - Resilient Rural Belize (Be-Resilient) - as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project Resilient Rural Belize (Be-Resilient) - as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,



Ms. Yvonne Hyde
Chief Executive Officer
National Designated Authority Focal Point – Belize

C: Mr. Jose Alpuche, Chief Executive officer, Ministry of Agriculture
Ms. Luisa Migliaccio, IFAD

Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	Resilient Rural Belize (Be-Resilient)
Accredited entity	International Fund for Agricultural Development (IFAD)
Environmental and social safeguards (ESS) category	Category B
Specific project location or target country/ies of the programme, as the case may be	Belize
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	2019-01-25
Language(s) of disclosure	English
Link to disclosure	https://www.ifad.org/documents/38711624/40206666/belize_2000001247_secap_review.pdf/2b126ee1-410f-89f4-99a8-2ac3b72f2de8 An ESIA consistent with the requirements for a category B project is contained in the Social, Environmental and Climate Assessment Procedures (SECAP) Note
Other link(s)	SECAP Disclosed Documents https://www.ifad.org/en/document-detail/asset/40206914
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	2019-01-25
Language(s) of disclosure	English

Link to disclosure	https://www.ifad.org/documents/38711624/40206666/belize_2000001247_secap_review.pdf/2b126ee1-410f-89f4-99a8-2ac3b72f2de8 An ESMP consistent with the requirements for a category B project is contained in the Social, Environmental and Climate Assessment Procedures (SECAP) Note
Other link(s)	SECAP Disclosed Documents https://www.ifad.org/en/document-detail/asset/40206914
Resettlement Action Plan (RAP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	N/A
Language(s) of disclosure	N/A
Link to disclosure	N/A
Other link(s)	N/A
Disclosure in locations convenient to affected peoples	
Date	2019-01-26
Place	Ministry of Agriculture H.M. Queen Elizabeth II Blvd., Belmopan City, Belize C.A

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP101

Proposal name:	Resilient Rural Belize (Be-Resilient)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country/(ies):	Belize
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
Significant efforts made in revising financing structure for GCF funding, from pure grants request to a mix of grants and concessional loans	Mobilization of the equity contribution by beneficiaries under the matching grant fund should be monitored and reported.
As an adaptation project, incrementality reasoning is provided in a qualitative manner, with GCF funding to be used for resilient irrigation systems, tropical greenhouses, drainage, and climate-proofing of infrastructure; while co-financing from the IFAD will support an increase in agricultural productivity	
Proven on-granting mechanisms, which have been tested with previously implemented IFAD projects; these will meet the needs of the communities while strengthening ownership of producer organizations	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the respective term sheet and addendum XVI, titled "List of proposed conditions and recommendations".

II. Summary of the Secretariat's assessment

2.1 Project background

3. The project aims to increase resilience of smallholder farmers in Belize to climate change impacts, mainly increased droughts, flooding, and significant rainfall pattern variations between seasons. These impacts have negative consequences on agricultural yields of important commodities for the country. In response to these climate impacts, the project will develop and climate-proof selected value chains of smallholder farmers, which strengthen economic stability and resilience.

4. The project will also increase access to markets through rehabilitation of critical infrastructure. It will deploy grant award and on-lending mechanisms and work with producers' organizations and local communities to meet their needs in addressing the impacts of climate change. Through this project, around 30,000 people in 23 communities in five districts will

benefit directly from increased access to climate information services for agricultural planning, climate-resilient agricultural practices and greater access to markets through resilient rehabilitated roads.

5. The project is structured in three components:
 - (a) Climate resilient value chains development (CRVC);
 - (b) Climate resilient rural infrastructure and assets development (CRRIA); and
 - (c) Cross-cutting activities and project management
6. The project requests USD 6 million in GCF grants and USD 2 million in the form of a concessional loan, which collectively is 40 per cent of the total project cost. Co-financing will be provided by:
 - (a) The Government of Belize, which will cover taxes and costs for the financing of road infrastructure and irrigation and drainage works;
 - (b) The International Fund for Agricultural Development (IFAD), which will provide loans for investment and technical assistance to improve production, value-added activities and improving market access. It will also cover the costs of strengthening organizations and public institutions; and
 - (c) Project participants, which will provide counterpart financing under the Matching Grand Fund.
7. Since it was first submitted, the funding proposal has gone through a rigorous review process and significant work has been done to strengthen it. The financing structure was revised and reallocated from a request purely for grants to one that mixes grants and concessional loans, which will support income-generating agricultural activities for adaptation. Greater clarity on the grant award and on-lending mechanisms has been provided, with a detailed project implementation manual annexed to the funding proposal.

2.2 Component-by-component analysis

Component 1: Climate resilient value chains development; (total cost: USD 7.9 million; GCF cost: USD 3.5 million, or 44 per cent)

8. Component 1 has three outputs: development of infrastructure and production plans; strengthening of producers' organizations; and development of value chains. GCF funding will be used for a climate vulnerability assessment, an analysis of value chains, and a market assessment as well as to develop value chains, including through a matching grant fund (MGF).
9. The need for an infrastructure and production plan based on a climate vulnerability assessment and market analysis is presented in a clear manner in the project structure. The proposal should, however, demonstrate how this plan will inform the implementation of other activities in the project, particularly the selection criteria for proposals submitted for the MGF and infrastructure funding award, so that development of the plan will have both short-term (project-level) and long-term contributions and effects.
10. The proposed interventions of combining climate-resilient agricultural commodities and technologies are assessed to be adequate to meet the needs of smallholder farmers in developing a value chain and accessing markets. The selected fruit and vegetable commodities have been produced in the target areas during the last three years (i.e. their production represents a business as usual scenario) and were selected for their high productivity and market potential. Upon request by the Secretariat, a climate rationale was provided by the accredited entity (AE) for the selection of vegetables, fruits and honey in relation to their resilience to climate impacts. The selected commodities are also assessed to be appropriate in

the context of climate impacts, given their stronger resilience to extreme weather events compared to tree-based fruits, which were not selected. While the selected commodities are still vulnerable to extreme weather events, combined with agronomic technological improvements, as presented in the funding proposal, the proposed agricultural interventions are expected to bring stable yields and production despite the anticipated impacts of climate change.

Component 2: Climate resilient rural infrastructure and assets development (total cost: USD 8.5 million; GCF cost: USD 4.1 million, or 48 per cent)

11. Component 2 aims to improve both agricultural productivity and market access through rehabilitation and provision of new roads as well as drainage and irrigation infrastructure. The component will increase agricultural production and access to markets.

12. The proposal states that GCF funding will be used to cover any additional costs for climate proofing of infrastructure. The budget details show that GCF funding will cover climate-proofing of rural and ancillary roads (30 per cent of the total cost of roads rehabilitation) and 70 per cent of small-scale irrigation and drainage infrastructure. During the review process, the AE strengthened the climate rationale by adding a description of the additional climate proofing activities that would be carried out under for this component, which was deemed satisfactory.

13. The component will be implemented through an infrastructure investment funding award, with a GCF contribution of USD 3.9 million. The fund will blend the IFAD loan and GCF grants, with the GCF funding dedicated to climate proofing of infrastructure. Given the full concessionality requested for the GCF funding, on-granting should be carefully monitored and reported to demonstrate that GCF resources are only used for adaptation activities for public goods.

14. The component also proposes to create a climate information system to provide accurate climate information to farmers to assist their agricultural planning. The project will need to report on the uptake of the climate information system by beneficiary farmers and demonstrate how such information is integrated into the national system and its interlinkage with the adoption of resilient agricultural practices.

Component 3: Cross-cutting activities and project management (total cost: USD 3.7 million; GCF cost: USD 0.3 million, or 8 per cent)

15. There are also cross-cutting activities, including capacity-building for public authorities, applied research and development, and policy engagement and consultation. These will have a total cost of USD 0.3 million, of which the GCF funding contribution is USD 0.13 million (43 per cent). The proposed activities are assessed to be adequate and needed to support components 1 and 2, as well as to increase awareness and encourage beneficiaries to engage and participate in the proposed project.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: N/A

16. As an adaptation project, the proposal reports an expected total number of direct and indirect beneficiaries. Beneficiary selection criteria is mainly driven by consideration of the poverty rate and potential for accessing and contributing to market and value chain development, with reference to climate change vulnerability. The number of direct beneficiaries was estimated using the number of poor households in rural Belize contained in the country's 2010 poverty assessment. More precision in making the estimation could have improved its accuracy.

17. Climate rationale and adaptation needs are elaborated and justified in a reasonable manner: Belize is a small island developing State located in the Caribbean hurricane belt that is highly vulnerable to seasonal variations and extreme weather events. The proposal provides some evidence to show climate change impacts (increased variability of precipitation patterns) on decline in yields and losses of agricultural production.
18. The project will combine adaptation measures and economic development through grant award and on-lending mechanisms, which will bring high impacts for participating beneficiaries. The MGF will include investments in renewable energy installation and efficient irrigation, which may have some mitigation potential (if replacing diesel-powered engines). This was not reported as a core indicator but should be reported as a co-benefit of the project.
19. The proposal is expected to reach 9.3 per cent of the population in Belize and targets 23 communities in five districts. Given the size of the country, the scale of the intervention is assessed to have substantial climate adaptation benefits.

3.2 Paradigm shift potential

Scale: N/A

20. Paradigm shift potential is demonstrated through a shift away from farming practices that are costly and environmentally unsustainable (due to the use of agrochemicals and fertilizers) to a more climate-resilient use of varieties and practices that will enhance the resilience of smallholder farmers while increasing incomes from productive agriculture. Proposed agronomic techniques are aligned with good practices in achieving adaptation results. These solutions have high potential to be scaled up and replicated in other regions of the country.
21. The proposal demonstrates good potential for scaling up and replication through using grant award and on-lending mechanisms that have been tested and proved through previous IFAD projects. During the implementation, the project should demonstrate that the expected financial contribution from beneficiary producer organizations has indeed been mobilized to implement their projects under the MGF and the infrastructure funding award. This will greatly increase ownership of the assets, technologies and infrastructure to be built by the farmers, thereby ensuring sustainability of the proposed project. Sustainability is also foreseen to be ensured through inclusion in the project of a variety of capacity-building and policy support activities.
22. A theory of change is presented in the funding proposal, which proposes solutions to support smallholder farmers in value chain development and market access. It shows clearly how existing constraints are expected to be addressed for smallholders and producers' organizations in building resilience to climate change. While expected benefits for beneficiaries are clear in the outcomes of the project, its adaptation results and impacts will need to be monitored during project implementation as they will depend on effective interlinkage between activities (e.g. development of a climate information system and production of resilient agriculture).
23. The exit strategy relies on a set of soft measures (institutional strengthening and knowledge building) as well as an expectation of private sector participation. During the review process, the AE was requested to provide more clarity on ownership of assets and financial planning for assets and infrastructure. While it is expected that project participants will take ownership of assets under the MGF, more concrete financial analysis and plans for the operation and maintenance of public infrastructure would be beneficial in the future.

3.3 Sustainable development potential

Scale: N/A

24. The proposal demonstrates high potential for sustainable development. Proposed interventions are proven solutions that are environmentally sound. The project should ensure careful use of pesticides and fertilizers as well as use of sound environmental practices if wells are drilled during the implementation to avoid any contamination of aquifers and unsustainable extraction of groundwater.

25. The project will generate substantial economic and social benefits for farmers through increased agricultural production and income. The project will bring a combination of increased income (i.e. poverty reduction) and increased capacity to address climate change challenges. Given the good potential for income generation, this will also bring positive social impacts for the society.

26. Gender considerations have been taken into account in the project design with an aim to mainstream women in agricultural development.

3.4 Needs of the recipient

Scale: N/A

27. The vulnerability of the country and target beneficiaries and their adaptation needs are generally explained throughout the proposal. The vulnerability of the target population is explained in terms of poverty levels. Given the significant dependence of the country's economy on agriculture, the needs of the beneficiaries are positively acknowledged.

28. Belize is a small island developing State located in the Caribbean hurricane belt that is vulnerable to the impacts of climate change. The project will deliver public environmental and economic goods along with increased incomes for producers' organizations. It will provide financial resources to overcome barriers that cannot be addressed by the market. There are currently few private sector investments in resilient agriculture in the target areas, but the proposal has good potential to bring in the private sector in the long term.

29. The proposal sufficiently explains and justifies the absence of alternative sources of financing. Elevated levels of public debt (100 per cent of gross domestic product) and the fact that most beneficiaries will have to turn to credit unions for capital to meet the co-financing requirements, justify the need for GCF grants and concessional loans.

3.5 Country ownership

Scale: N/A

30. The proposal shows that the proposed project contributes to the country's Nationally Determined Contribution and the National Agricultural Sector Adaptation Strategy (2015). The proposal is part of a broader strategy for the agricultural sector, with many international organizations involved. The project demonstrates complementarity with other donors' projects and initiatives to achieve targets in the agriculture sector in line with national climate change strategies.

31. The Ministry of Economic Development (MED) has been selected as executing entity due to its long and successful partnership with IFAD. However, its track record of managing investments of a similar size was not provided. A financial management assessment of MED has been undertaken in accordance with a policy of IFAD and has been submitted in an annex to the funding proposal.

32. The proposal demonstrates that project design and site selection was done in a participatory manner with the relevant ministries and stakeholders, ensuring participation of women. The project requires technical support from the Ministry of Agriculture, which will require inter-ministry coordination.

3.6 Efficiency and effectiveness

Scale: N/A

33. The proposal demonstrates adequate cost-effectiveness, with an economic internal rate of return (IRR) of 18.8 per cent over 15 years and a financial IRR of 18.8 per cent.

34. The funding proposal demonstrates financial adequacy and minimum concessionality. The choice of a mix of grants and a concessional loan is assessed to be adequate considering the proposed activities and their corresponding financial instruments. Grants are for the activities that have a low and long-term profitability and for which costs cannot be fully recovered by the beneficiaries. The concessional loan is for activities under the MGF, which is expected to produce incomes for participating producer organizations.

35. There is a mismatch in tenor and grace periods between the IFAD and GCF loans. The AE provided two justifications for the mismatch: (a) it was decided by the Government of Belize as they are sovereign backed loans; and (b) the higher concessionality of the GCF loan terms reflects the high vulnerability of Belize to climate change and limits the country's short-term burden of debt. This justification will be strengthened by targeting GCF loans to climate adaptation activities, and this consideration should be reflected in the selection criteria of the MGF.

36. Long-term financial viability will be ensured if the project delivers financial returns of 15 per cent as projected. There should be sufficient capital to operate and maintain the investments.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

37. **E&S risk category:** The AE has classified the project as having moderate environmental and social risks and potential impacts equivalent to GCF's category B risk categorization. The AE's risk categorization has considered potential medium-level adverse environmental and social risks and impacts related to improvement of rural roads, construction of community based agro-processing facilities, use of natural resources such as surface and groundwater, and development of small-scale irrigation and drainage systems. The AE's screening also reviewed the country's categorization which identified the proposed project activities to fall within Schedule II categorization (i.e. small-scale projects with impacts that may not be as serious as Schedule I projects depending on size, location and other considerations) aligned with IFAD's category B projects. The SECAP indicates that no activities that will qualify above the current categorization will be undertaken. To support this, the SECAP provided for an exclusion list of activities that will not be considered under the project.

38. **Safeguard instrument and information disclosure:** The AE has submitted a social, environmental and climate assessment procedures (SECAP) review note as the safeguards instrument for the project. It has identified the anticipated potential environmental and social risks and impacts which are listed against the project's components. As the project areas are not yet completely defined, there is a need to conduct site-specific environmental and social impacts assessment (ESIA) or limited level environmental study (LLES), as appropriate, when the areas have been defined and finalized. The AE has disclosed the SECAP on its website in English (as English is the official language in Belize).

39. **Key environmental and social risks and impacts and mitigation measures:** The SECAP identified and assessed the potential risks and impacts of the project and provided for project level mitigation measures through the environmental and social management plan (ESMP) which can serve as guidance in the implementation of the project. The ESMP provides

for mitigation, monitoring and reporting measures on its performance including institutional and organizational arrangements and indicative budget. It also provides for measures for information disclosure including the process for continued consultation and participation of affected people, grievance redress mechanism, and is subject to refinements during project implementation. A more focused due diligence will be conducted once the specific locations and interventions are finalized and will contribute to the development of specific measures that will be integrated into the ESMP.

- (a) *Labor and working conditions and community health, safety and security:* Since workers are expected to be mobilized in implementing the project components, the project is expected to comply with national employment and labor laws and promote safe and healthy working conditions. Rehabilitation works will need to be implemented over existing rural secondary roads and small bridges which could lead to potential exposure of local communities to risks related to traffic, noise and dust exposure, or untoward incidents with private security personnel. The project will assess the potential site level health, safety and security (HSS) concerns on the affected communities and ensure management of such risks in accordance with the SECAP and guided by the AEs Guidance Statement 14 on Community Health.
- (b) *Resource efficiency and pollution prevention:* Small-scale irrigation and drainage systems (i.e. no single irrigation scheme will be larger than 50 hectares) will include intake structures, pumping stations, rain harvesting ponds, main and secondary distribution networks, drainage networks and flood protection structures, among others. Local hydrological studies will be conducted to ensure that mitigation measures to reduce associated impacts will need to be incorporated in the localized ESMP and good practice irrigation technologies (e.g. drip irrigation, water harvesting practices including capture of run-off where feasible) will be implemented and not adversely affect multiple water users. On-farm irrigation/drainage systems for connecting to main or secondary networks (e.g. drip, sprinkler, gravity ditches or field open drains) will be provided by the project to irrigate the plots. The project will promote the combined use of surface and groundwater including recharge activities.
- (c) The SECAP has identified pollution resulting from the use of agro-chemicals and pesticides as a potential impact of the project since it may indirectly encourage the use of agro-chemicals due to increased quality of farm inputs or water availability for irrigation. To manage this, the project plans to engage extension agents to assess viable options to chemical pesticides (e.g. organic alternatives and bio-pesticides) in line with national and international standards. The project also aims to promote Integrated Pest Management (IPM) approaches and has provided for developing a pest management plan. Notwithstanding, the project should conduct training to project beneficiaries during the implementation phase on handling, storage, application and disposal of pesticides and agrochemicals in accordance with the Food and Agriculture Organization's International Code of Conduct on the Distribution and Use of Pesticides or other good international industry practice (GIIP).
- (d) *Land Acquisition:* The SECAP indicates that the project will not acquire new lands in the implementation of the road rehabilitation and agro-processing facilities or any other facilities since it is only expected to improve on the existing land used for such existing activities/facilities. Hence, no relocation and/or loss of shelter or livelihoods of the households or communities are expected at the implementation stage. The project provides for the conduct of an environmental and social screening to ensure that activities requiring acquisition of land or that could lead to physical and/or economic displacement will be excluded from being supported under the project.

- (e) *Biodiversity and Cultural Heritage:* By design, the project will not have any intervention inside protected areas and in natural habitats and critical areas. The SECAP also excludes the implementation of activities in buffer zones that can adversely affect biodiversity and local habitats. Since the project's infrastructural interventions will be on existing facilities, the project is also not expected to affect cultural heritage sites and will exclude activities in such areas.
- (f) *Indigenous peoples:* While the project does not anticipate adverse impacts on lands and resources of recognized or self-identified indigenous peoples, the SECAP incorporates an Indigenous Peoples Planning Framework (IPPF) that provides for the process for ensuring informed consultation and participation in implementing the activities and developing an Indigenous Peoples Plan, including free, prior and informed consent (FPIC), as needed. The IPPF is also supported by the AEs guidance documents on FPIC process.
40. **Institutional arrangements including monitoring and evaluation:** The Ministry of Economic Development (MED) is identified as the lead executing agency that will ensure the creation of a project management unit (PMU). The PMU will host the multidisciplinary team of specialists (including safeguards specialist) who will take on the day-to-day implementation of the project. The site level environmental assessment of the proposed design solutions will be carried out by the Environment Department of the Ministry of Agriculture, Fisheries, Forestry, Environment and Sustainable Development (MOA) which will also undertake the operations and maintenance of irrigation and drainage systems through the Water Users' Associations. The Ministry of Works will undertake the operations and maintenance of the roads over the life of the project. Extension agents will be engaged, and farmer-to-farmer training mechanism will also be established. A project monitoring and evaluation system which includes monitoring of the ESMP is included in the SECAP with broad project parameters that will need to be monitored and managed. During project implementation, further elaboration of the ESMP and the monitoring parameters at the site level will need to be prepared to include site specific measures.
41. **Stakeholder engagement:** Consultations undertaken with various stakeholders in the preparation stage identified the issues raised are presented in the SECAP. The AE has also included an approach for stakeholder engagement for which a plan will be developed with details of how the AE will ensure that information related to the project will be disclosed and stakeholders will be consulted in later stages of the project. Stakeholder engagement will also be undertaken during the ESIA process.
42. **Grievance redress mechanism:** Information on the AE's institutional-level grievance redress mechanism is included in the SECAP with details of the processes that are followed in managing complaints and how the mechanism can be accessed by the stakeholders. A project-level grievance redress mechanism will also be developed and information on how it can be availed and the process for lodging a complaint will be shared to various stakeholders throughout the project life.

4.2 Gender policy

43. The AE has submitted a gender assessment. Therefore, the funding proposal complies with the operational guidelines of the GCF gender policy and action plan. The gender assessment has outlined the general context of gender issues in Belize. Roles of men and women in the sector of intervention have also been included in the gender assessment. Women are actively involved in agricultural activities that contribute to food security of households, in addition to participating in agricultural value chains though they are underrepresented in small-farmer producer organisations, whilst men are more involved in commercial production of agricultural products.

44. Consultations conducted by the AE in January 2019 in the project area identified the differential needs of men and women in the context of the project. It identified that there are differentiated needs for agricultural inputs and equipment, accessing markets and value addition to their produce.
45. A gender action plan has also been prepared by the AE. The gender action plan includes indicators and targets against each of the project's activities. The AE is advised to ensure that gender-related activities in the action plan are described in more detail. For example, clarity and articulation is required of the specific activities that will be undertaken to ensure better representation and participation by women in producer organisations as part of output 2. The AE is recommended to articulate in the gender action plan how the needs and priorities of men and women that were identified during consultations will be addressed. For instance, there is a need to articulate how the matching grant fund will be employed to that effect, and the incentives that will be used to ensure women get access to it.
46. An indicator with sex disaggregated targets has been included in the logic framework of the funding proposal at the fund-level impacts and at the outcome level. Sex-disaggregated targets have also been included for some of the activities in the gender action plan. The baseline study which will be undertaken at the beginning of the project as indicated in the gender assessment should collect information on gender matters that can be used to rationalise targets that have been provided and include sex-disaggregated targets where they have not been included.
47. Responsibilities for the implementation of the gender action plan are assigned to the project management unit. The AE is recommended to ensure that the unit allocates human resources with gender expertise for the continuous implementation, monitoring and evaluation of the gender action plan.
48. Costs is not included in the gender action plan. The AE is advised to ensure financial resources are allocated from the project budget for the implementation of gender-related activities. Timelines need to be aligned with the timeframe, in years, of the implementation of the project as indicated in the funding proposal.
49. The AE is recommended to provide a revised gender action plan taking into consideration the Secretariat's assessment findings before commencing the implementation of the project.

4.3 Risks

50. **Overall programme assessment (medium risk):**
 - (a) GCF is requested to provide approximately USD 8 million to increase the resilience of smallholder farmers. The overall financing structure is balanced, with GCF financing proposed with a split of instruments (76 per cent in grants and 24 per cent in the form of a loan). The programme is co-financed by the AE with investments targeting increased productivity (machinery and facilities), providing an additional USD 8 million loan, together with the Government grant contribution (USD 3.2 million). The programme will be further supported by at least 15 per cent equity contributions from the beneficiaries;
 - (b) The programme targets smallholder farmers and has to be executed in six years with interventions in the value chain (producers' organizations) and an upgrade of public infrastructure necessary for production (roads, drainage and information systems). Good coordination by the implementing entity (i.e. MED) and effective governance will be crucial for successful implementation; and

- (c) The grants resources will be dedicated to the part of the project involving costs that cannot be supported by matching revenues (technical assistance, training) and will cover the incremental costs of climate proofing productive investments to address erratic weather patterns caused by climate change. The GCF grants are therefore directly targeting additional costs for agricultural and rural development in the country brought about by climate change.

51. **Accredited entity/executing entity capability to execute the current programme (low risk):**

- (a) IFAD, the AE, has long and proven experience in policy advisory and financing investments in agriculture, and working experience with the Government of Belize; and
- (b) The executing entity is MED and the Ministry of Finance (MOF) is the representative of the borrower for all financial and legal aspects of the programme. MED has robust experience in capital budgeting and national economic planning as well as oversight in the implementation of projects and coordinating activities with financial institutions and multilateral development banks.

52. **Programme specific execution risks (medium risk):**

- (a) Governance risk (medium): IFAD will have to ensure that MED and other ministries (e.g. the National Climate Change Committee and National Climate Change Office under the Ministry of Forestry and the Ministry of Rural Development) take ownership of the project through the Programme Oversight Committee so that successful execution is widely supported. A potential weakness in coordination of the ministries may delay the project or offer suboptimal technical advice to farmers;
- (b) IFAD will have to ensure that the Project Management Unit (PMU) receives adequate support. Although several stakeholders may be involved, IFAD is familiar with such arrangements and should be able to replicate it effectively as in other supported operations;
- (c) Implementation risk (medium): the recipients of interventions are smallholder organizations that currently have difficulties in accessing markets or establishing stable production. They will be technically advised to execute productive projects and incorporate new technologies to enable income stability on their farms. As nine different production models are targeted for more than 6,000 rural households with less than 25 acres of land, the reach of the project will have to be wide and effective to cover adequately the knowledge transfer, follow the implementation and assure stability of results over time. This sustained effort by the implementing entities has to be assured for 15 years to achieve the forecasted incremental net flows of benefits and reflect consolidation of the activities and new technologies;
- (d) Equity contribution risk (low): the programme relies on technical advisers and Monitoring & Evaluation Managers to verify the beneficiary equity contributions of at least 15 per cent of the total cost for all the approved subinvestments. Market demand for the proposed technologies is expected to be high on average; however, some of the poorest beneficiaries might be unwilling or unable to contribute with the required level of equity. GCF may consider a covenant to be included in the term sheet to ensure that equity contributions are verified before the related GCF funds are disbursed to the subinvestments;
- (e) Financial viability (medium): the financial model is carried out for nine products for a 15-year period. The model assumes the adoption of improved technologies in production systems for farmers who have applied similar technologies or have seen them operating on other farms. The financial viability shows a financial IRR of 15.74 per cent and a net present value discounted at 10 per cent of 9.4 million Belize dollars

(BZD). This level of return may be lower as it depends on the programme effectiveness to enable a more profitable farm yield quickly (higher income generation). With regard to road infrastructure investments, benefits assume effective reduction in transportation costs (e.g. fuel, maintenance) and a reduction in damages and losses of the products transported;

- (f) Economic viability (medium): the project economic viability shows an economic IRR of 18.97 per cent and a net present value discounted at 6 per cent of BZD 29 million. IRR assumes positive externalities that could be better demonstrated (e.g. increased jobs, better quality agri-products). The cash flows use shadow prices to eliminate the effect of taxes and subsidies, market distortions, unemployment and the exchange rate that may be further elaborated and reported as the project progresses; and
- (g) Both financial and economic analyses demonstrate a sufficient resilience in case of stressed scenarios (e.g. 10 per cent decrease in output prices and 10 per cent increase in costs, one-year delay of project implementation).

53. **GCF portfolio concentration risk (low risk):**

- (a) In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of the result area and single proposal is not material.

54. **Compliance (medium risk):**

- (a) The counterpart contributions element of the project may pose a higher risk of money-laundering/terrorist financing due to the commingling of funds from unverified sources. Credit unions are generally not regulated/supervised for anti-money-laundering/counter-terrorist financing (AML/CFT) controls. Furthermore, the funding proposal does not indicate the application of strict fund sourcing standards and does not contain a description of the measures/strategy the AE will adopt to identify and verify the source of funding for the counterpart contributions; and
- (b) While the proposal does not provide information on how this risk will be addressed, the accreditation master agreement (AMA) contains mandatory AML/CFT provisions, to which the AE is required to adhere. These include the obligation to apply international best practice standards regarding AML/CFT controls throughout the project's implementation. The AE is further required to cascade and ensure the implementation of appropriate AML/CFT due diligence and controls in activities conducted by executing entities and other subsidiary third parties involved in the project cycle. Therefore, the elevated risk derived from the counterpart contributions aspect may in effect be mitigated by the AML/CFT obligations of the AE under the AMA.

55. **Conclusion (medium risk):**

- (a) It is recommended that Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall programme	Medium	• Coordination of the different ministries is critical for successful implementation; • Specialized and effective technical assistance is key to enable productive and commercial planning among beneficiaries; and • Experienced accredited entity with similar programmes can ensure
Accredited entity/executing entity capability	Low	
Project specific execution	Medium	
GCF portfolio concentration	Low	

Compliance	Medium	adequate monitoring and effective implementation
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4.4 Fiduciary

56. As the AE, the role of IFAD will be to supervise the project, provide support and resolve issues that may arise during implementation. It will ensure that all fiduciary requirements are met, and that the programme's fiduciary risk is kept at a minimum while monitoring the adequacy of the financial management arrangements of the PMU.

57. MED, the executing entity, will be designated as lead programme agency with overall responsibility for programme implementation. As part of the design, and in accordance with IFAD financial management policies and guidelines, a financial management assessment was undertaken for the proposed programme's arrangements within the lead programme agency, MED.

58. MED will ensure proper financial management and implementation of the project through the creation of a dedicated PMU. The principal mandate of the PMU will be to carry out the overall programming and budgeting and to take the lead for overall project implementation, including working with service providers, Government ministries and departments, individual beneficiaries, producers' organizations and the municipalities of the project's target areas.

59. Two USD accounts will be opened in the Central Bank of Belize, to be maintained by MOF, into which the IFAD and GCF funds will be deposited. The PMU will submit requests to MOF for the transfer of IFAD and GCF funds from the designated accounts to the operating account(s). The Government of Belize counterpart contribution will also be deposited to the operating account(s) in quarterly tranches.

60. The programme's external audit will be conducted by a private audit firm according to IFAD Guidelines on Project Audits. The PMU will prepare the terms of reference for the auditor for review by IFAD on a no objection basis. Procurement for the proposed project would be carried out in accordance with IFAD guidelines and the provisions stipulated in the IFAD General Conditions for Financing.

4.5 Results monitoring and reporting

61. This project focuses on adaptation and spans three result areas – most vulnerable people/communities, health/well-being, food and water security and ecosystem/ecosystem services. The expected direct and indirect beneficiaries are 30,000 (12,000 females and 18,000 males) and up to 95,296 (50/50 gender split), respectively.

62. The revised implementation timetable in section C.8 of the funding proposal incorporated comments related to elaborating activity description. It will further benefit by addressing another comment mentioned previously regarding showing milestones for deliverables associated with the output and/or activity (e.g. timing of submission of an assessment report regarding: 1.1 develop a climate vulnerability assessment).

63. Regarding the logical framework, the revised funding proposal addressed most of the comments and suggestions provided by the Secretariat. It is also noted that the baseline figures will be revised at the start of the project when the baseline survey and subsequent studies are undertaken. The AE is recommended to further improve the logic framework with the guidance provided by the Secretariat during the second-level due diligence.

4.6 Legal assessment

64. The Accreditation Master Agreement was signed with the Accredited Entity on 24 September 2018, and it became effective on 9 November 2018.

65. The Accredited Entity has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project. It is recommended that, prior to submission of the funding proposal to the Board: (a) the Accredited Entity has obtained all its internal approvals; and (b) the Fund has received a certificate or legal opinion from the Accredited Entity in form and substance satisfactory to the Fund confirming that all final internal approvals by the Accredited Entity have been obtained, and that the Accredited Entity has the authority and capacity to implement the project.

66. The proposed project will be implemented in Belize. The GCF has signed a bilateral agreement on privileges and immunities with Belize.

67. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Delivery by the Accredited Entity to the Fund of a certificate or legal opinion within 120 days of the Board approval confirming that it has obtained all its internal approvals;
- (b) Signature of the funded activity agreement in form and substance satisfactory to the Secretariat within 180 days from the date of Board approval or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all its internal approvals and has the capacity and authority to implement the project, whichever is later; and
- (c) Completion of legal due diligence to the satisfaction of the Secretariat.

Independent Technical Advisory Panel's assessment of FP101

Proposal name:	Resilient Rural Belize (Be-Resilient)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential *Scale: Medium*

1. According to the funding proposal, the purpose of the planned activities is to increase the adaptive capacities of poor and vulnerable smallholder farmers against climate change (CC) threats in Belize. The targeted farmers live in 23 communities distributed across five selected priority districts (Belize, Cayo, Orange Walk, Stann Creek and Toledo) in Belize. The three main pillars of sustainable development – environmental, economic and social – are tackled in parallel by the funding proposal to increase the resilience of smallholder farmers which ultimately should increase the target group's capacity to adapt to CC variations and extreme weather events. The outcomes of the funding proposal will contribute to the GCF result areas: most vulnerable people and communities; health and well-being, food and water security; and infrastructure and the built environment.

2. Based on the United Nations categorization, Belize is a small island developing State. According to the Poverty Assessment Report 2010,¹ rural poverty grew to 55 per cent of the total population in 2009, compared with 28 per cent in urban areas. This inevitably has led to a decrease in the adaptive capacities of smallholders, most of whom fall within the poverty range and, as a consequence, are more vulnerable to CC exposure.

3. The project has two components and several cross-cutting activities belonging to both.

4. Component 1 is mainly for strengthening smallholder participation in select value chains (VC) through the promotion of climate-resilient and environmentally sustainable production methods, product diversification, and related innovations. Initially, activities planned in this component will focus on the development of six vegetables (tomatoes, sweet peppers, hot peppers, cabbages, carrots and onions), one fruit (pineapples), and beekeeping products (principally honey). These products were selected from over 20 commodities identified by the Government of Belize and community stakeholders through a participatory process. This component comprises:

(a) Technical support and the physical presence in the project districts of climate-resilient agriculture specialists working with extension agents of the Ministry of Agriculture;

(b) Establishment of a competitive matching grant fund (MGF), funded by a GCF loan and the Government of Belize, is part of this component. It will be structured to support climate-resilient production and value-chain development investments. Producer organizations and individual members receiving support from technical areas of the project will be eligible for funding;

¹ Funding proposal, page 14.

- (c) Increase food security, self-consumption and healthy food choices through the implementation of backyard gardens.
5. Component 2 aims to support public climate-resilient infrastructure enhancing smallholder farming business/rural enterprise opportunities while serving the largest number possible of direct and indirect project participants. It includes:
- (a) Providing investments for rehabilitation and provision of climate-proof roads and drainage and irrigation schemes;
 - (b) Establishing a nationwide climate information system;
 - (c) Formation of Water user associations and providing technical assistance for institutional development and management of irrigation and drainage schemes.
6. Both components are supported by cross-cutting activities: capacity-building, research and development (R&D), and policy engagement/consultation.
7. The total project cost is USD 20 million, of which USD 8 million will be provided by the GCF – USD 6.09 million in the form of a grant and the rest from a senior loan. The senior loan will be a sovereign loan for the government, used to fill the MGF.²
8. As it reported in the funding proposal, the expected number of direct beneficiaries of the project is 30,000 people (7.8 per cent of the country's total population), including 12,000 women and 18,000 men. Some 3,500 rural households in the villages and communities make up the priority areas jointly identified by Government of Belize and the International Fund for Agricultural Development (IFAD), the accredited entity (AE), for programme implementation. Another 2,500 households from non-priority areas will be identified during implementation, totalling 6,000 households. According to the country poverty assessment, poor households in rural Belize comprise five people or more, which equals a total of 30,000 direct affected project participants. Given that 30 per cent of farmers in Belize are women, farming is a family business in which women engage in several farming and marketing activities, and women are the main group responsible for backyard gardening, around 40 per cent of all project participants will be women and 60 per cent will be men. It is assumed that the project will have an indirect impact on an additional 23,243 households engaged in farming in Belize. The number of indirect beneficiaries is calculated with the assumption that the average household size is 4.1 people per household, which leads to 95,296 indirect beneficiaries.
9. The indicators for assessment of the project impacts are as follows: (i) the expected number of rural households engaged in climate vulnerability assessments and trained in climate resilience practices and strategies; (ii) the number of households with improved agricultural productivity and market opportunities; (iii) the number of rural people with strengthened resilience (male and female); (iv) the number of households that have increased resilience and food security through backyard gardens; (v) an increase in the number of people (male and female) with access to agricultural water supply to ensure water supply during the growing season; (vi) kilometres of road constructed, rehabilitated or upgraded to all-weather status; the number of regulatory innovations and/or mechanisms for climate-responsive planning and development; (vii) the number of households reporting adoption of environmentally sustainable and climate-friendly technologies or practices; (viii) the number of facilities (covered structures, storage facilities) rehabilitated/constructed for CC; and (ix) the number of people receiving climate information for decision-making.
10. The impact of activities planned in the funding proposal is assessed as medium at this stage because only a pre-feasibility study on CC risks and relevant adaptation measures is

² The MGF will cover – including a project participant contribution (15 per cent) – investments in sustainable and climate-resilient production, such as hurricane-adapted low-cost greenhouses, renewable energy pumps, agroforestry systems and technical assistance for climate-resilient practices.

provided in the submission. This will be updated to a feasibility study after the project initiation, and some identified CC risks, smallholder farmers' needs, and relevant measures may be revised later. Training materials, which should be based on a final feasibility study, are also not provided at this stage of assessment. In addition, the independent Technical Advisory Panel (TAP) is concerned that (based on the budget distribution) too much focus is placed on the development of training materials and study tours over investment activities.

1.2 Paradigm shift potential

Scale: High

11. There are several activities in the project concept that could be considered as transformational for increasing the CC resilience of agriculture and rural development in Belize. These are:

- (a) Consideration of CC-related risks when planning agriculture and rural developments. Assisting smallholder farmers in assessment of the CC impact on the value chains and in development with business proposals for their MGF application;
- (b) Support of loan application activities for financing the adaptation activities through MGF;
- (c) Introduction to renewable energy technologies in the agriculture sector, such as solar panels and solar pumps;
- (d) Conducting special study to assess mechanisms which provide Belizean smallholders with climate disaster insurance, allowing them to make production investments with greater financial security³. This study will be conducted within the cross-cutting component of the funding proposal, an applied R&D. Potential mechanisms include an index-based or conventional indemnity insurance scheme (subcomponent 1.2, cross-cutting activities policy); and
- (e) Assessment of feasibility of using ecosystems payment services to stimulate maintenance and restoration of local ecologies (e.g. native forests) which provide essential environmental services but do not currently generate revenue for smallholder farmers and other landowners will be also conducted within the same R&D component.

12. Potential for transformational changes is high, though the independent TAP considers that practical implementation of R&D initiatives considered under d) and e) would be more appropriate target for GCF.

1.3 Sustainable development potential

Scale: High

13. The sustainable development potential of the funding proposal is high, contributing to six Sustainable Development Goals:

- (a) Goal 1: No poverty - "Economic growth must be inclusive to provide sustainable jobs and promote equality." The poorest people and communities in Belize are predominantly rural, and their livelihoods depend heavily on climate-sensitive sectors such as small-scale agriculture, fishing, and tourism (which impact their food security and livelihoods). Moreover, a large proportion of the poor living in coastal areas are exposed to the risks of hurricanes and storm surges, with those living near watersheds prone to flooding. These areas generally have lower levels of protective infrastructure,

³ A mechanism for smallholder farmers to access a disaster recovery fund for Caribbean countries – launched by the Caribbean Development Bank and Caribbean Catastrophe Risk Insurance Facility within the Integrated Sovereign Risk Management in the Caribbean project – is not yet defined.

and housing is of lower quality, thus increasing their vulnerability to hurricane and flood risks. Rural poverty is dominated by agricultural workers. The project mainly targets poor smallholder farmers and will ensure their inclusiveness in the decision-making process;

- (b) Goal 2: Zero hunger – “The food and agriculture sector offers key solutions for development, and is central for hunger and poverty eradication.” As described in the funding proposal, in general, poverty in Belize is associated with living from hand to mouth (and some days experiencing hunger), a lack of jobs, inadequate or absent amenities, poor housing, a lack of good-quality water, education and/or enough money to send children to school, and a general struggle to survive. This project aims to increase the adaptive capacity of poor smallholder farmers and reduce poverty and hunger by supporting them to increase the productivity and quality of their agricultural products and increase accessibility to market;
 - (c) Goal 5: Gender equality – “Gender equality is not only a fundamental human right, but a necessary foundation for a peaceful, prosperous and sustainable world.” As the Ministry of Agriculture is one of the few Government of Belize ministries with a gender focal point, the project will support activities designed/led by the focal point team. The projects’ strategy for social inclusion and gender is based on the mainstreaming of both concepts throughout most activities in order to increase the participation of women and young people;
 - (d) Goal 8: Economic growth – “Decent work and economic growth.” The project is oriented around economic growth in rural areas and, in particular, the agriculture sector, aiming to increase the resilience of rural infrastructure and agriculture activities to CC and economic shocks;
 - (e) Goal 13: Climate change – “Educate young people on climate change to put them on a sustainable path early on.” The project is dedicated to developing climate risk-related information and providing training to the different stakeholder groups (e.g. government, extension centres/officers, local smallholder farmers, etc.); and
 - (f) Goal 15: Biodiversity, forests, desertification – “Plant a tree and help protect the environment.” This includes improvement of soil management and an increase in the resilience of crops and many other activities that contribute to this goal.
14. The sustainable development potential of the funding proposal is high as far as, conceptually, the project’s purpose is to change business-as-usual development to sustainable development of agriculture sector.

1.4 Needs of the recipient

Scale: High

15. As a small island developing State in the Caribbean hurricane belt, Belize is highly vulnerable to the impacts of CC. While the most extreme and visible effect is the increased intensity of tropical storms and hurricanes, CC also manifests itself through increased temperatures and sea-level rise as well as the documented trends of increased droughts, flooding and significant rainfall pattern variations. Belize is one of the most vulnerable countries to CC in the world. An assessment of individual countries’ vulnerability to climate threats, prepared by the non-governmental organization Germanwatch, revealed that over the 20-year period 1996–2015 Belize was ranked twenty-sixth out of 181 countries in the Global Climate Risk Index.⁴ This research identified important vulnerabilities in the food sector, affecting food security, nutrition and livelihoods.

⁴ Available at <<https://germanwatch.org/de/download/16411.pdf>>.

16. According to a recent systematic country diagnostic by the World Bank Group, Belize is within the group of countries most adversely affected by weather-related events and other natural hazards. As such, Belize incurs annual losses of close to 4 per cent of gross domestic product due to natural disasters (Carneiro, 2016).
17. Several studies, including the country's second and third national communications state that an increase in temperature has an adverse effect on economically important crops to the country, such as sugarcane and citrus. Furthermore, higher temperatures resulting from CC have already caused soil aridity, a proliferation of pests and diseases and put pressure on water resources for water irrigation purposes. Sea-level rise has already intensified flooding and saline intrusion leading to soil salinization.
18. The new agricultural policy framework (National Food and Agricultural Policy 2015–2030) and the National Adaptation Strategy to Address Climate Change in the Agriculture Sector (launched in 2015) emphasize the importance of the sector to Belize's economy and highlight the need to implement adaptation measures and increase product value to ensure maintenance of food security. Both policy documents also stress the vulnerability of small-scale/family farmers and their crucial role in rural development.
19. In particular, the National Adaptation Strategy to Address Climate Change in the Agriculture Sector in Belize reports that CC and climate variability will impact agriculture systems and practices such as soil fertility and land preparation; pest and disease control; and water requirements (excess and deficits). Higher temperatures will cause increased stress on current livestock breeds, and crop types and varieties. It is very likely that climate change and climate variability will result in reduced rainfall overall, but the most detrimental effect is likely to come from the variation in the seasonal distribution of rainfall, leading to more periodic droughts and floods.
20. A preliminary survey of different stakeholders (policy makers, agriculture producers and support institutions) revealed that pests/diseases and soil nutrition management were considered the most critical adaptation needs for Belize.
21. In the regions considered by the funding proposal, the following CC-related threats and adaptation needs were identified:
 - (a) **Toledo.** This is the district with the highest risk of increased drought frequency and severity. Land productivity dynamics revealed that 68,304 hectares (ha) reported declining productivity, an additional 11,963 ha presented early signs of decline, and 29,707 ha are currently stable but stressed. The selected VC, pineapple, is sensitive to rainfall variability but highly adaptable with irrigation. There is a high willingness among local farmers to adopt different crop varieties that better adapt to droughts;
 - (b) **Stan Creek.** This district is at risk of increased drought frequency and severity, and hurricanes. Land productivity dynamics revealed that 59,566 ha reported declining productivity, an additional 7,202 ha presented early signs of decline and 35,877 ha are currently stable but stressed. The selected VC, hot pepper, is highly sensitive to the salinity of soil and waters, and water stress. It is better adapted to hot climates. There is a high willingness of local farmers to adopt different crop varieties that better adapt to droughts;
 - (c) **Cayo.** This district reported the greatest decline in productivity and is at risk of an increase in flood/drought frequency. Land productivity dynamics revealed that 194,046 ha reported declining productivity, an additional 5,086 ha presented early signs of decline and 75,947 ha are currently stable but stressed. The selected VC, sweet pepper, is highly sensitive to salinity of soil and waters, pests, diseases and water stress. Another selected VC, cabbage, is irrigation dependant, highly sensitive to water limitation and sensitive to flooding, but can benefit from warmer temperatures. The third selected VC, tomato, is moderately sensitive to reduced irrigation and salinity and can benefit from

warmer temperatures within a tolerance range. It is sensitive to flooding in high temperatures;

- (d) **Belize.** This district is at risk of increased flooding frequency and severity, and most at risk from increased frequency/intensity of hurricanes. Land productivity dynamics revealed that 101,969 ha reported declining productivity, an additional 7,961 ha presented early signs of decline and 55,756 ha are currently stable but stressed. The selected VC, onion, is irrigation dependent, and sensitive to saline soils, warmer temperatures and excessive rainfall. It is adaptable with shading/protective screens, controlled irrigation and adequate drainage systems. Other selected VCs are tomato and sweet pepper, already described above; and
- (e) **Orange Walk.** This district is at risk of increased flooding frequency and severity, and most at risk of increased frequency/intensity of hurricanes. Land productivity dynamics revealed 175,655 ha reported declining productivity, an additional 2,872 ha presented early signs of decline and 50,711 ha are currently stable but stressed. The vulnerability of selected VCs, onion and carrot, are described above.

22. The target group of the project is smallholder farmers with fewer than 25 acres of land, who are engaged in part-time or full-time farming or are agricultural workers and have the willingness and potential for improving productivity and access to markets. This group can be defined as poor, those whose income is below the poverty line, and vulnerable rural families, whose incomes are 25 per cent over the poverty line but likely to fall into poverty as a result of increased CC effects (e.g. drought, floods) and economic external shocks (changes in prices).

23. Based on the information provided in the national strategies and communications, and the adaptation needs of selected regions identified through the pre-feasibility study, the independent TAP concludes that the need for further investigation into problems and implementation of adaptation needs is high for the country and for selected districts.

1.5 Country ownership

Scale: High

24. In its nationally determined contribution (NDC), the Government of Belize considers adaptation to CC as a high priority. In addition to its vulnerability to natural disasters and climate-related shocks, Belize's small and open economy, geographical location and the lack of vital resources constrain its capacity to adapt to the effects of CC and variability in the short and long term.

25. The NDC for Belize is consistent with the overall goal of the Growth and Sustainable Development Strategy (GSDS) of the country, which encompasses medium-term (2016–2019) economic development, poverty reduction, and longer-term sustainable development. The GSDS is the nation's primary planning document and outlines critical success factors for the country's development and to ensure a better quality of life for all Belizeans, living now and in the future. This planning document also provides detailed guidance on priorities and on specific actions to be taken during the planning period, including actions that contribute to longer-term development objectives beyond 2019.

26. The National Climate Resilience Investment Plan (2013) provides the framework for an efficient, productive and strategic approach for building economic and social resilience and development. Special importance is given to building climate resilience and improving disaster risk management capacities across all sectors.

27. The National Climate Change Policy, Strategy and Action Plan, 2015-2020, provides policy guidance for the development of an appropriate administrative and legislative framework, in harmony with other sectoral policies, for the pursuance of a low-carbon development path for Belize.

28. In all of the listed documents, agriculture is considered one of the most vulnerable sectors of the economy to CC. Adaptation measures considered in the NDC highlight the need to improve both crop and livestock husbandry practices; increase access to drought tolerant crops and livestock breeds; adopt better soil and water management practices; reduce post-harvest losses; and provide early-warning/meteorological forecasts and related information to be competitive in the region.
29. Capacity-building efforts around adaptation planning is a priority. In addition, resource needs are expected to be significant over the period used to implement the adaptation actions that are specified in the national CC action plan. Some specific CC adaptation needs in the sector plans include the need to educate different stakeholder groups about CC adaptation measures and to help them develop their capacity to research, develop and implement adaptation strategies. Some institutions have recognized the need to shift from general public awareness and education to a community-based approach for environmental education programmes, which would address specific issues and concerns. This is key concept of the funding proposal.
30. All the strategies listed above and in other documents, such as the national communications to the United Nations Framework Convention on Climate Change, demonstrate a very deep analysis of the CC vulnerability of agriculture and rural infrastructure as well as the country's adaptation needs. The quality and detail of the reports are very high, which demonstrates the country's clear ownership of CC adaptation problems. The activities planned in the funding proposal directly contribute to this priority area of the country's GSDS and to the approach to shift from general public awareness and education to a community-based approach for environmental education programmes. Therefore, country ownership is considered as high.

1.6 Efficiency and effectiveness

Scale: Medium

31. Similar to the impact assessment, the opinion of the independent TAP is that a better balance between the investment component and technical assistance (training and consultancy) would better fit GCF policy.
32. Through different national documents and a pre-feasibility study provided by the accredited entity, it was identified that soil fertility and pests/diseases are the main CC risks for the agriculture sector in selected districts considered in the funding proposal. However, the funding proposal considers rehabilitation of public roads and irrigation systems as key adaptation activities for investment against floods and droughts, and for the increase of the adaptive capacity of smallholders. Consequently, it is the opinion of the independent TAP that, after a full and inclusive feasibility study is conducted for these districts, some risks and adaptation measures might be changed.
33. In addition to the above, the efficiency and effectiveness of the project should be assessed based on a feasibility study for adaptation measures and training materials developed. The pre-feasibility study and draft outline of training materials submitted at this stage of the assessment do not provide enough evidence on which to categorize the efficiency of the implemented activities as high. Therefore, these criteria are assessed as medium.

II. Overall remarks from the independent Technical Advisory Panel

34. Taking into consideration the high climate change vulnerability index of the country, needs of target groups and urgency of actions, the iTAP recommends approval of this FP subject to the following condition:

Prior to the first disbursement of funds by GCF under the funded activity agreement, the AE shall deliver in a form and substance satisfactory to the Secretariat:

- (a) Updated Appendix 11 of the Funding Proposal - Programme Implementation Manual including the criteria for the selection of sub-projects (climate change, financial, social, environmental) eligible to receive financing by MGF (Matching Grant Fund) and IIFA (Infrastructure Investment Funding Award), such eligibility criteria shall take into consideration sector/measure specificities and climate change dimension;

Prior to the second disbursement of funds by GCF under the funded activity agreement, the AE shall deliver in a form and substance satisfactory to the Secretariat:

- (a) Final outline and the first draft of the training materials, prepared on the basis of the feasibility study, to be used during the project implementation, which shall be delivered by the Accredited Entity to the Fund before the start of the training activities;
- (b) A document containing the description of some sub-projects for each sector (increase road resilience, irrigation systems, renewable energy use in irrigation systems, increase crop resilience or fertility of soil, etc.), which are considered as potential adaptation measures.

Response of the accredited entity to the independent Technical Advisory Panel's assessment (FP101)

Proposal name: Resilient Rural Belize (Be-Resilient)

Accredited entity: International Fund for Agricultural development (IFAD)

Impact potential

IFAD acknowledges ITAP's assessment. Recommendations have been included in FP and will be further refined in PIM. The detailed vulnerability/value chain analysis at the regional level will be conducted in Year 1 of project implementation with project stakeholders. This participative methodology will directly lead to on-site investment to increase smallholders' resilience to climate change. A revised training materials outline is provided as per ITAP recommendation, and materials will be fully developed in Year 1 of the project as agreed with ITAP and GCF. Project funding is largely targeted towards investment (80% of GCF funding and 64% of project total), noting that TA is essential to promote a shift towards climate resilient agriculture.

Paradigm shift potential

IFAD acknowledges ITAP's assessment

Sustainable development potential

IFAD acknowledges ITAP's assessment

Needs of the recipient

IFAD acknowledges ITAP's assessment

Country ownership

IFAD acknowledges ITAP's assessment.

Efficiency and effectiveness

IFAD acknowledges ITAP's assessment and agrees that final vulnerability assessments and value chain analysis will identify site specific risks and adaptation measures that will refine the list of options provided in the FP. It is IFAD's position that the final development of these assessments is best done in a fully participative manner with project stakeholders. For increased efficiency and continuity of project activities, this assessment is planned as step 1 of project implementation, to be followed up directly with investment in the proposed adaptation measures.

Overall remarks from the independent Technical Advisory Panel:

IFAD is in full agreement with ITAP's project assessment and confirms that it is able to comply with the stipulated conditions.



Investing in rural people

Belize

Resilient Rural Belize (Be-Resilient)

Gender Assessment and Action Plan

GCF Additional Financing

Part I: Gender Analysis/Assessment:

Belize has a total population of 395,882 inhabitants according to the Labor Force Survey of April 2018. The rural population is slightly higher than the urban population (55% and 45% respectively) and equally slight is the difference between the percentage of men and women (56% and 54% respectively).

Belize has taken important steps in fulfilling its international commitments in favour of gender equality and equity and has renewed its Gender Policy in 2013. The government and civil society participation was part of this renewal, which generated a consensual document, with important modifications in relation to the previous one. Compliance with the Gender Policy requires efforts on the part of different sectors and governmental, non-governmental and civil society entities, which are still incipient. However, the progress made in central issues such as health and education are a sign that it is on the right track.

The Government of Belize, recognizing that rural women are the most affected by poverty and lack of access to services and resources for personal development and for their contribution to the development of their communities and territories, participates in the framework of the Agricultural Council Central American (Council of Ministers of Agriculture of the Central American Integration System Region and the Dominican Republic) in the development of an Agenda for the Economic Empowerment of Rural Women and has a focal point in the Ministry of Agriculture and in the Women and Family Support Department (Ministry of Human Development) to implement this agenda at the national level and give a better response to these women. Human, material and financial resources are currently insufficient and it will be necessary to strengthen the institutional ones in order to generate the expected results.

General information at National level

Maternal mortality rate	Maternal mortality rate was 28 per 100,000 live births, estimated in 2015 Source: World Bank. https://datos.bancomundial.org/indicador/SH.STA.MMRT Consulted 12/20/2018																
Infant mortality rate (per 1,000 live births)	The percentage of infant mortality per 1,000 live births is 13.4 for men and 11.0 for women. (2017) Source: World Bank. https://datos.bancomundial.org/indicador/SH.STA.MMRT Consulted 12/20/2018																
Educational status of girls and boys	The satus of formal school attendance shows that boys and girls between 2 and 16 years, segregated by age groups, show almost equal attendance levels and that the decrease in attendance in the group from 13 to 16 is a common phenomenon for men as women. <table><tr><th colspan="4">Formal* School Attendance Status (2010)</th></tr><tr><th>Age</th><th>Total</th><th>Males</th><th>Females</th></tr><tr><td>2 - 4</td><td>33,1</td><td>32,6</td><td>33,6</td></tr><tr><td>5 - 12</td><td>95,9</td><td>96</td><td>95,9</td></tr></table>	Formal* School Attendance Status (2010)				Age	Total	Males	Females	2 - 4	33,1	32,6	33,6	5 - 12	95,9	96	95,9
Formal* School Attendance Status (2010)																	
Age	Total	Males	Females														
2 - 4	33,1	32,6	33,6														
5 - 12	95,9	96	95,9														

	<table><tr><td>13 - 16</td><td>75,9</td><td>75,5</td><td>76,4</td></tr></table> Source: The Statistical Institute of Belize. Belize Population and Housing Census. 2010 Country Report 2013	13 - 16	75,9	75,5	76,4
13 - 16	75,9	75,5	76,4		
Adult literacy rate (disaggregated by sex),	Literacy covers almost the entire population of Belize, reaching 97% among women and 95% among men (2015) Source: Unicef. 2017. Child Rights Monitoring: Reporting on Children in the SDGs. drawn from the 2015 MICS5 Survey.				
Poverty rate	Poverty in Belize affects 40% of households and 41.3% of its inhabitants. 13% of non-poor households are in a vulnerable situation and 14% of the population is in that situation. The sex of the individual has little influence on poverty rates with men having slightly higher poverty rates (42% of men are poor compared to 40% of women). Source: IFAD. 2009. Poverty Assessment Report				
Labour force participation rate (disaggregated by sex),	By April 2018, the participation of the population over 14 in the labor force is 65.5%, Disaggregated by sex, 78.3% of men and 52.9% of women are part of the labor force. labor force, which gives a difference close to 25 percentage points in favor of men. However, these data do not take into account the contributions to domestic work and care that women perform. Source: The Statistical Institute of Belize. Labour Force Survey Tables Abril 2018. http://sib.org.bz/statistics/labour-force/ consulted 07/01/2019				
Unemployment rate (disaggregated by sex),	By April 2018, female unemployment is 15% while male unemployment is 6%. In rural areas, female unemployment rises to 16%. While only 1.5% of the unemployed are male heads of household, 7.7% are women. Source: The Statistical Institute of Belize. Labour Force Survey Tables Abril 2018. http://sib.org.bz/statistics/labour-force/ consulted 07/01/2019				
Political participation rate (disaggregated by sex),	Belize has one of the lowest percentages of women holding seats in parliament. In 2017, the share of seats in parliament held by women was 11.1%. The House of Representatives is made up of 32 members, of which 3 are occupied by women (9.38%), The Senate consists of 13 members and women occupy 2 seats (15.38%) (2018)				

	Data from 2012 show that women achieve higher levels of participation in political positions such as municipal councils, village councils and mayors. In total, women were 25.9% of councilors and 22.2% of seniors. Source: UNDP. http://hdr.undp.org/en/data consulted 07/01/2019, Union Interparlamentaria- Cámara de Representantes: Belice, 2018. Consulted 02/03/2018 at http://archive.ipu.org/parline-e/reports/2031 and UNDP. 2015. Were are the women. A Study of Women, Politics, Parliaments and Equality in the CARICOM Countries.
Life expectancy (disaggregated by sex)	In 2016, life expectancy for women is higher than men in approximately five and a half years. While women are expected to live 73.4 years, men barely reach 67.74 years. Source: World Bank. https://databank.worldbank.org/data/source/gender-statistics

Country of intervention

What is the legal status of women in the country of intervention?
The government of Belize is a signatory to all the main international conventions setting out rights for women. Entre ellos, the Convention on the Elimination of all Forms of Discrimination Against Women (CEDAW) passed in 1979; this is the first international instrument that is not neutral in terms of gender, and promotes the creation of affirmative action measures to increase participation opportunities for women in all sectors. Mainly art. 14 refers to rural women. At the national level, the legislation enshrines a series of principles and guarantees for all inhabitants. However, there are criticisms from groups of feminists and intellectuals that the national constitution is conservative and invisible to women. Belize has a Gender Policy from 2013, which updates the previous Policy (designed in 2002 by the National Commission for Women, and approved by legislative bodies) and emphasizes the diversity of the Belizean population and the need for coordination and harmonization of strategies with other stakeholders, especially government institutions and civil society organizations. Additionally, it has the Domestic Violence Law of 2007, the revision of the Penal Code of the year 2000 and the Law of Protection against Sexual Harassment of 1996, revised in the year 2000.
What are commonly held beliefs, perceptions, and stereotypes related to gender
Traditional gender roles are maintained in Belize. Women continue to be responsible for domestic tasks and care. The stereotyped assignment of these roles assumes that their abilities lie in the performance of those tasks, limiting their opportunities to the private sphere of the family, affecting their participation in economic activities. Men, on the other hand, are assigned the tasks of economic character in the public sphere.

Despite advances in this area and evidence of women's work at the community level and in productive activities, these roles are reproduced in the family, school and other mechanisms that maintain these cultural patterns, mainly in rural areas.

What is the division of labour among women and men?

From the analysis of the activities carried out by men and women during the 24 hours a day, the imbalance in the allocation of domestic and care activities was evidenced, which are carried out in their totality by the women who dedicate an average of 8 hours to the day to its execution, while the men do not dedicate time to them (there was only one case of a man who invests 1 hour a day in going to pick up his children to school). This imbalance mainly affects the hours of rest (about 6 hours for men and 4 hours for women) and the participation of women in productive activities (10 hours for men and 7 hours for women). On average, both sleep the same amount of hours (7 hours).

Source: Consultations conducted in 2 workshops in the project area (January 10 and 14, 2019)

What is the participation between women and men in the formal/informal economy?

5 out of 10 women and 8 out of 10 men of working age are integrated into the workforce in Belize. The majority of both are services and sales workers (19% males and 34% females) or they work in elementary occupations (23% males and 17% females) *. The informal economy is also important in the country and it is women more than men who seek opportunities to generate income in this sector, although the amounts are low, the risks are high and the legal and social protection is very weak.

Source: The Statistical Institute of Belize. Labour Force Survey Tables April 2018. [http://sib.org.bz/statistics/labour-force/consulted 07/01/2019](http://sib.org.bz/statistics/labour-force/consulted%2007/01/2019)

Project footprint area

What is the situation of women and men?

In the rural areas of Belize, families live mainly from agriculture and the provision of services, especially related to tourism. Most families involve more than one of their members in the activities of production and income generation. In the case of agricultural or other activities that take place close to the home, women are actively involved although they are usually unpaid family workers or simply are not recognized in their work, are not paid and are not considered part of the labor force.

While men are more linked to the activities of commercial production, women make a significant contribution to the food security of families and participate in the links of agricultural value chains related to the availability of raw materials (such as nurseries) and value aggregation (such as packaging and transformation).

Domestic tasks and care are performed by women and are not remunerated or socially recognized as a contribution to the family or national economy.

The sex of the individual has little influence on poverty rates with men having slightly higher poverty rates (42% of men are poor compared to 40% of women). However, the hardships caused by poverty are no equal on men and women; due to the responsibilities assigned by society to women, they will often experience the sharp end

of having to deal with the lack of adequate funds to maintain their household, particularly where they have no independent source of income (CPA, 2009).

In terms of the proposed project/program, will there be any anticipated differences in men's and women's vulnerability and adaptive capacity to climate change? If so, what are these?

There are differences. The effects of tropical storms, hurricanes and other events such as droughts and floods are devastating for women, more so than for men. Due to women's role within the family, particularly as care givers, women assume much of the stress caused by these phenomena and suffers the effects of the lack of necessary resources for her attention.

The group most vulnerable to climate are women in conditions of poverty such as those who live in the project area, who are the most prone to suffer malnutrition due to the tendency to prioritize men for food consumption in the face of shortages. Chauvinist attitudes are also common, limiting the participation of women in decision-making about the handling of emergencies and the distribution of resources, including financial and material resources to reactivate agricultural and other activities, related to the generation of income.

Women have a different capacity to adapt than men, as a result of their roles in the family and the community. This capacity must be increased through access to decision-making and the necessary resources to face the emergencies caused by climate change and extreme climatic events, in its different manifestations

Are there existing gender inequalities that may be exacerbated by climate change impacts in the proposed project/program footprint area?

The impacts of climate change could exacerbate gender inequalities in different ways, including:

- there may be an increase in women's time dedicated to family care activities, given the limitations on access to resources and necessary services for their attention, among which the availability of food stands out,
- the economic losses produced by disasters can put at risk the availability of productive assets and household, making it more difficult for women to maintain or replace them due to their limitations of access to financing,
- if assistance to address the imbalances caused by climate change is limited and there is little participation of women in decision-making about its distribution, the likelihood of this resulting greater access by men to the detriment of women is quite high.

What are some of the inequalities that exist between different social groups in the project/program footprint area? How do these inequalities affect people's capacity to adapt to climate change?

Some of the inequalities that manifest themselves in the project area are:

Income inequality. Women receive less income for their work than men, even among those in the informal sector, placing them in a more vulnerable position due to their inability to acquire or restore their assets to adapt to climate change.

Inequality in access to goods and services. The supply of goods and services provided by institutions usually has a gender bias, relegating women's access to the background after men. The limitations on access to land,

technology, technical assistance and other services limit their ability to adapt to climate change. Inequality of access to information. Women are less exposed than men to the media that disseminate climate information or information about resources and services available to cope with the effects of climate-induced disasters. The availability of smartphones by the majority is an option to get information to women. Inequality in capacity building opportunities. Although the formal education of women is slightly higher than that of men, their possibilities of receiving non-formal training on issues related to climate change is lower due to their low participation in existing organizations, mainly in productive ones in rural areas. and ignorance of their role in facing emergencies in the family and the community.
What roles women and men are anticipated to play in the context of the project/program? What will these entail in terms of time commitment and need for mobility?
Mechanisms will be established for women to participate in all the activities proposed by the project in an average of 40% of the total beneficiaries. Because women are currently underrepresented in small-farmer economic and productive organizations, they will act on the modification of traditional patterns of role assignment for men and women, so that they have more time to achieve the expected participation. The project will provide training in gender equality at all levels, provide resources for affirmative actions that help address existing barriers in organizations and communities that will benefit from the project, monitor their active participation in each of the project activities and build roads to facilitate the mobility of products and people.
What resources (economic, financial, physical, natural, other assets) do women and men have access to? Who manages or controls access to these resources?
According to the results of the consultations held during the formulation of the project, women, like men, consider that both have access to resources or assets of the home, resources for production and natural resources. However, although most women also consider that the control of these resources is in the hands of both, some believe that men have exclusive control over natural resources and in the case of men, although most believe that women in terms of shared control of resources, some consider that they have control over all resources. It emphasizes that neither men nor women assign control of resources in a unique way to women, as happens in the case of men. Source: Consultations conducted in 2 workshops in the project area (January 10 and 14, 2019)
Do women and men from vulnerable communities have equal access to information and opportunities necessary to participate and benefit fully from the anticipated outcomes of the project/program?
Access to information on productive aspects and climate change is limited for both men and women, but mainly for women. In response to this limitation, the project will develop a climate information system and will support the improvement of market information technologies to make this information available to the rural communities that need it most. Small-farmers will be direct beneficiaries including men and women alike. Information and dissemination about events and activities of the project will be disseminated through means that guarantee the participation of men and women in the expected proportions.
Do women have equal access to education, technical knowledge, and/or skill upgradation?

93% of women and 91% of men aged 15-24 living in Belize are literate. Women and men have equal access to formal education at all levels and before age 16 have equal percentages of attendance at formal education in different age groups. Women of working age have higher educational levels than men: they are less among the people with only the primary level (46% vs 40%) and they are the majority with the secondary level or more (44% vs 39%) .

According to information from the LFS of April 2018, 32545 people from rural areas mentioned having received some type of training for work and of them 31% were women while in the urban area they are 40%. In the case of women, 61% received training related to Clerical / Service / Sales workers (25% of the men) while 53% of the men were trained in Skilled Agric / Craft / Plant & Machinery (and only the 18% of women). This information allows us to reflect on the lack of training for the work received by women in the topics of greatest interest to the project and justifies the efforts to bring the services related to agricultural value chains and climate change closer to women as well as to women. men gives rural families small-farmers.

Source: Source: The Statistical Institute of Belize. Belize Population and Housing Census. 2010 and Labor Force Survey April 2018

Will services and technologies provided by the project/program be available and accessible to both women and men?

The Project will create mechanisms to guarantee the participation of women and men in the defined percentages. It will invest in technical assistance and training for better performance of both in prioritized value chains and markets, including the use of technologies that promote resilience and adaptation to climate change. They will have access to climate information systems and markets and prices of agricultural products. For their participation in POs, they will also have access to capacity building for organizational strengthening.

To what extent do women and men from vulnerable communities participate in decision – making processes? What type of decisions are made by women? What are the constrains (social, cultural, economic, political) that restrict women’s active participation in household and community level decision – making processes?

Women are responsible for decision-making at home. Within their communities, where possible, they occupy decision-making positions in organizations related to health, school and others related to their traditional domestic and care role. Their participation is limited in the decisions on political, productive and financial aspects that are considered of the scope of action of men

Are there any opportunities to promote the leadership of women in local governance/political systems and formal/informal institutions? If not, what are some of the constrains that hinder women from assuming leadership roles?

According to the available information, women have greater opportunities to occupy leadership positions in local politics than at the national level. Additionally, the 2011 Agricultural Census reveals that 30% of farmers are females who are in charge of taking decisions in all activities in their farms and more than 25% of women work in irrigation and water management.

In spite of the above, women in rural areas have little participation in producer organizations and their participation in value chains and adaptation to climate change has not been made visible, so their opportunities

for leadership are diminished . The project proposes to increase the participation of women in OPs and their presence in leadership positions that can then be scaled up to institutional and political levels, as has been the experience in other countries with projects executed by IFAD. It will also provide more information about the participation of women in value chains and climate change, to identify niches of opportunity for their participation in decision-making in those fields.

What are the differential needs/priorities of women and men in the context of the project/program? Will the project/program be able to address their respective needs and priorities? If so, how?

Once informed about the objectives of the consultations and provided the central information of the project, the participants analyzed their needs and priorities. For this, they focused on the activities they carry out in the agricultural value chains in which they participate. The priority needs common to men and women are: irrigation systems, road access to farm, cover structures, seeds to plant, chemicals and tools. The men also prioritized processing for vegetables, machinery and equipment and women, supplies for canning, small machineries, cold storage and nurseries.

The project's strategy is based on the confluence of the building resilience to climate change, extreme climatic events and the agricultural value chain approach, which allow integrating the needs and priorities expressed by the people during the consultation. The project will make available to the producer organizations resources of the Matching Grant Fund (MGF) for the financing of initiatives framed in value chains prioritized from the link of raw material to the market and will have technical assistance and training for their development considering climate change. Additionally, this fund will allow investing in affirmative actions that favor the participation of women in the initiatives that are financed.

Source: Consultations conducted in 2 workshops in the project area (January 10 and 14, 2019)

Have the needs of specific (and vulnerable) sub-groups been taken into account by the project/program (e.g. children, girls, women and men with disabilities, the elderly, widows)?

The project especially takes into account women and young people, integrating them as direct beneficiaries of the project by 40% and 20% respectively. Other subgroups are not differentiated as direct beneficiaries of the project, although because they are part of the families served directly and indirectly, they will benefit from the implementation of the project.

Has the project/program recognized the distinct vulnerabilities of women and men and developed specific response strategies for each target group?

The Programme recognizes the importance of addressing the gender restrictions and inequalities that mainly affect women and adopts measures so that producer organizations not only increase the participation of women but also identify participatively the restrictions that affect their participation and act to eliminate them.

Are the specific knowledge and skills of women and men, especially from vulnerable groups, being utilised to contribute to project/program outcomes and solutions?

The meeting of knowledge and specific skills of men and women, with the expertise, technological innovations and financing provided by the project will make it possible to achieve the expected results and impacts on the families and communities of the selected area. The different characteristics, forms of participation and

experiences in the production and adaptation to climate change of men and women, will be used to their own benefit by providing opportunities for economic empowerment and increasing their resilience to adverse climate events.

Has the project/program identified opportunities to challenge gender stereotypes and increase positive gender relations through equitable actions? If so, what are these opportunities and actions?

The project considers training in gender equality as the best resource to modify the sociocultural aspects that act in the stereotyped assignment of roles and values for men and women. This training will be carried out to all the groups directly linked to the implementation of the project, namely PMU, MED and MOA, consultants, field technical teams, and project participants. This training, as far as project implementers are concerned, should provide practical tools for its action with a focus on gender equality, according to specific responsibilities. The effects of this training will be monitored constantly in order to reinforce when necessary.

Additionally, the project contemplates the allocation of economic resources for the development of practical actions that favor the participation of women in the project and can be actively integrated into the different activities.

Another important action is the assignment of women's participation goals that mean an increase in relation to the situation existing at the beginning of the project, seeking greater equality between men and women.

Strategy for Gender Equality and Social Inclusion

This assessment including the focus groups developed, as well as the preliminary study presented in appendix 2 particularly sections A and C, in relation to gender and the situation of women and youth served as the basis for the formulation of the project strategy gender equality and social inclusion.

Following IFADs participatory approach and operational model, the project will use three instruments throughout project implementation to gather the additional information to feed development of gender sensitive IPPs: rolling baseline studies at project beginning, the Climate Vulnerability Assessment and the Value Chains Analysis and Market Assessment. These will in turn be utilized to shape project specific action plans under strict guidelines to be included in the Programme Implementation Manual.

In the case of baseline studies, the indicators generated will be gender sensitive and differentiated by age groups between young people and adults. (PIM, 4.7). In addition, through the studies included in component 1 (Appendix 4, 1.1.), specific information on climate vulnerability and value chains will be obtained. Specifically, The Climate Vulnerability Assessment, as part of its bottom-up approach, will provide information on the vulnerability and adaptive capacity of different social groups, including groups of women and youth, providing specific information for each of the 5 priority areas. The Climate Vulnerability Assessment will translate these findings into gender-sensitive guidelines to prioritize management strategies for climate change adaptation. Value Chains Analysis and Market Assessment will provide particular attention will be given to analysing the role of women and youth. The methodology should specifically consult, and identify opportunities for both women and men. This information will be used to improve efficiencies in the production, processing and marketing of the products and promote youth and women's participation in all stages of the value chain through the implementation of the business plans.

Complementing the Development of Infrastructure Production Plans (IPP), built for each area based on the Climate Vulnerability Assessment and the Value Chains Analysis and Market Assessments, an analysis and consolidation of the information collected on gender and social inclusion will be carried out. Thereby supporting definition of opportunity niches for increased and strengthened participation of women and young people in climate resilient actions and value chains. The elaboration of Plans of Action on Gender (and social inclusion) in each area, based on the information obtained from the studies mentioned above, during the first year of implementation of the programme, will serve to achieve greater specificity and concreteness in the proposals of the Strategy.

The Strategy of Social Inclusion and Gender is based on the mainstreaming of both concepts throughout most of the activities, in order to guarantee the increase in the participation of women and young people, the strengthening of their capacities in different fields and their access to all the benefits offered by the program. Some specific activities are carried out as affirmative measures and have a differentiated budget.

- In addition to the above, component 1 includes Producers Organizations Strengthened (Subcomponent 1.2) that includes training. For this, Organizational Development Plans will be developed. They will take into account gender differences, as well as the specific needs of women and young people. This program includes among its themes: gender equality and social inclusion, which will be given to all beneficiaries. It also contemplates the realization of Equality -Gender learning route for beneficiaries and a "pilot" Program that includes the preparation of young men and women (50% women) from the communities to be managers of the Local Organizations
- As part of Subcomponent 1.3: Climate Resilient production and marketing, organizations will design value chain Business Plans (BPs). One of the principles that will guide the development of BP will be: Social Inclusion and Gender: The organization must show the expected number of women and youth partners, (close to 40% women and 20% youth). Based on a participatory analysis, each organization will define the measures to be taken to encourage the participation of women and youth and make affirmative action be necessary. These actions are part of the PB of the organization.
- For the selection and financing of productive initiatives through the Matching Grant Fund though (MGF), measures are included such as:
 - o The Calls for proposals for the Matching Grants Fund, will explicitly mention the Program's target group and specifically invite young people and female farmers to participate.
 - o Promote the participation of women and young people and their level of participation in the organization will receive different scores. If they meet the percentages of 40% and 20% or more, they will receive the highest score.
 - o One cross cutting sub-category of the MGF investment will be affirmative action in favor of women and youth or other at need groups. For example, domestic load relief, and other specialized courses. Regulations to be specified in PIM
- Regarding Technical Assistance, they will guarantee equal participation of women in the same proportion as that of their involvement in POs for Climate Resilient production planning and marketing. The MoA Climate staff and extension agents (PMU) will receive training in gender equality concepts and practical tips as to how to ensure equal male and female participation in extension, training, farmer to farmer training, and technical assistance

- While women are less involved in marketing, the Program marketing support will take into consideration that between 14-18% of marketing of grains, permanent crops, and vegetables are carried out by women, and they must be part of the participants in training, fairs, buyer / seller encounters.
- Backyard gardens for Increased resilience and food-security, contemplate a greater participation of women and will be promoted to approximately 60%.
- Component 2 Climate Resilient Rural Infrastructure and Assets (CRRIA) (Sub-component 2.2) considers that the participation of women should be promoted during the creation and strengthening of The water users' groups (WUGs). The program will facilitate the participation of women as members and as leaders involved in decision making in organizations on water use
- Research and Development is considered part of the Cross Cutting Activities. In these studies, special attention will be given to providing technological options adapted to the needs of women and young people, according to their participation in value chains and backyard gardens. It will also take advantage of the study of productive conversion possibilities of cane producers in the district of Corozal, to identify opportunities for women and young people to be integrated into value chains with alternative basic products. This action will reinforce the inclusion and gradual incorporation of Corozal District communities that meet the criteria established by the Program.
- Project will finance capacity building for GoB staff through its participation in a learning route to look at aspects related to climate resiliency, rural organization strengthening, and gender equality and social inclusion.

Impact:

The Program will increase the resilience to climate change of poor rural households to minimize the impacts of climate and economic events on small farmers in 23 communities in five Districts of Belize, while supporting sustainable market access for their products. The main criteria for identifying communities include the presence of both male and female smallholder farmers.

Outcome:

The direct beneficiaries will be 6,000 rural households and an estimated 30,000 people of which 12,000 will be women (40%) and 6,000 will be young (50% women). All of them will benefit from the climate vulnerability assessment and mitigation plans that will be developed in a participatory manner. Approximately 2,000 of these households will also benefit from support in the establishment or improvement of backyard gardens as one of the strategies to increase climate change resilience (60% of direct beneficiaries will be women). 2,000 small-scale farmers will benefit from climate resilient training and technical assistance, investments through the (MGF), organizational strengthening, and market access support. This group of 2,000 small-scale farmers would be made up of 40% "poor farmers" and 60% "vulnerable farmers" and 30% of the beneficiaries will be women. Specific outcomes are: strengthened institutional and regulatory systems for climate-responsive planning and development, increased generation and use of climate information in decision-making and strengthened adaptive capacity and reduced exposure to climate risks.

Output(s) and activities:	Indicators and Targets	Timeline	Responsibilities
1. Infrastructure and Production Plan (IPP)	23 IPPs developed at community level	Q8	
Activity 1.1. Develop a Climate Vulnerability Assessment	Climate Vulnerability Assessment includes information on vulnerability and adaptability of women and youth	Q1 - Q6	PMU
Activity 1.2. Develop a value-chain analysis and market assessment	Identified the roles, potentialities and opportunities for participation of women and youth throughout the value chains and markets	Q1 - Q6	PMU
Activity 1.3. Develop the Infrastructure and Production Plan (IPP)	The IPP defines the opportunity niches for the participation of women and young people and the actions necessary to strengthen their participation in value chains, the basis for development of Organizational Development Plans for each Producers Organizations, Business Plans (BPs) proposed by the POs, including specific measures to strengthen the participation of women and youth actions.	Q3 – Q8	PMU

	This IPP will be accompanied by a summary of the findings on women and youth and with it draw up a Gender Plan which incorporates the strategy of social inclusion and gender project		
2. Producers Organizations Strengthened	450 rural formal members of organizations supported. (180 mujeres and 90 youth)	Q20	PMU
Activity 2.1. Training	Capacitation included in The Organizational Development Plans take into account gender differences, as well as the specific needs of women and young people	Q2 – Q8	PMU
	The training program includes among its themes: gender equality and social inclusion, which will be given to all beneficiaries	Q2 – Q8	PMU
	Equality - Gender learning route for beneficiaries developed	Q10	PMU
Activity 2.2. Develop internal administrative instruments	The Pilot Local Managers Program prepared 36 young people (50% women) to be managers of the local organizations	Q2 – Q8	PMU
3. Climate resilient production and marketing	700 persons (280 women and 140 youth) trained in business management and marketing.	Q20	PMU
Activity 3.1. Design of Business Plans (BPs)	The organization meets the expected number of women and youth partners (close to 40% women and 20% youth) to be a candidate for a BP .	Q4 – Q11	POs
	Affirmative actions to encourage the participation of women and youth are part of the PB of the organization. They are justified by a participatory analysis, and each organization define the actions to be taken.	Q4 – Q11	POs

Activity 3.2. Select and finance through Matching Grant Fund (MGF)	Select criteria and finance organizations favor the participation of women and young people: i.e. If they meet the percentages of 40% women and 20% youth, the proposal will receive the highest score.	Q6 – Q20	PMU
	Affirmative actions in favor of women and youth or other at special groups is a cross cutting sub-category of the MGF investment. Regulations to be specified in PIM.	Q6 – Q20	PMU
Activity 3.3. Technical assistance	Equal participation of women in the same proportion as that of their involvement in POs for Climate Resilient production planning and marketing.	Q4 – Q20	PMU
	The MoA Climate staff and extension agents (PMU) were training in gender equality concepts and practical tips as how to ensure equal male and female participation in extension, training, farmer to farmer training, and technical assistance	Q4 - Q12	PMU - MoA
4. Rural Roads Improvements and Small-scale Irrigation and Drainage (SSID)	2820 households directly benefitting from improved public infrastructure	Q20	PMU
Activity 4.4. Institutional development	Participation of women in organizations on water use is 40% of the members and 30% of them occupy positions of decision making	Q6 – Q12	PMU
5. Climate Information System (CIS).	30,000 households members directly benefitting from climate information system. 12,000 women and 6,000 youth	Q20	PMU
6. Government Strengthened	30 Specialized trainings developed including Gender equality in all of them	Q20	PMU

Activity 6.3. Develop studies	The studies on Agricultural Adaptation Strategies, Ecosystem Services Payments, and Smallholder Cane Farmer Production Diversification include the gender equality approach.	Q2 – Q10	PMU
Activity 6.4.Capacity building for MoA	30 Specialized trainings developed including gender equality in all of them	Q3 – Q20	PMU
	Supported technical activities identified by the Gender Focal Point in MoA, such as studies, workshops, etc. related to issues related to BeResilient.	Q3 – Q20	Gender Focal Point MoA
7. Backyard gardens for increased resilience and food-security	2000 households developing backyard gardens. Participation of women will be 60% of direct beneficiaries.	Q20	PMU